



- Equipment**
- 0102H [GCs+HWJ {R.29}*]
 - 0108A [Add+USB {R.26}]
 - 0109A [Network customer RJ45 {R.25}]
 - 0202H [AX5 X/Xa Safety OCT 2x6A {R.31}]
 - 0302A [Exhausting, Remote control {R.29}]
 - 0407E [2xCabinet W600H800D400+MP {R.29}]
 - 0410A [ECO mounting bundle kit {R.26}]
 - 0411A [Basic Parts TSc {R.31}]
 - 0605C [Emergency cord {R.29}]
 - 1102A [Main Cabi supply S+N switched {R.29}]
 - 1201C [Additional Safe-Stop 2 {R.29}]
 - 1501A [Cabinet cooling, Door, 75W {R.25}]
 - 0202G [L1-AX5 Y OCT Safety 3A {R.31}]
 - 0406A [L1-Plasma Interface Cabinet {R.29}]
 - 0501B [L1-SL200 {R.29}]
 - 0904I [L1-HYP XPR170core+SwBox {R.30}]
 - 1001A [L1-Laser Pointer {R.25}]
 - 1107A [L1-Transform 36V4x100VA {R.25}]
 - 1110A [L1-EP PowerBox 4x4A {R.29}]
 - 1301A [L1-Havarie, Torchholder, Magnetic {R.25}]
 - 1301B [L1-Havarie tactil lift up {R.25}]

Messer Cutting Systems GmbH

Headquarters

Otto-Hahn-Straße 2-4
D-64823 Groß-Umstadt / Germany



Customer:
Blocotelha

projekt 289
machine 289 MultiTherm Eco
length / width 80 / 40
power 9 kVA
Toolbox Toolbox_31
drawing No. 289001273_Blocotelha

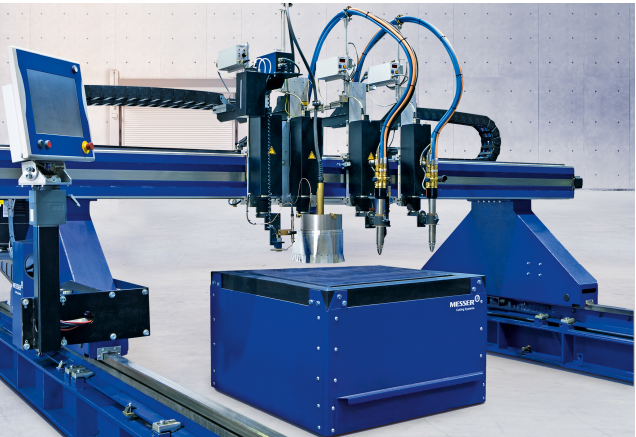
editor Thomas.Lehle
building year 2021
regulation EN60204-1

shop floor: **EU**

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Number of pages 146

Eplan-version 2.6.3
10713

machines number: 289

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0	27.01.2021	marcel.gruber					This version was created by ABIGAIL
Doku	31.03.2021	Thomas.Lehle					Customer
Index	Date	Name	Date	Name	Date	name	Remarks
Revision	Created on		reviewed		Approved		

Documentation evaluation

directory to Modul Overview location aware

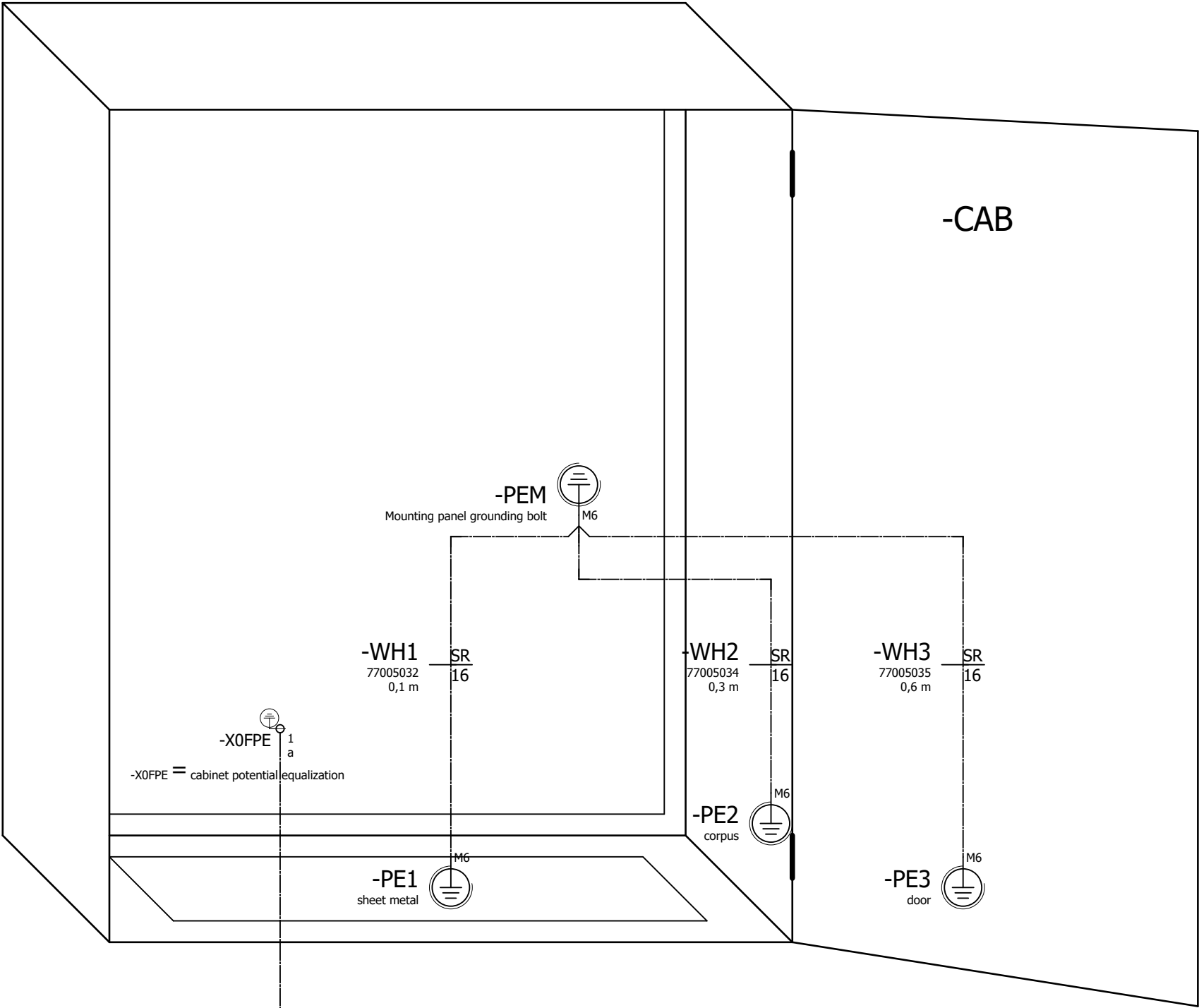
\$-FOS-F06-3A

module		description		group box	page
[=]	1B	table of contents		Module Overview	2...4
[=]	4A	topology EtherCAT		Bus Kabel;SPS-Kartenübersicht	12...18
[=]	Z6A			layout plan	103...109
[=]	Z6B			connection plan	110...143
[+]	B (gantry)				
[=]	3CB	grounding bar		connection points	9
[=]	BI	DC power supply interface	EP power - distributor	24VDC, EtherCat;rack overview	42;43
[=]	IB	interface box	Fieldbus star hub	power supply, EtherCat;rack overview	65;66
[+]	E (end truck left)				
[=]	ELC	Not-Halt Reißleine	Reißleine vorne	Safety circuit	61
[=]	ELD	Not-Halt Reißleine	Reißleine hinten	safety circuit	62
[=]	MX	motor X	AM8531 OCT X	connection motor encoder	76
[=]	OB	operation panel Keyboard	hand wheel - Joystick	EtherCAT;rack overview	80;81
[=]	OP	operation panel	Operation interface	EtherCAT;emergency stop, operating elements;operating elements;rack overview;supply, Akku	82...86
[=]	PG1	plasma gas console	type core	cooling gas connection console;power supply gas;rack overview	90...92
[=]	UTB	unit temperature	climatisation	backplane operation panel GCs	102
[+]	G (end truck right)				
[=]	EA	emergency stop additions	emergency stop	connection	55
[=]	MXa	motor Xa	AM8531 OCT Xa	connection motor encoder	77
[+]	HB (cabinet main)				
[=]	1A	cover sheet	cover sheet	Messer	1
[=]	3B	grounding cabinet		grounding points in cabinet	5
[=]	3BB	grounding cabinet		grounding points in cabinet	6
[=]	3D	grounding machine	grounding machine	overview	10
[=]	AA	AC power supply cabinet	AC operating voltage	main switch, PE/N;potential distribution L1/L2/L3;protective conductor / neutral conductor	19...21

			Toolbox_31	Option:	1			module	table of contents	Page	
			Toolbox version	customer:	Blocotelha						
			automated	commission:	289001273						
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle		01.04.2021				
							Module Overview				
							sector	location	+A	=1B	
										2	

directory to Modul Overview location aware

			Toolbox_31	Option: 2			module		table of contents		Page	
			Toolbox version	customer: Blocotelha								
			automated	commission: 289001273								
Modification	Date	Name	evaluation has been generated	Ed. by: Thomas.Lehle	01.04.2021		Module Overview	sector	location	+A	=1B	3



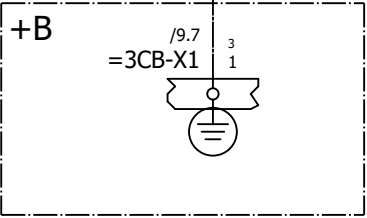
-CAB

-CAB			
parts list // LN folder 72881342			
piece	article	ID	
1	77071312	V	corpus, single door, 600 x 800 x 400, mounting plate

+B
gantry

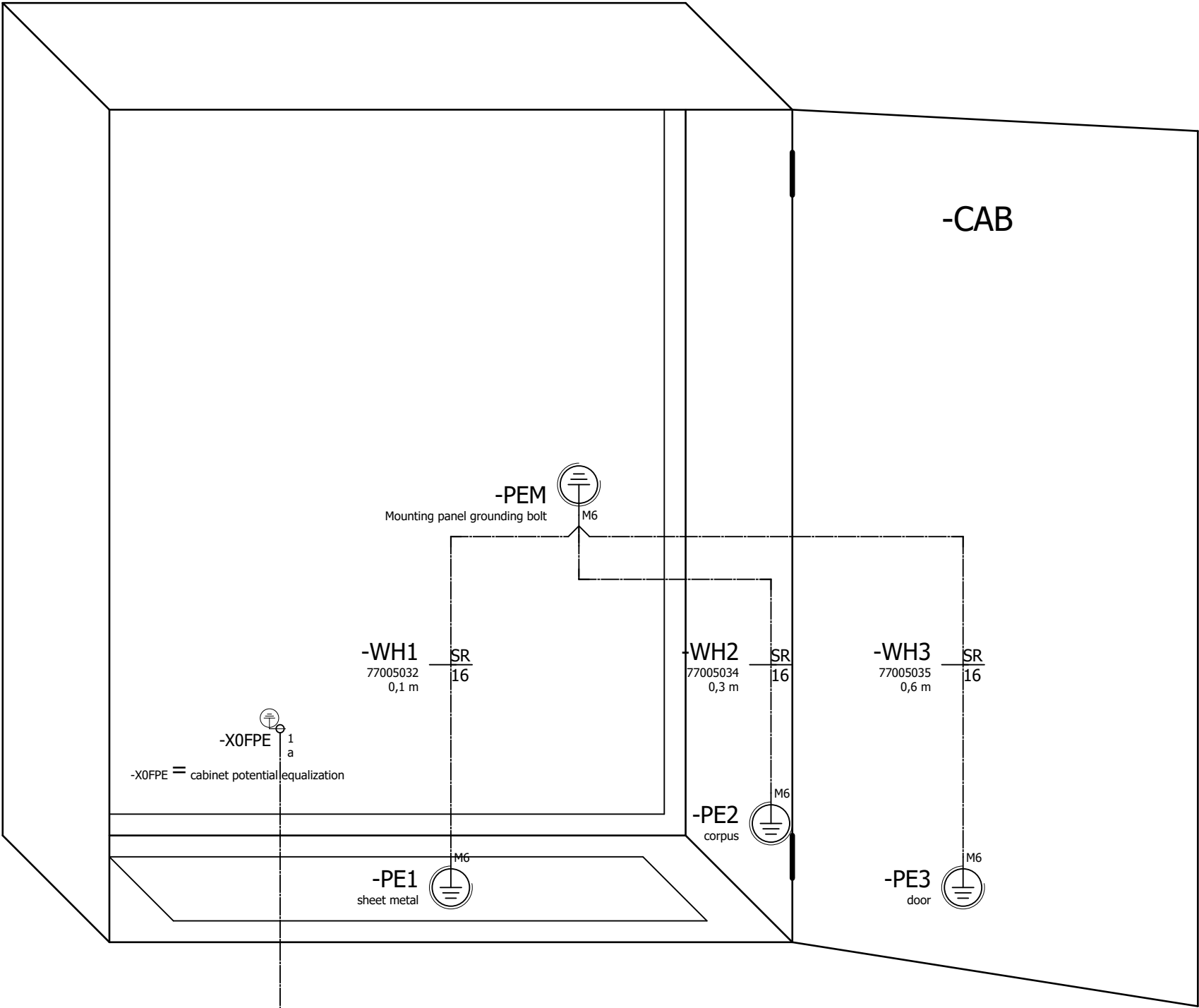
-A

+B-A			
parts list // LN folder 72881345			
piece	article	ID	
1	72161810	V	mounting material
piece	article	ID	
1	77005344	V	Sticker, transport monitoring



[2]600x800x400+MP

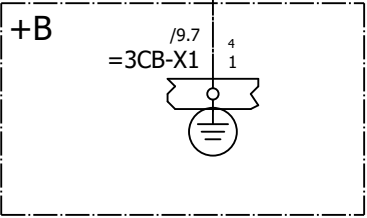
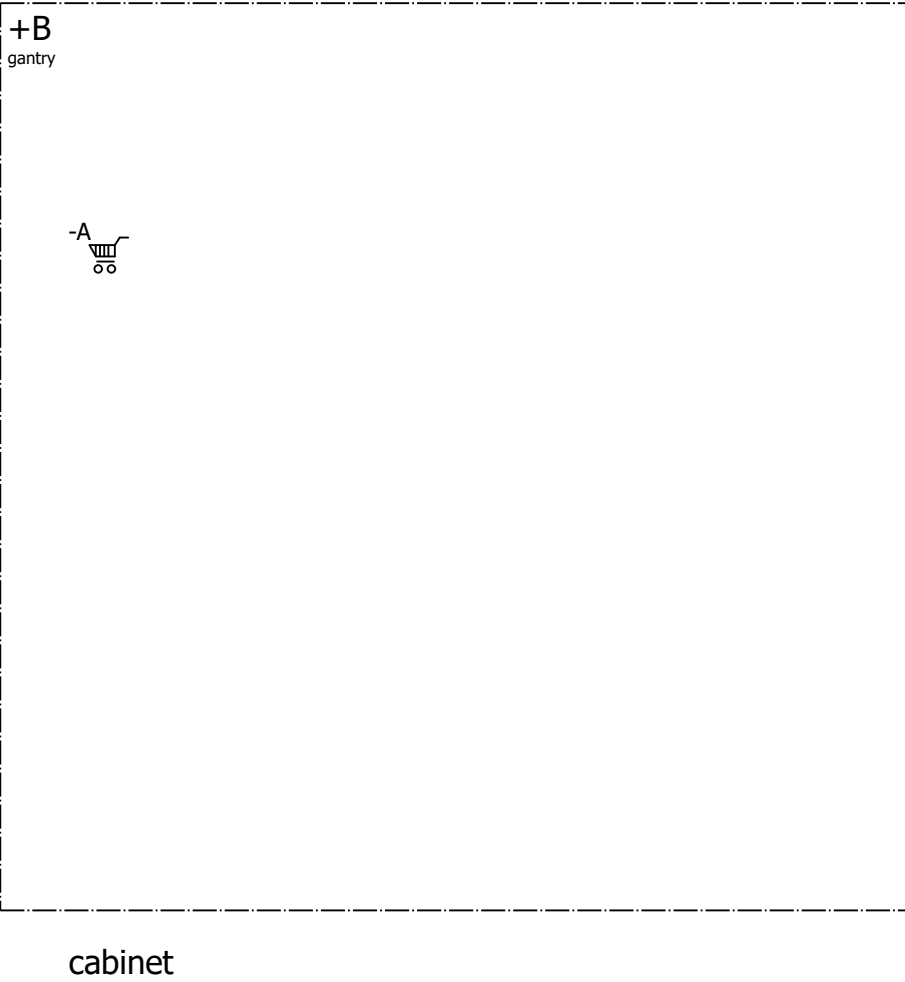
cabinet

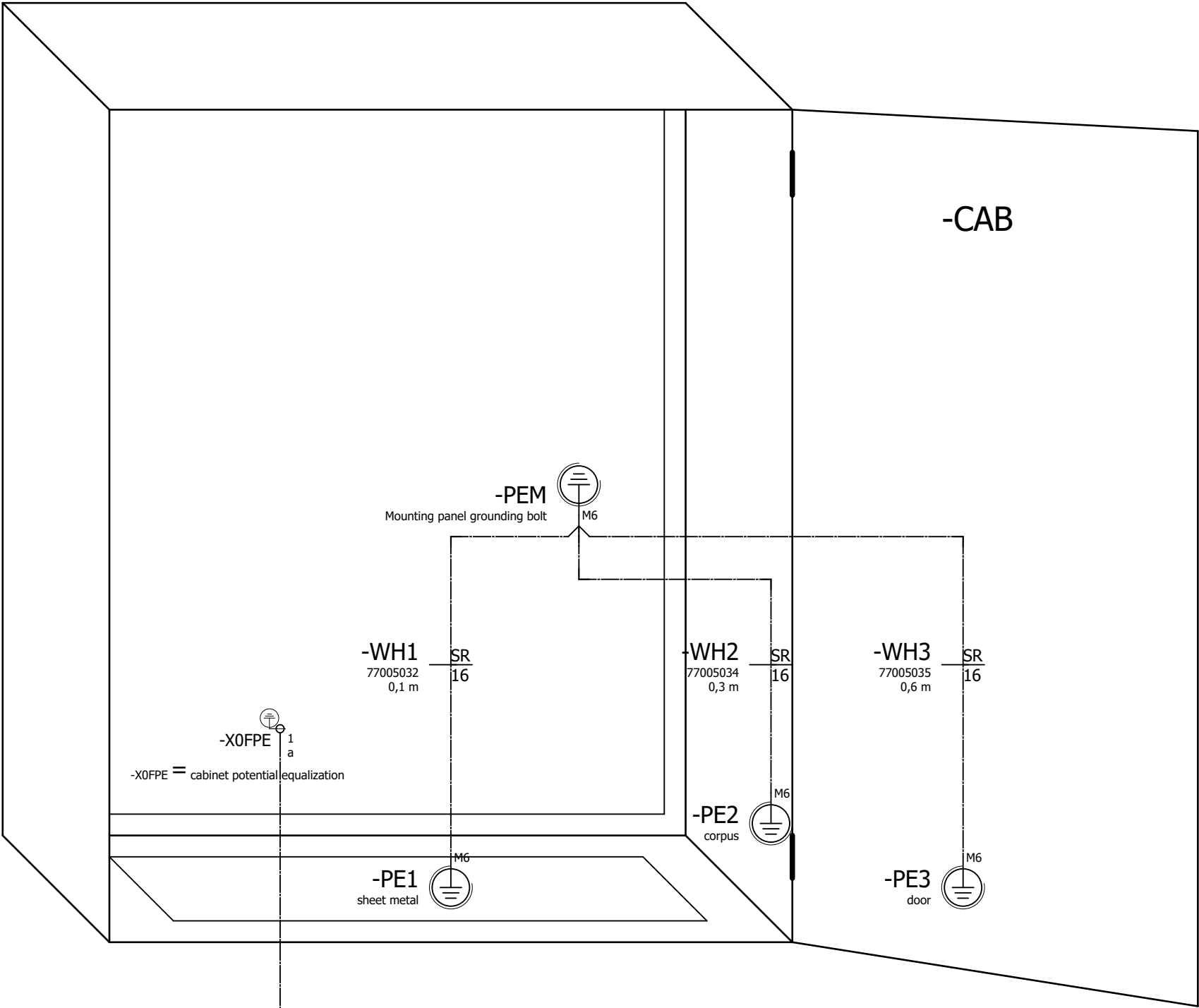


-CAB

parts list // LN folder 72881342

piece	article	ID	corpus, single door, 600 x 800 x 400, mounting plate
1	77071312	V	





-CAB

-CAB		
parts list // LN folder 72881342		
piece	article	ID
1	77005896	V

+B
gantry

-A

[8]500x500x210

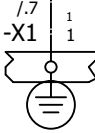
cabinet



-W1
CA0065
14 m



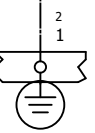
GNYE
70



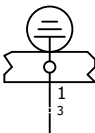
-W2
CA0066
14 m



FPE
70



-W12
CA0061
0,2 m



FPE
70

-GND
table



X1

-X1 = potential equalization machine



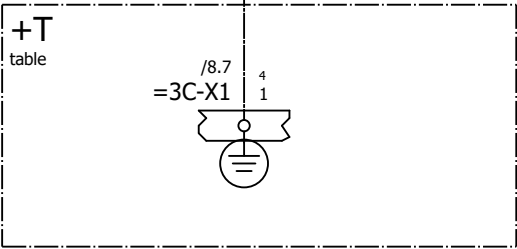
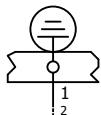
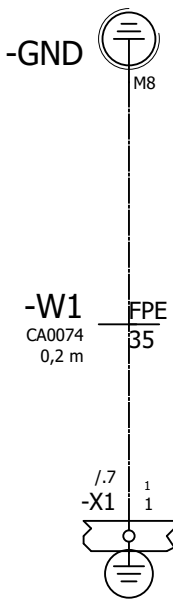
jumper

•	1 - grounding customer / /8.1	G00000775
•	2 - functional grounding customer / /8.2	
•	3 - grounding table / /8.4	
•	4 - grounding table / =3CB+B/9.3	
•	5 - cabinet / =3BC+HP1/7.1	
•	6 - torch connection console grounding / =PB1+L1/89.9	
•	7 - plasma power source functional ground / =PS1+P/95.8	
•	8 - /	
•	9 - /	
•	10 - /	
•	11 - /	
•	12 - /	

grounding
customer

functional grounding
customer

grounding
table



grounding
jumper

grounding
table

X1

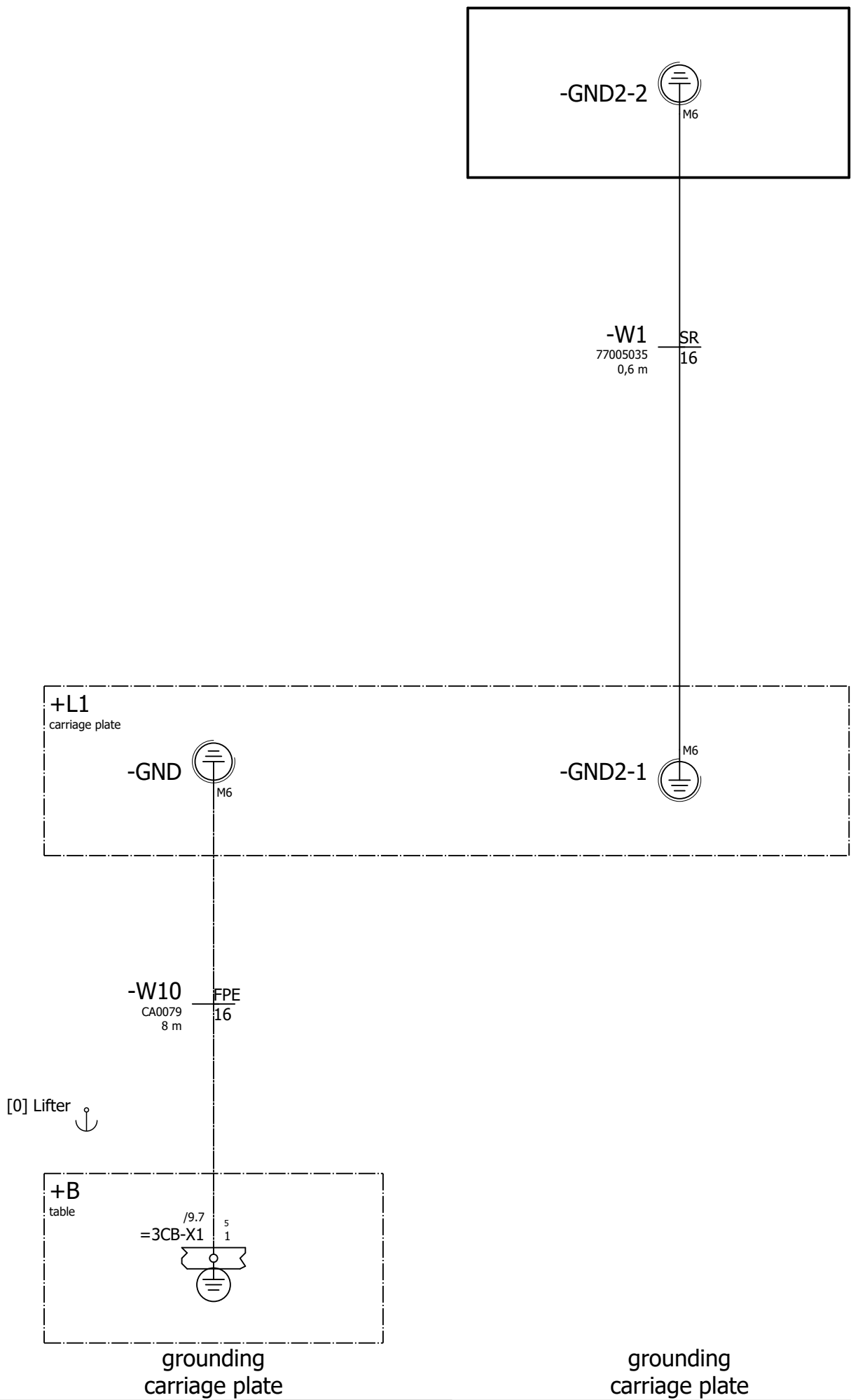
-X1 = potential equalization machine

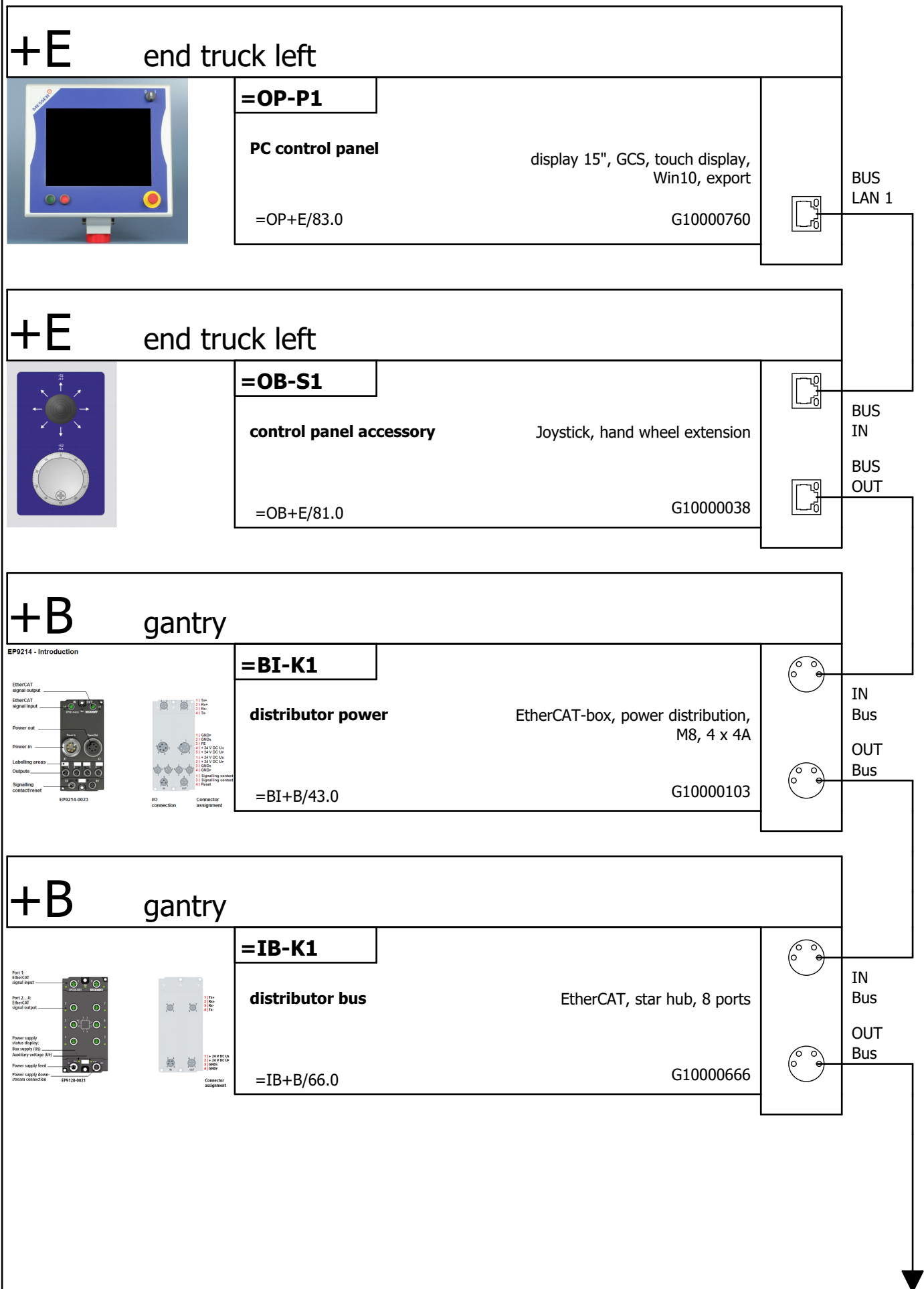


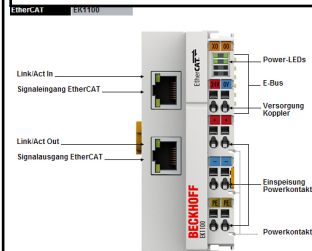
jumper

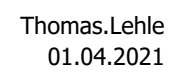
•	1	- grounding jumper / /9.1	G00000775
•	2	- grounding table / /9.3	
•	3	- cabinet / =3B+HB/5.1	
•	4	- cabinet / =3BB+HB/6.1	
•	5	- grounding carriage plate / =3E1+L1/11.1	
•	6	- panel grounding / =OP+E/84.9	
•	7	- gas console grounding / =PG1+E/91.9	
•	8	- /	
•	9	- /	
•	10	- /	
•	11	- /	
•	12	- /	

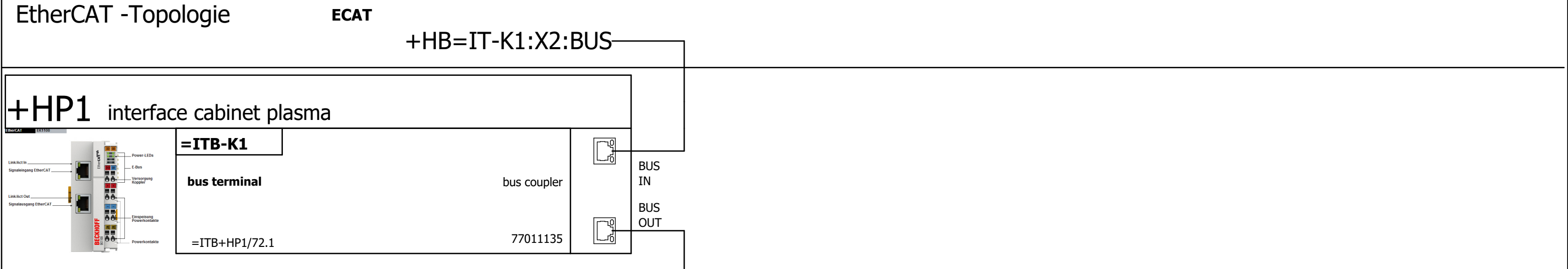
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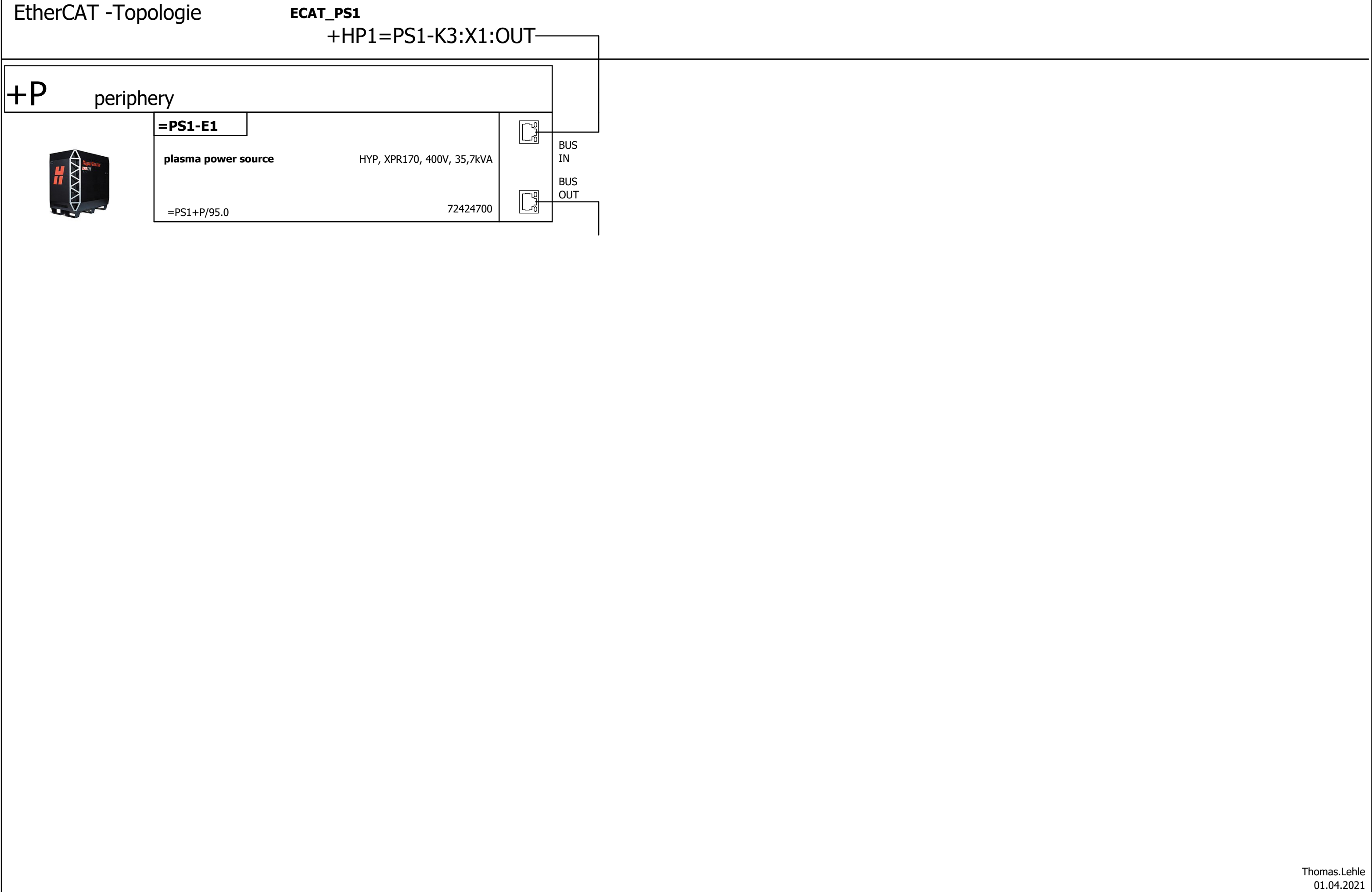












+HB=IT-K1	+HB=IT-K2	+HB=IT-K3	+HB=IT-K4	+HB=IT-K5	+HB=IT-K6	+HB=IT-K7	+HB=IT-K8	+HB=IT-K9
77011135 EK1100	77011261 EL1904 Twin Safe ON X X X X X X X X 12345678 ON X X X X X X X 12345678	77011261 EL1904 Twin Safe ON X X X X X X X X 12345678 ON X X X X X X X X 12345678	G10000441 EL1859 ↓ ↕ ↓	77011261 EL1904 Twin Safe ON X X X X X X X X 12345678 ON X X X X X X X X 12345678	77011261 EL1904 Twin Safe ON X X X X X X X X 12345678 ON X X X X X X X 12345678	77011262 EL2904 Twin Safe ON X X X X X X X X 12345678	77011262 EL2904 Twin Safe ON X X X X X X X X 12345678	77011260 EL6900 Twin Safe ON X X X X X X X X 12345678
24L+ 24L- PE	24L+ 24L-	24L+ 24L-	24L+ 24L-	24L+ 24L-	24L+ 24L-	24L+ 24L-	24L+ 24L-	24L+ 24L-
=IT+HB/67.1	=IT+HB/68.1	=IT+HB/68.6	=IT+HB/69.1	=IT+HB/69.6	=IT+HB/70.1	=IT+HB/70.6	=IT+HB/71.1	=IT+HB/71.6

Safety address: Dez. <11>

Safety address: Dez. <1>

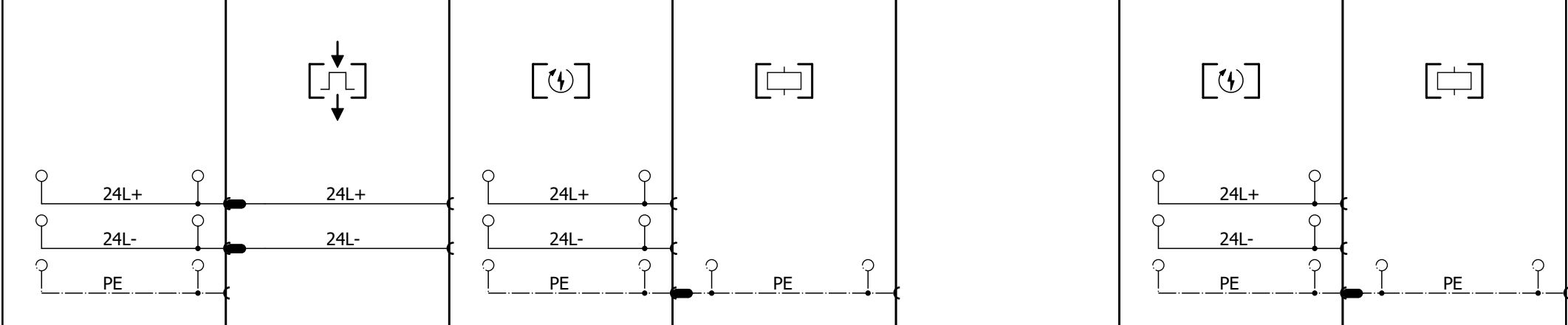
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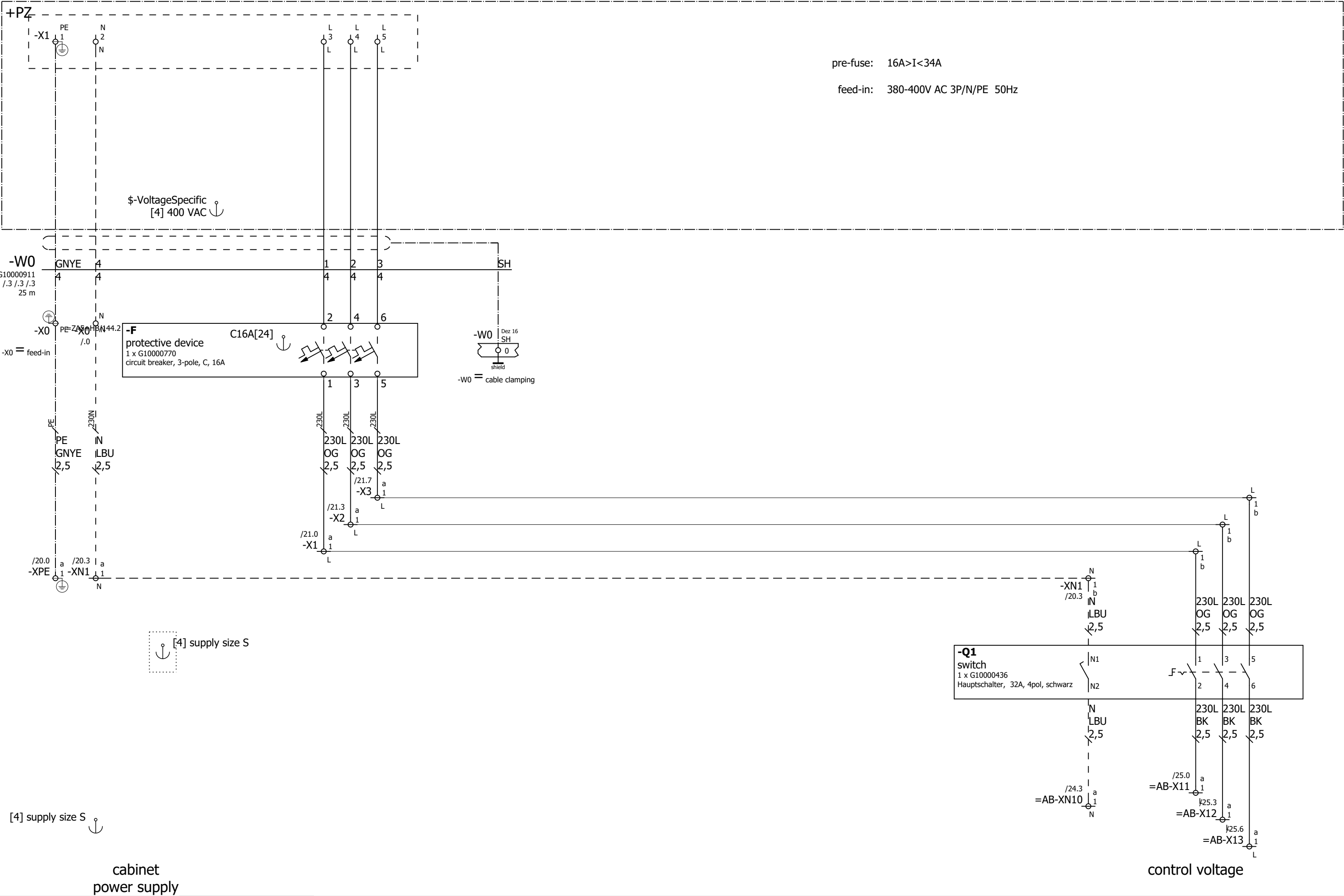
Safety address: Dez. <36>

Safety address: Dez. <7>

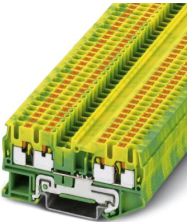
Safety address: Dez. <6>

Safety address: Dez. <2>

+HP1=ITB-K1	+HP1=ITB-K2	+HP1=PS1-K1	+HP1=PS1-K2	+HP1=PS1-K3	+HP1=UE-K1	+HP1=UE-K2
77011135 EK1100	G10000441 EL1859	77011215 EL9410	77011138 EL2622	G10000781 EK1122	77011141 EL9100	77011139 EL2612
						
=ITB+HP1/72.1	=ITB+HP1/73.2	=PS1+P/95.1	=PS1+P/95.4	=PS1+P/95.7	=UE+P/99.7	=UE+P/99.5



XPE



-XPE = potential distribution
protective conductor

jumper

•	1	77070158
	a - service power socket	/19.0
	b - service power socket	=AS/30.5

XN1



-XN1 = potential distribution
neutral conductor

jumper

•	1	G10000679
	a - control voltage	/19.0
	b - control voltage	/19.8
	c - service power socket	=AS/30.3

[4] supply size S

potential distribution
protective conductor

potential distribution
neutral conductor

X1

-X1 = potential distribution L1

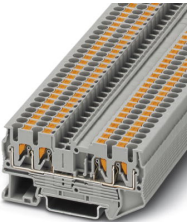


jumper

•	1	77070157
	a - cabinet power supply /19.2	
	b - control voltage /19.8	
	c - service power socket =AS/30.3	
•		

X2

-X2 = potential distribution L2



jumper

•	1	77070157
	a - cabinet power supply /19.2	
	b - control voltage /19.8	
•		

X3

-X3 = potential distribution L3



jumper

•	1	77070157
	a - cabinet power supply /19.2	
	b - control voltage /19.8	
•		



[4] supply size S



[4] supply size S

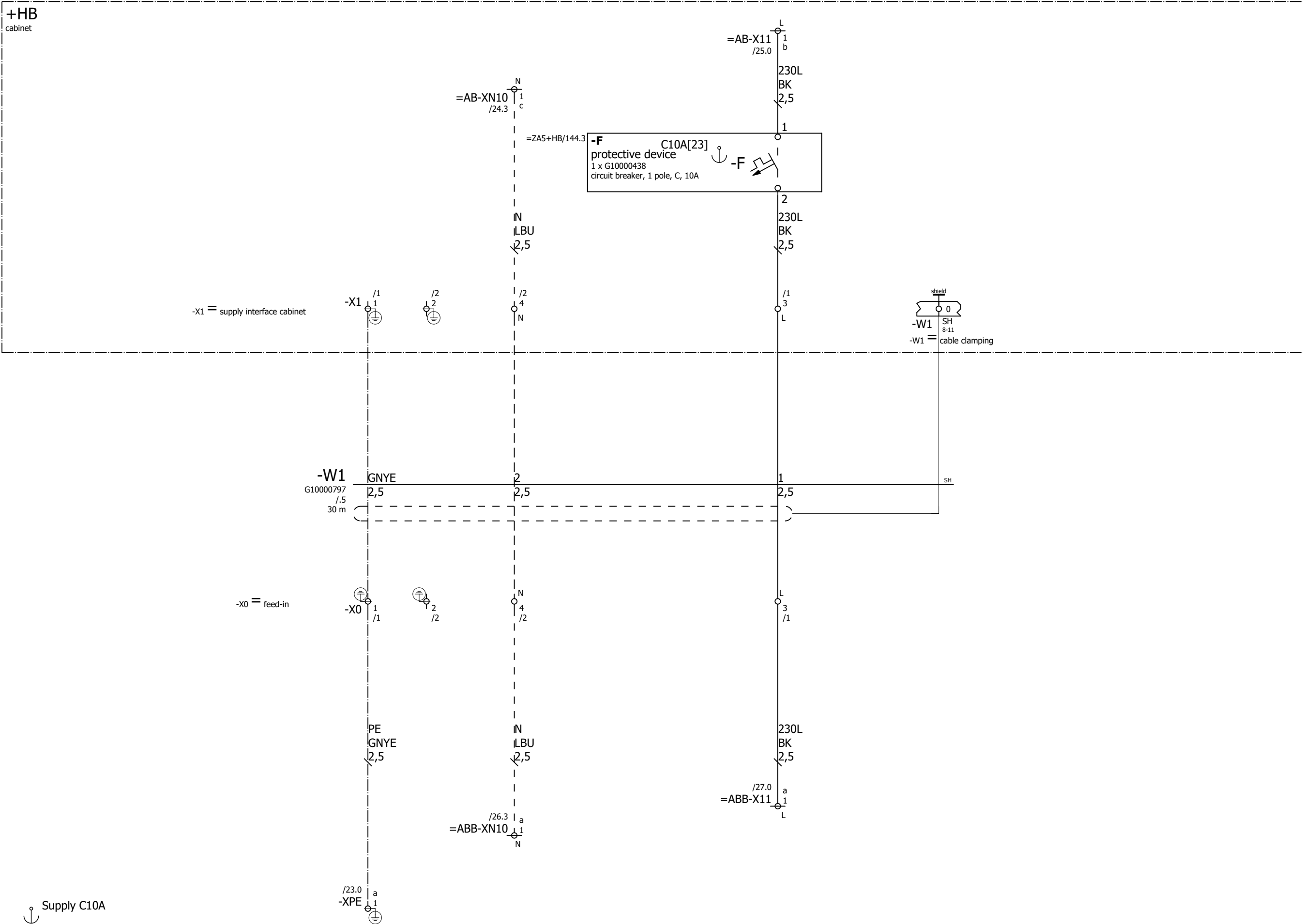
#

potential distribution
phase L1

potential distribution
phase L2

potential distribution
phase L3

#05	01.04.2021	Thomas.Lehle	Toolbox_31	Option: customer: commission:	Main Cabi supply S+N switched {R.29}		AC operating voltage		module AC power supply cabinet		
#03	25.03.2021	Thomas.Lehle	Toolbox version		Blocotelha		potential distribution L1/L2/L3				
#01	15.03.2021	Thomas.Lehle	AA2 - ABI - 29.29.08 - CC: EU		289001273						
Modification	Date	Name	Toolbox module designation	Ed. by.	Thomas.Lehle	01.04.2021			sector 1C2G location	+HB	=AA



+HB
cabinet

-X1 = supply interface cabinet

-W1
G10000797
/2,5
30 m

-X0 = feed-in

interface cabinet
supply

-F
C10A[23]
protective device
1 x G10000438
circuit breaker, 1 pole, C, 10A

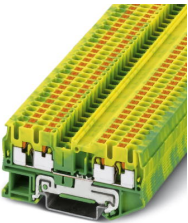
-W1
SH
8-11
cable clamping

Supply C10A

interface cabinet
mounting plate

01	28.01.2021	Adeline.Wegang	Toolbox_31	Option:	L1-Plasma Interface Cabinet {R.29}		feed-in		module		AC power supply cabinet		Page 22
			Toolbox version	customer:	Blocotelha		X0 , PE/N/1~						
			AA5 - ABI - 29.29.01 - CC: EU	commission:	289001273								
Modification	Date	Name	Toolbox module designation	Ed. by.	Thomas.Lehle	01.04.2021			sector	3E7P location	+HP1	=AAB	

XPE



-XPE = potential distribution
protective conductor

jumper

•	1	77070158
	a - interface cabinet supply	/22.2
	b - filter fan grounding / PE	=UE+P/99.6

Supply 10A

potential distribution
protective conductor

XN10

-XN10 = potential distribution
operating voltage N



jumper

•	1	G10000679
a - control voltage =AA/19.8		
b - 230N =BA/32.4		
c - interface cabinet supply =AAB+HP1/22.3		
d - transformer supply / =AL/29.1		

 [4] supply size S

operating voltage
neutral conductor

X11

-X11 = potential distribution
operating voltage L1



jumper

•	1	77070157
a - control voltage =AA/19.8		
b - interface cabinet supply =AAB+HP1/22.5		
c - controller power supply AC / L1 =CXXA/46.1		
d - controller power supply AC / L1 =CY1/49.1		

X12

-X12 = potential distribution
operating voltage L2



jumper

•	1	77070157
a - control voltage =AA/19.8		
b - transformer supply =AL/29.1		
c - controller power supply AC / L2 =CXXA/46.2		
d - controller power supply AC / L2 =CY1/49.2		

X13

-X13 = potential distribution
operating voltage L3



jumper

•	1	77070157
a - control voltage =AA/19.8		
b - operating voltage power supply 24VDC =BA/32.6		
c - controller power supply AC / L3 =CXXA/46.2		
d - controller power supply AC / L3 =CY1/49.2		

[4] supply size S

[4] supply size S

#

operating voltage
line L1

operating voltage
line L2

operating voltage
line L3

#05	01.04.2021	Thomas.Lehle	Toolbox_31	Option: customer: commission:	Main Cabi supply S+N switched {R.29}	MESSER	AC operating voltage	module	AC distribution	Page 25
#03	25.03.2021	Thomas.Lehle	Toolbox version		Blocotelha		lines L1/L2/L3			
#01	15.03.2021	Thomas.Lehle	AB1 - ABI - 29.29.08 - CC: EU		289001273					
Modification	Date	Name	Toolbox module designation	Ed. by.	Thomas.Lehle	01.04.2021		sector 1C2G location	+HB =AB	

XN10

-XN10 = potential distribution
operating voltage N



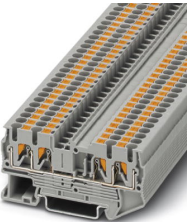
jumper

•	1	G10000679
	a - interface cabinet supply	=AAB/22.3
	b - plasma power source switch-off	=PA1+HJ1/87.8
	c - 230N	=BAB/35.4

Supply C10A

operating voltage
neutral conductor

X11



-X11 = potential distribution
operating voltage

jumper

•	1	77070157
a -	interface cabinet supply	=AAB/22.5
b -	=BAB/35.6	
c -	=PA1+HJ1/87.6	



Supply C10A

operating voltage
line

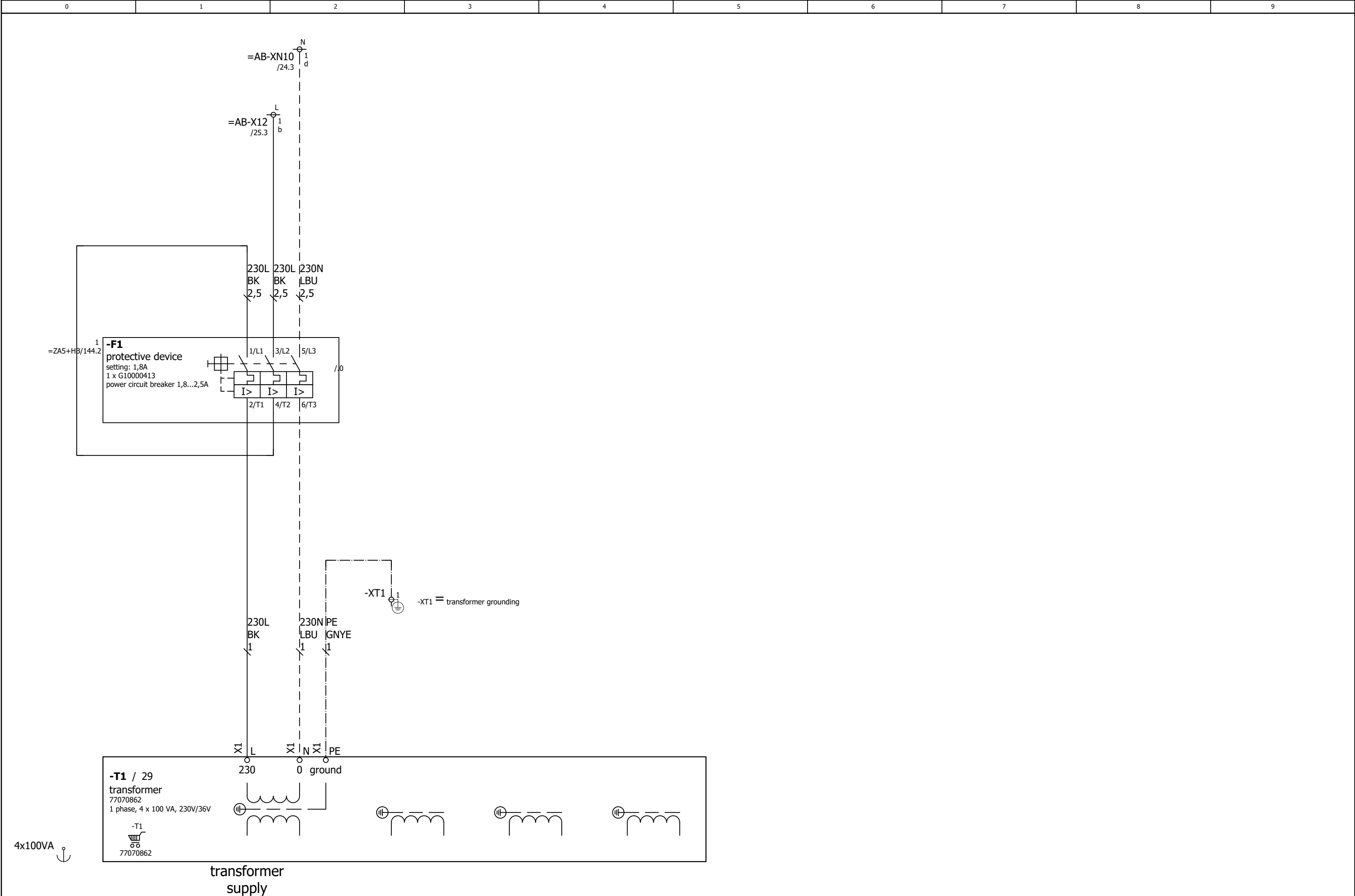


77070862

transformer

-T11 phase, 4 x 100 VA, 230V/36V

X1		
L	230	transformer supply /29.1
N	0	= /29.2
PE	ground	= /29.2
X2.1		
L	36	SmartLifter AC power supply =LI1+L1/75.3
N	0	= =LI1+L1/75.2
X2.2		
L	36	
N	0	
X2.3		
L	36	
N	0	
X2.4		
L	36	
N	0	





G10000827

protective device

-F1 electronic circuit breaker 10A

Contact

13

NO

14

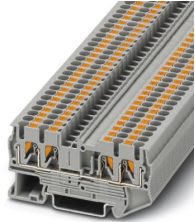
NO

IN+

1

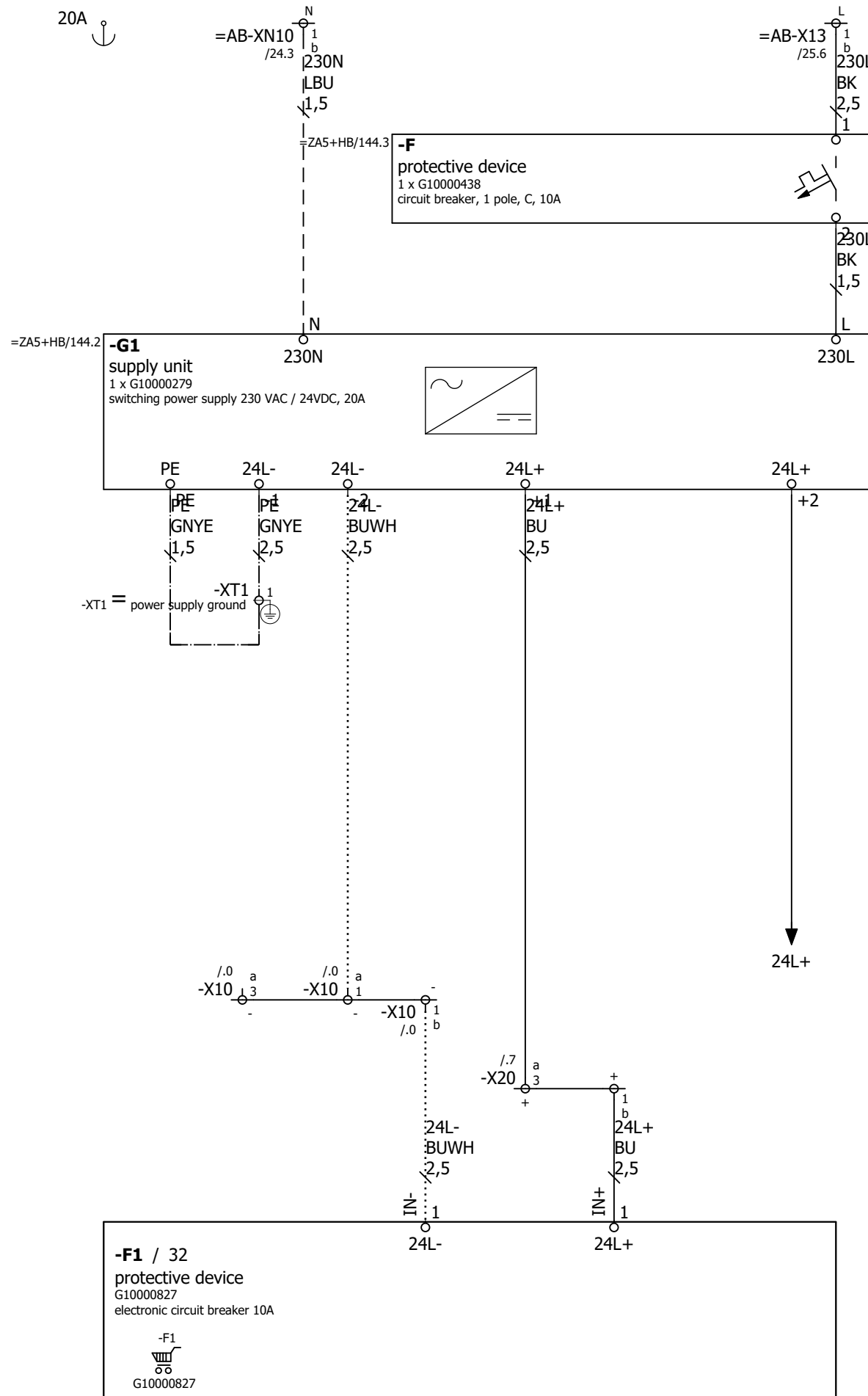
24L+

$\times 10^{-3}$ = potential distribution
operating voltage 24L-

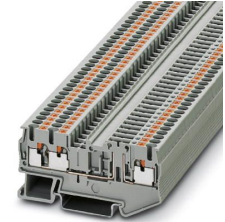


jumper

1	77070157
a - operating voltage power supply 24VDC / 24L-	/32.4
b - 24L-	/32.4
c - distribution 24 VDC external	=BI+B/43.1
d - control voltage 24Lbuffered / 24L-	=BU/44.4
3	77070157
a - operating voltage power supply 24VDC	
d -	
control voltage active /	=ER/64.8
2	77070157
a - potential distribution limiting Y+	=DL1+L1/53.3
b - sensor limit Y-	=DL1B+L1/54.3
c - Safety interface external	=EI/58.1
4	77070157
a - EtherCAT coupler operating supply / 24L-	=IT/67.6
b - EtherCAT coupler powerbus / -	=IT/67.7
c - SmartLifter DC operating voltage	=LI1+L1/75.5
d - operation panel supply /	=OP+E/84.4
5	77070157
a - operating voltage power supply 24VDC	=UT/101.2
b - cabinet cooling	=UT/101.3
c - operation panel cooling	=UTB+E/102.2



-X20 = potential distribution
operating voltage 24L+



jumper

1	G10000602;G10000459;541021	a - operating voltage power supply 24VDC =UT/101.1 b - operating voltage power supply 24VDC / 24L+ /32.5
2	G10000602;G10000459;65541022	a - control voltage power distributors =BB/39.5 b - SmartLifter DC operating voltage =LI1+L1/75.5 c - SmartLifter DC control voltage =LI1+L1/75.7
3	G10000602;G10000506	a - operating voltage power supply 24VDC / 24L+ /32.5 b - distribution 24 VDC external =BI+B/43.3 c - control voltage 24L+ buffered / 24L+ =BU/44.5
4	G10000602;G10000459;65541022	a - control voltage 24L+ channel 2 =BB/40.6
5	G10000602;G10000459;65541022	a - control voltage 24L+ channel 3 =BB/41.4
6	G10000602;G10000459;65541022	a - control voltage 24L+ channel 4 =BB/41.6
7	G10000602;G10000459;541021	a - cabinet cooling =UT/101.3
8	G10000602;G10000459;65541022	a - control voltage 24L+ channel 1 =BB/40.4

operating voltage
24L-

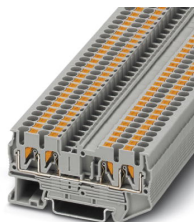
operating voltage
power supply 24VDC

operating voltage
24L+

#05	01.04.2021	Thomas.Lehle	Toolbox_31	Option: customer: commission:	Main Cabi supply S+N switched {R.29}		operating voltage	module	DC power supply	Page 32
#03	31.03.2021	Thomas.Lehle	Toolbox version		289001273		power supply			
#03	25.03.2021	Thomas.Lehle	BA1 - ABI - 29.29.00 - CC: EU		Ed. by. Thomas.Lehle 01.04.2021		sector 1C2G location	+HB	=BA	
Modification	Date	Name	Toolbox module designation							

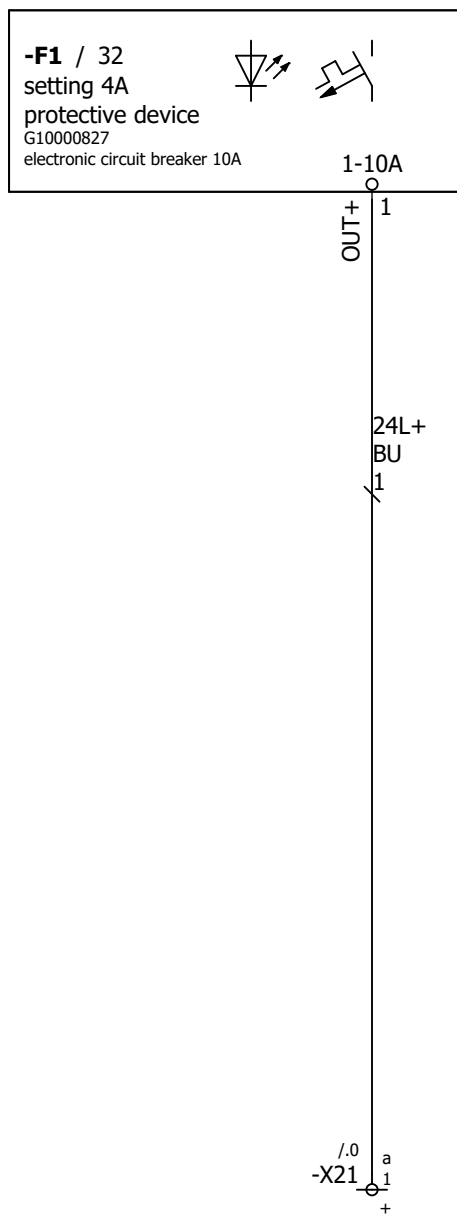
X21

-X21 = operating voltage 24L+
channel 1

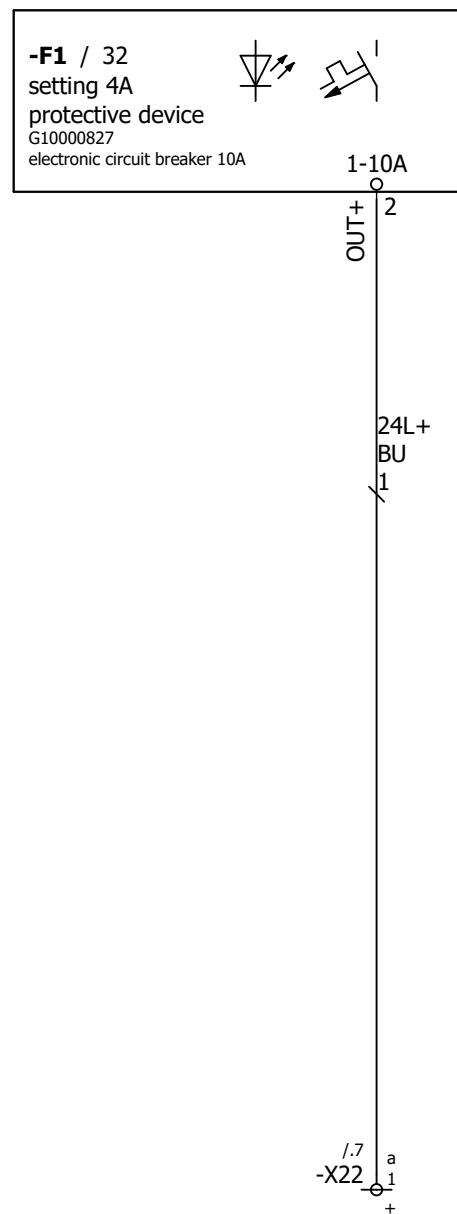


jumper

•	1	77070157
	a - operating voltage 24L+ channel 1 / 1-10A	/33.4
	b - potential distribution limiting Y+ =DL1+L1/53.3	
	d -	
	EtherCAT coupler operating supply / 24L+ =IT/67.5	



operating voltage 24L+
channel 1



operating voltage 24L+
channel 2

X22

-X22 = operating voltage 24L+
channel 2



jumper

•	1	77070157
	a - operating voltage 24L+ channel 2 / 1-10A	/33.6
	b - sensor limit Y- =DL1B+L1/54.3	
	d -	
	EtherCAT coupler powerbus / + =IT/67.6	

X23

-X23 = operating voltage 24L+ channel 3



jumper

•	1	77070157
a - Safety interface external / 1-10A /34.4		
b - Safety interface external =EI/58.1		
d - operation panel supply / =OP+E/84.3		

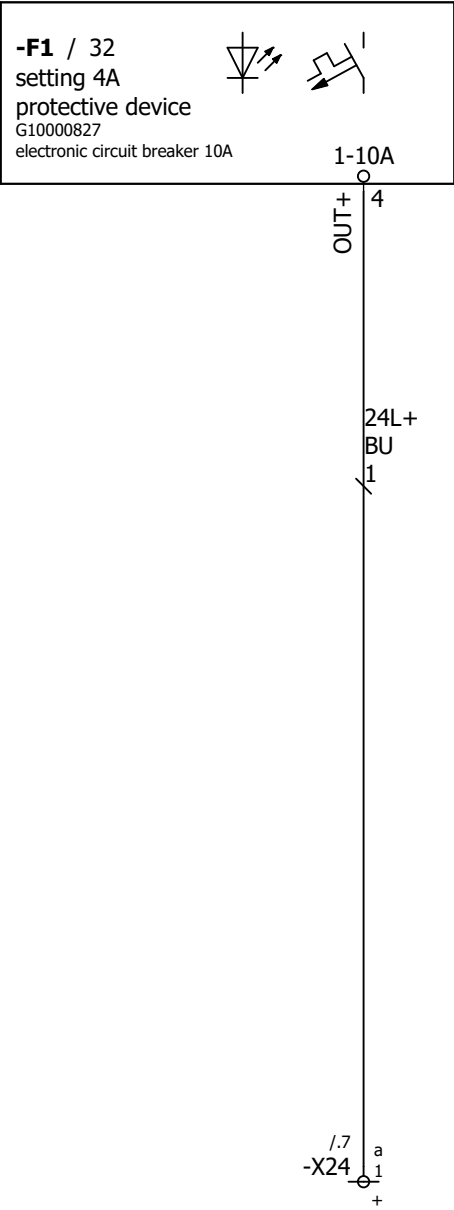
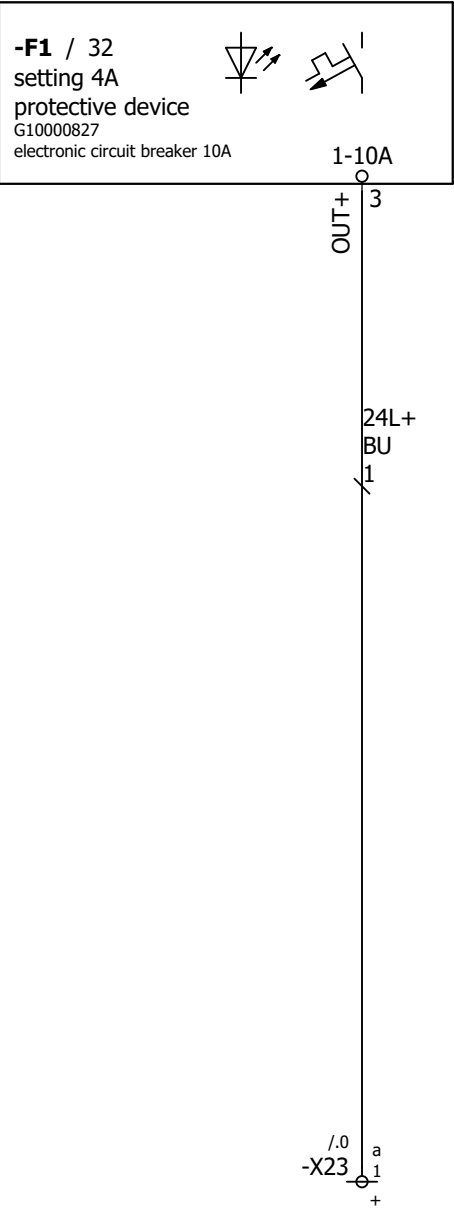
X24

-X24 = operating voltage 24L+ channel 4



jumper

•	1	77070157
a - operating voltage 24L+ channel 4 / 1-10A /34.6		
b -		
c - control voltage active =ER/64.8		



20A

operating voltage 24L+ channel 3

operating voltage 24L+ channel 4

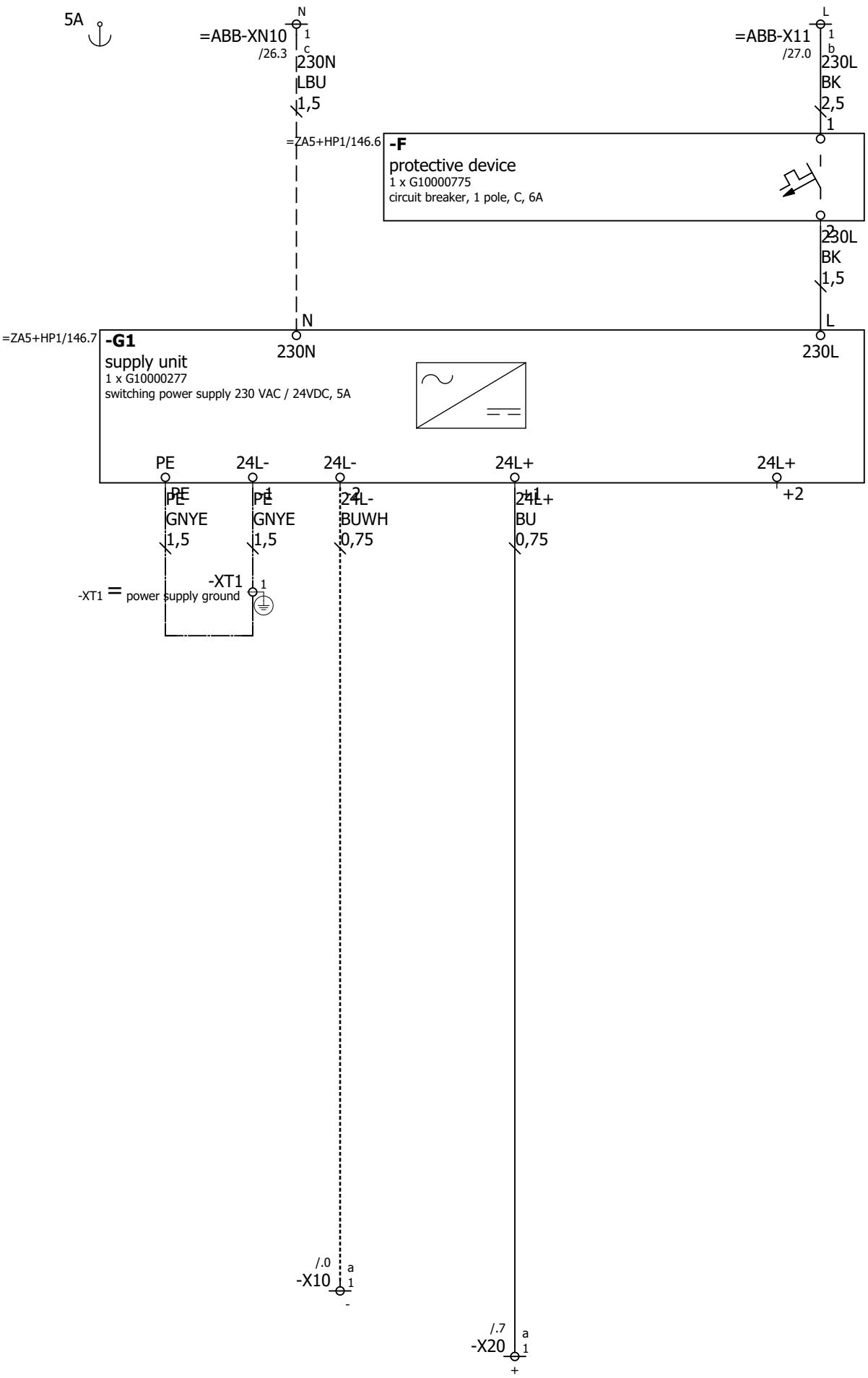
X10



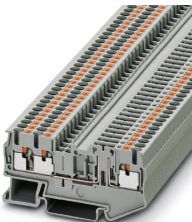
operating voltage
-X10 = 24L-

jumper

1	77070157
a - power supply 24 VDC operating voltage / 24L- /35.4	
b - Safety interface external =EIB/60.1	
c - EtherCAT coupler operating supply / 24L- =ITB/72.6	
d - EtherCAT coupler powerbus / - =ITB/72.7	
2	77070157
a - feed-in / 24L- =PS1+P/95.2	



X20



operating voltage
-X20 = 24L+

jumper

1	G10000602;G10000506
a - power supply 24 VDC operating voltage / 24L+ /35.5	
2	G10000602;G10000459;0510267
a - operating voltage 24L+ channel 1 /36.4	
3	G10000602;G10000459;0510267
a - operating voltage 24L+ channel 2 /36.6	
4	G10000602;G10000459;0510267
a - operating voltage 24L+ channel 3 /37.4	
5	G10000602;G10000459;0510267
a - operating voltage 24L+ channel 4 /37.6	

operating voltage
24L-

power supply 24 VDC
operating voltage

operating voltage
24L+

X21



-X21 = operating voltage 24L+ channel 1

jumper

•	1	77070157
---	---	----------

- a - operating voltage 24L+ channel 1 /36.4
- b - Safety interface external =EIB/60.2

X22

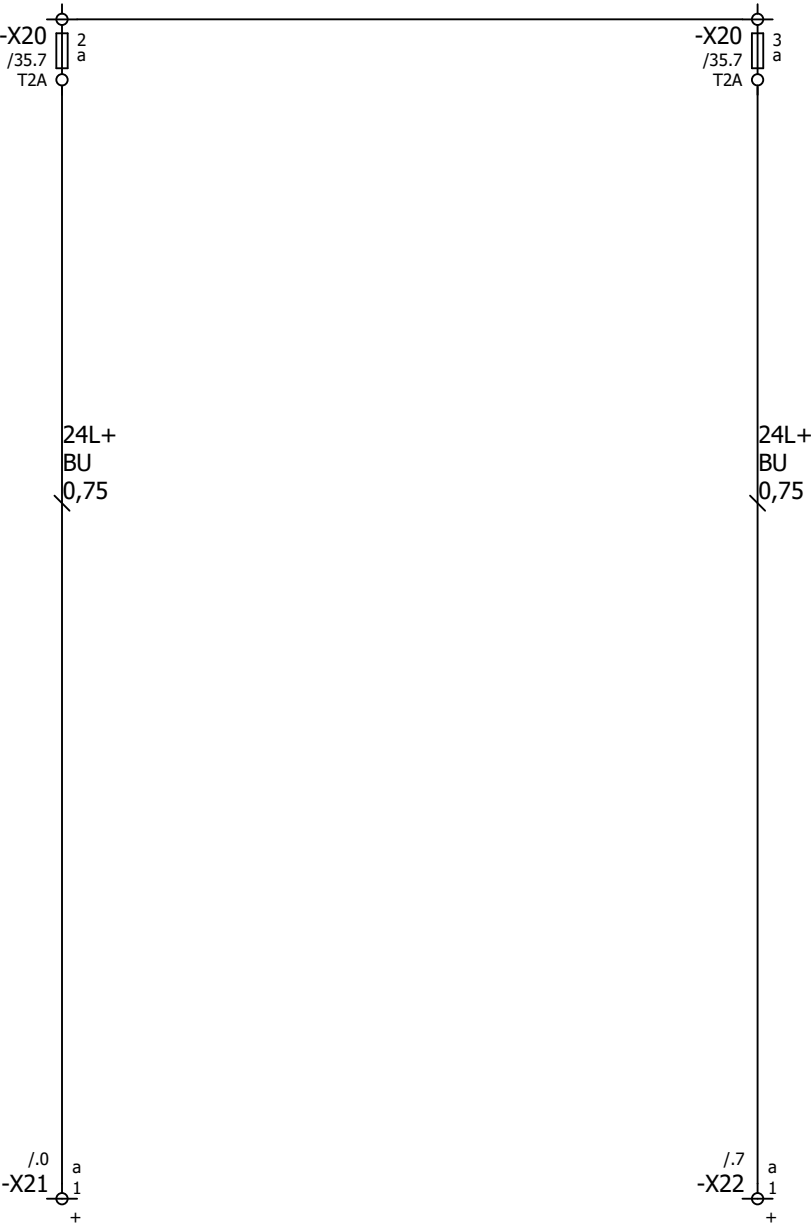


-X22 = operating voltage 24L+ channel 2

jumper

•	1	77070157
---	---	----------

- a - operating voltage 24L+ channel 2 /36.6
- b - EtherCAT coupler operating supply / 24L+ =ITB/72.5
- c - exhaustion active =SP+P/97.5

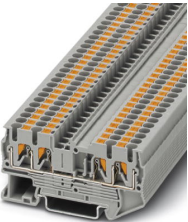


5A

operating voltage 24L+ channel 1

operating voltage 24L+ channel 2

X23



-X23 = operating voltage 24L+ channel 3

jumper

•	1	77070157
a - operating voltage 24L+ channel 3 /37.4		
b - EtherCAT coupler powerbus / + =ITB/72.6		

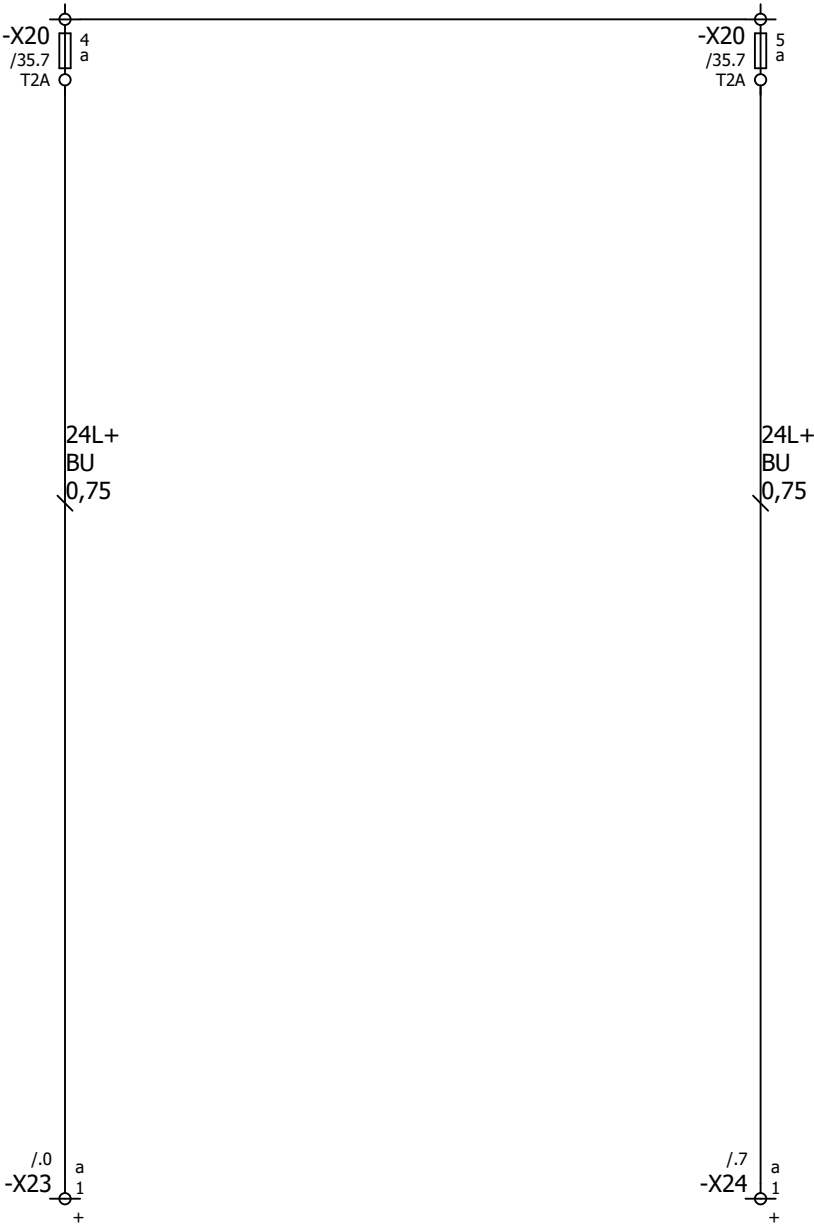
X24



-X24 = operating voltage 24L+ channel 4

jumper

•	1	77070157
a - operating voltage 24L+ channel 4 /37.6		
b - plasma power source supply 24VDC / 24L+ =PS1+P/95.		



5A

operating voltage 24L+ channel 3

operating voltage 24L+ channel 4



G10000109

relay

-F1 extension module, 24V DC

72a		
72b		
A1a		
A1b	control voltage active	=ER/64.8
A2a		
A2b	control voltage active	=ER/64.8

Logic

11a	control voltage return	/39.1
11b		
12a	control voltage return	/39.1
12b		
23a	control voltage 24L+ channel 1	/40.4
23b		
24a	control voltage 24L+ channel 1	/40.4
24b		
33a	control voltage 24L+ channel 2	/40.6
33b		
34a	control voltage 24L+ channel 2	/40.6
34b		
43a	control voltage 24L+ channel 3	/41.4
43b		
44a	control voltage 24L+ channel 3	/41.4
44b		
53a	control voltage 24L+ channel 4	/41.6
53b		
54a	control voltage 24L+ channel 4	/41.6
54b		
63a	control voltage power distributors	/39.5
63b		
64a	control voltage power distributors	/39.5
64b		
71a		
71b		

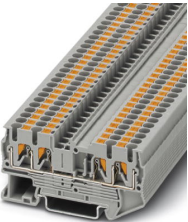


control voltage
24L+

MESSER 

Page
39

X21



-X21 = control voltage 24L+ channel 1

jumper

•	1	77070157
a - control voltage 24L+ channel 1 /40.4		

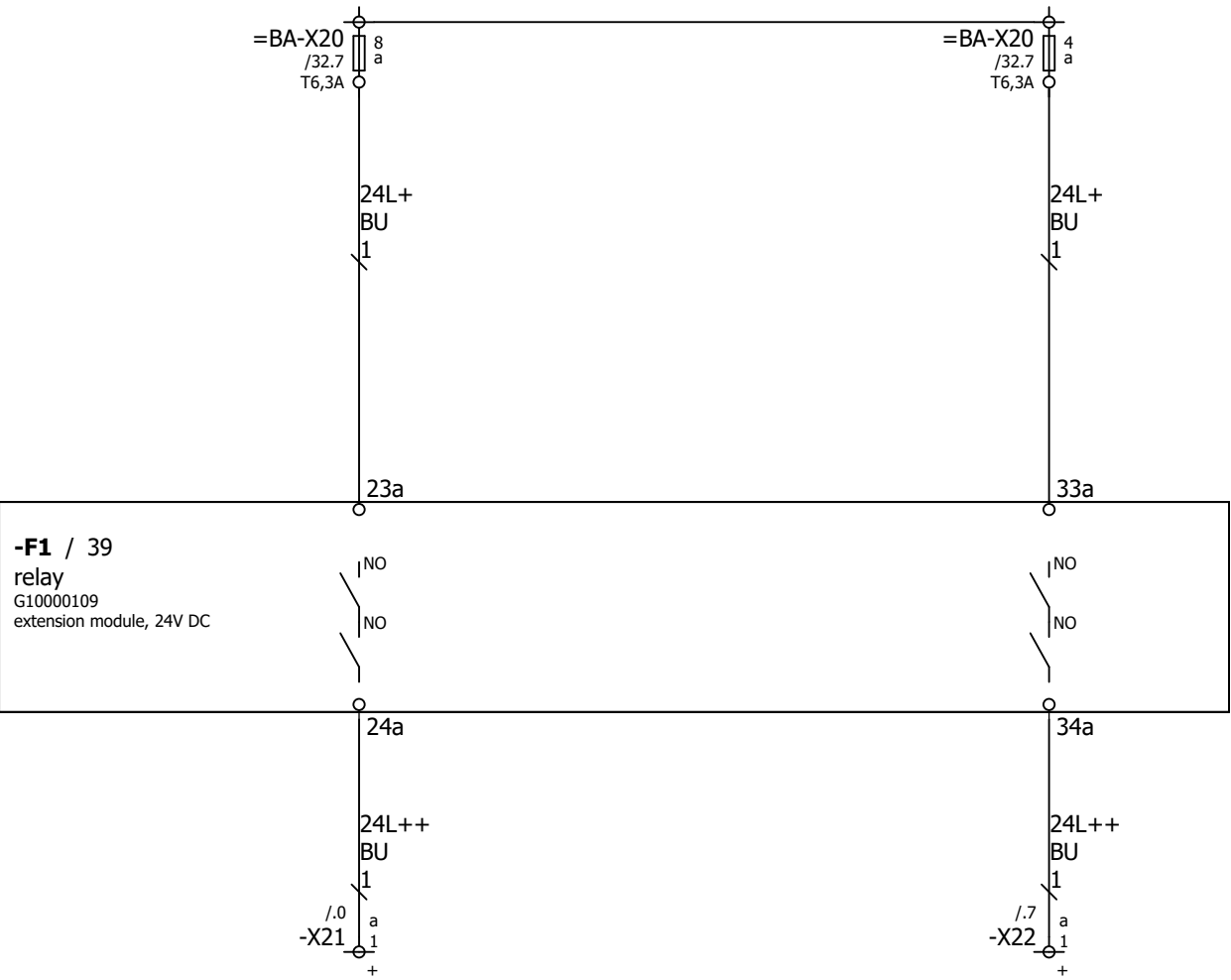
X22



-X22 = control voltage 24L+ channel 2

jumper

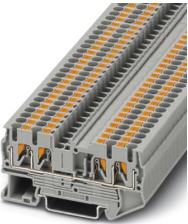
•	1	77070157
a - control voltage 24L+ channel 2 /40.6		



control voltage 24L+
channel 1

control voltage 24L+
channel 2

X23



-X23 = control voltage 24L+ channel 3

jumper

•	1	77070157
a - control voltage 24L+ channel 3 /41.4		

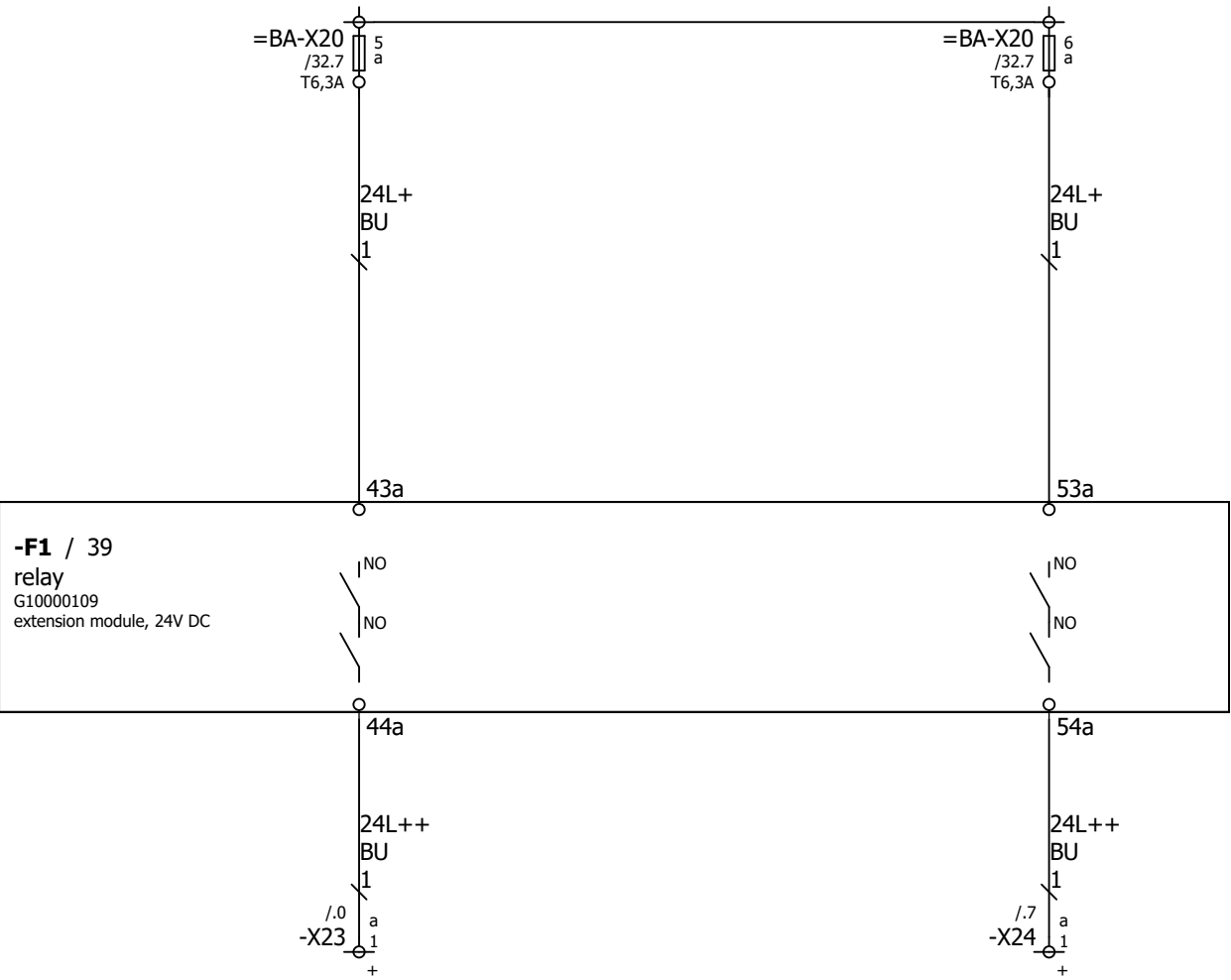
X24



-X24 = control voltage 24L+ channel 4

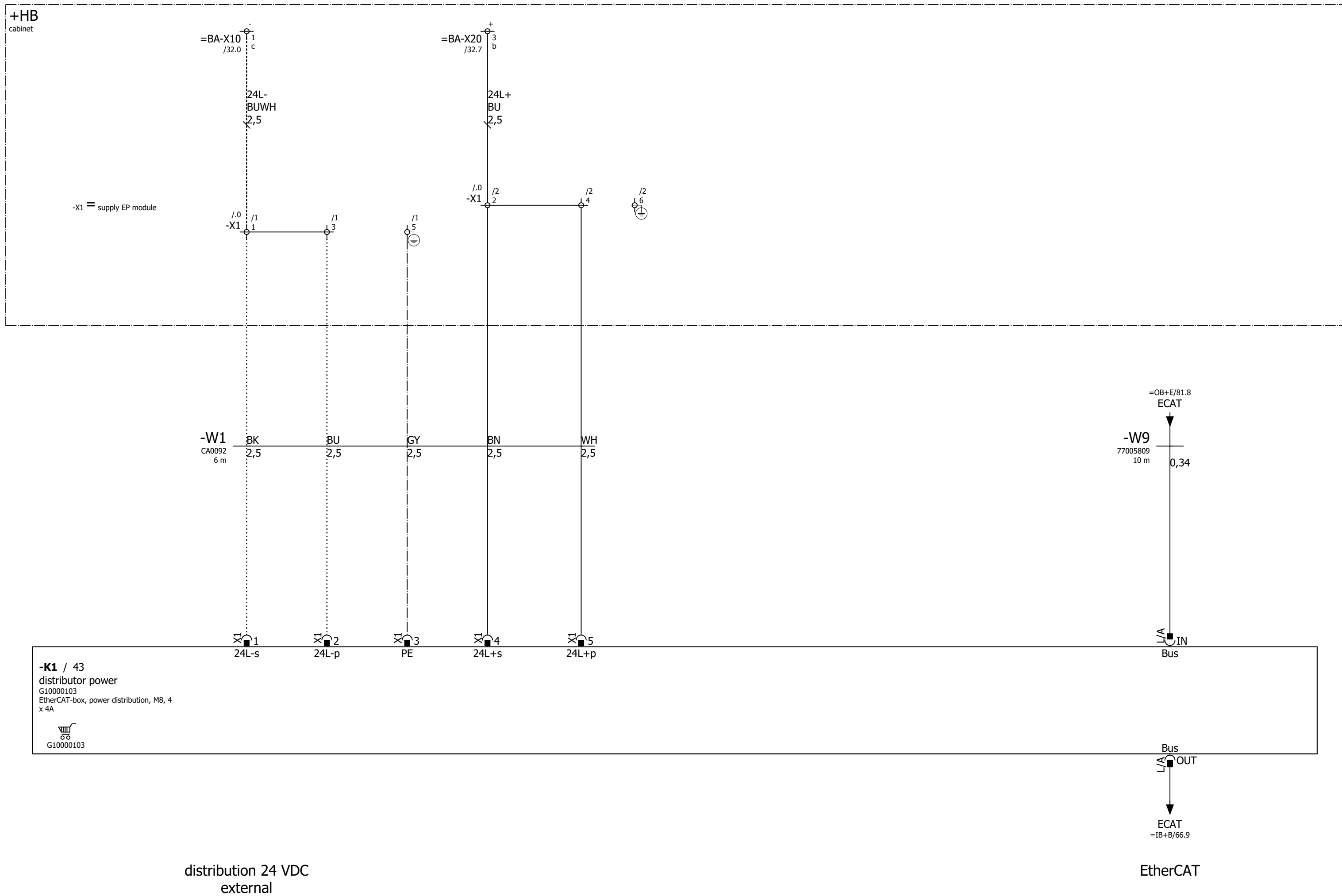
jumper

•	1	77070157
a - control voltage 24L+ channel 4 /41.6		



control voltage 24L+ channel 3

control voltage 24L+ channel 4



			Toolbox_31	Option:	L1-EP PowerBox 4x4A {R.29}		EP power - distributor	module DC power supply interface		Page 43
			Toolbox version	customer:	Blocotelha		24VDC, EtherCat			
			BI1 - ABI - 29.30.00 - CC: EU	commission:	289001273					
Modification	Date	Name	Toolbox module designation	Ed. by:	Thomas.Lehle	01.04.2021	sector 1B1B location +B =BI			



G10001010

controller

AX5206, 2 channel, AC 2x 6A,
FW >=2.10

-K1

GND

grounding
M6 ground

Safety

PCI
PCI Bus

X01

supply grid
1 L1
2 L2
3 L3
PE GND

X02

DC Link
1 DC+
2 DC-

X03

supply DC
1 Up+
2 Us+
3 GND

X04

Bus
1-4 Bus-IN

X05

1-4 Bus-OUT

X06

digital I/O's
0 input
0V 24L-
1 input
2 input
3 input
4 input
5 input
6 input
7 in/out
24 24L+

X11

Hiperface
1 SIN+
2 GND-9V
3 COS+
4 n.c.
5 Data+
6 Us-9V
7 n.c.
8 n.c.
9 REFSIN
10 n.c.
11 REF COS
12 n.c.
13 Data-
14 n.c.
15 n.c.
SH shield

X12

Resolver
1 Temp.
2 AGND
3 COS-
4 SIN-
5 REF-
6 n.c.
7 n.c.
8 n.c.
9 Temp.GND
10 COS+

X12

11 SIN+
12 REF+
13 n.c.
14 n.c.
15 n.c.
SH shield

X13

Motor
PE GND
U Motor
V Motor
W Motor

X13/X14

SH shield

X14

Temp./brake
B+ brake +
B- brake GND
PE shield
T+ Temp.+
T- Temp.-

X21

Hiperface
1 SIN+
2 GND-9V
3 COS+
4 n.c.
5 Data+
6 Us-9V
7 n.c.
8 n.c.
9 REFSIN
10 n.c.
11 REF COS
12 n.c.
13 Data-
14 n.c.
15 n.c.
SH shield

X22

Resolver
1 Temp.
2 AGND
3 COS-
4 SIN-
5 REF-
6 n.c.
7 n.c.
8 n.c.
9 Temp.GND
10 COS+
11 SIN+
12 REF+
13 n.c.
14 n.c.
15 n.c.
SH shield

X23

Motor
PE GND
U Motor
V Motor
W Motor

X23/X24

SH shield

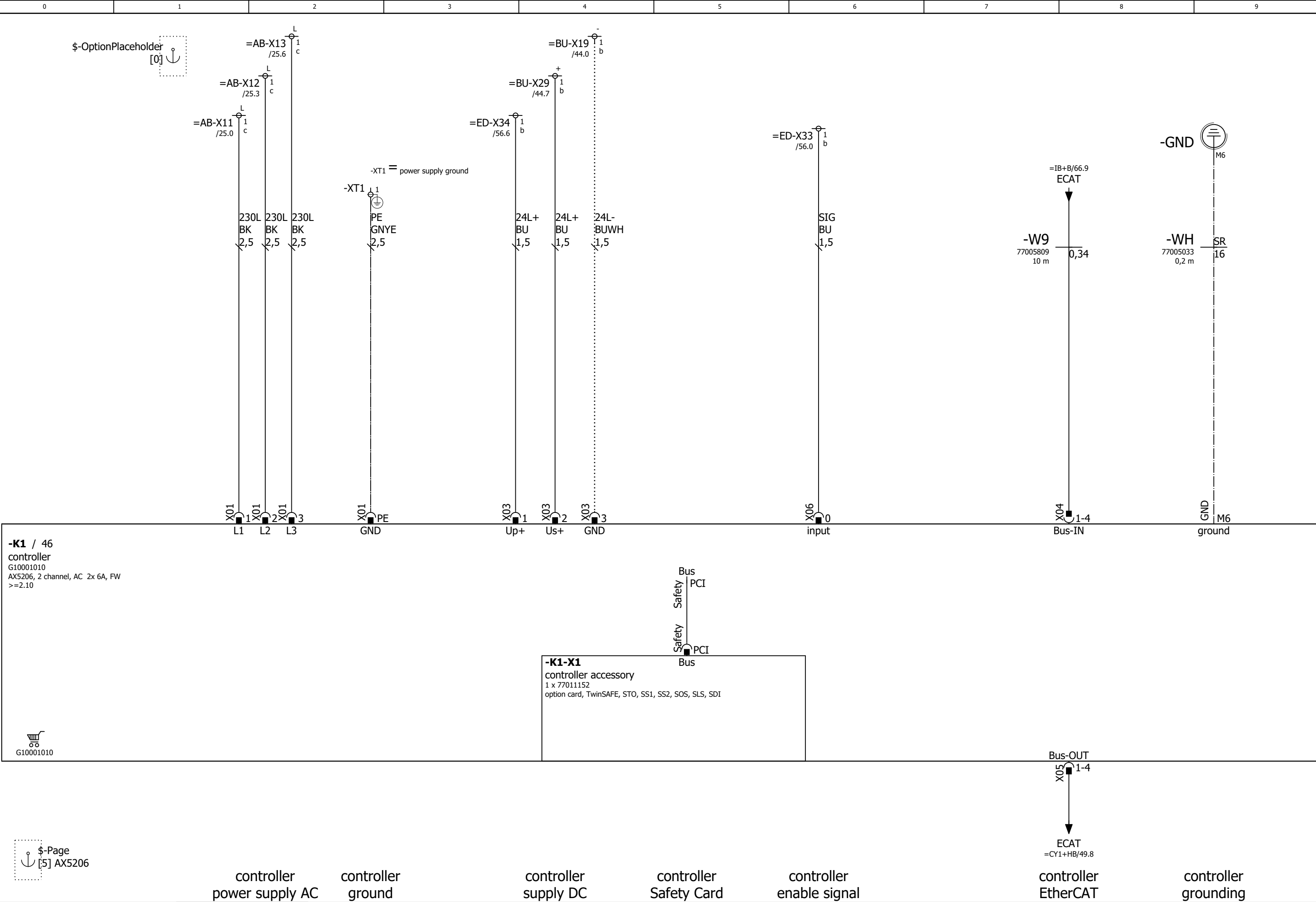
X24

Temp./brake
B+ brake +
B- brake GND
PE shield
T+ Temp.+
T- Temp.-

Safety adress: HEX -<c8> / Dez: <200>

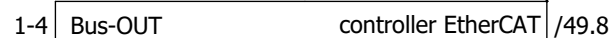
ON

			X		X	X			
X	X	X		X	X		X	X	
1	2	3	4	5	6	7	8	9	10



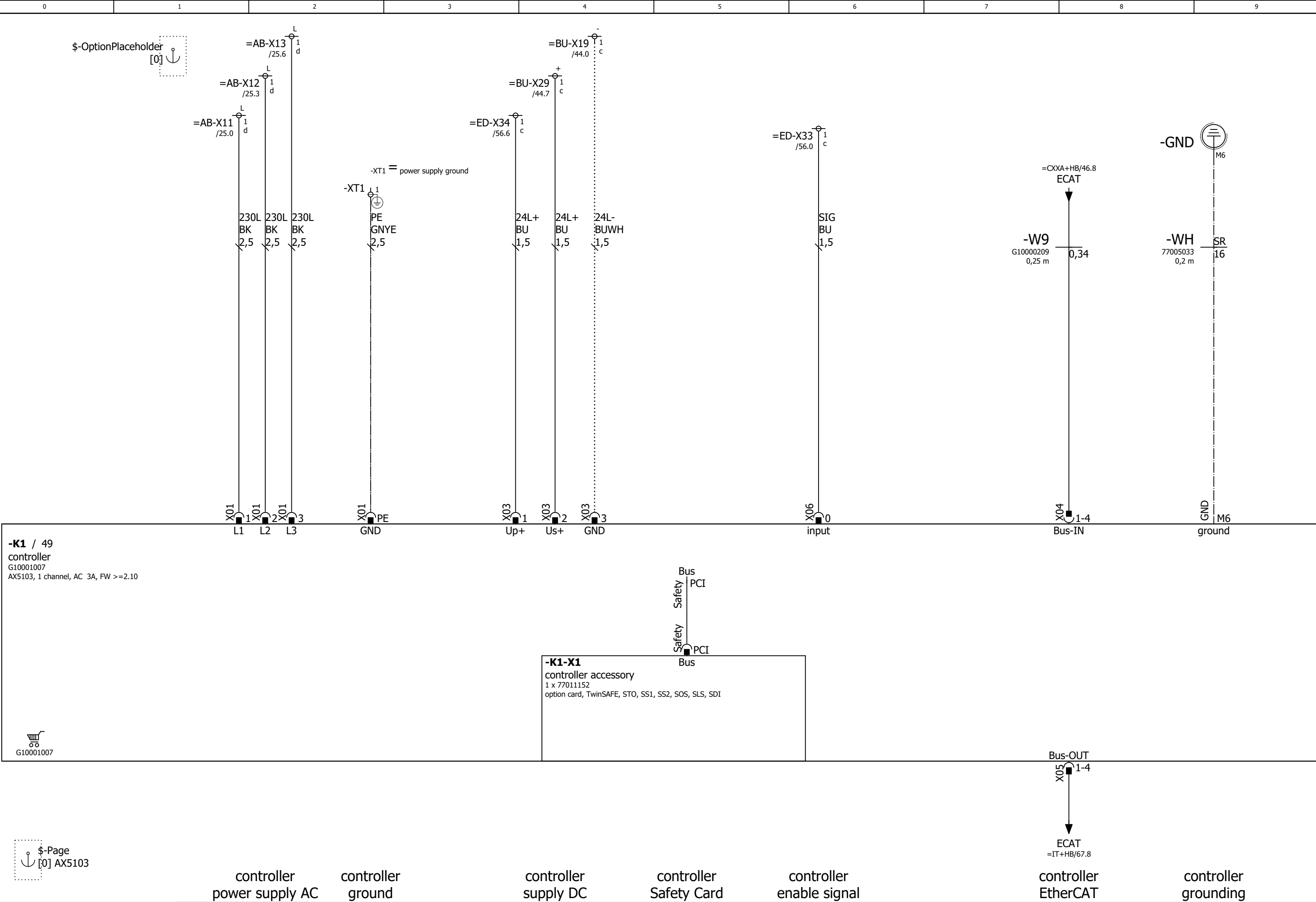
Documentation evaluation

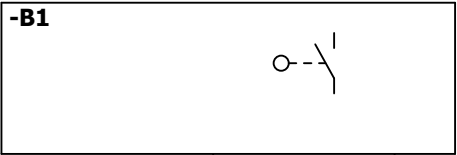
\$-FOS-F19-PLC



○ N

		X			X	X			
X	X		X	X			X	X	X
1	2	3	4	5	6	7	8	9	10





BN

BU

-WB1
2 m

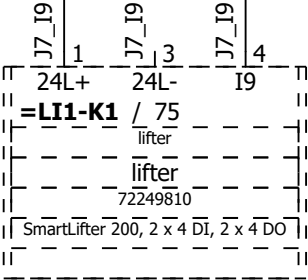
BN

BU

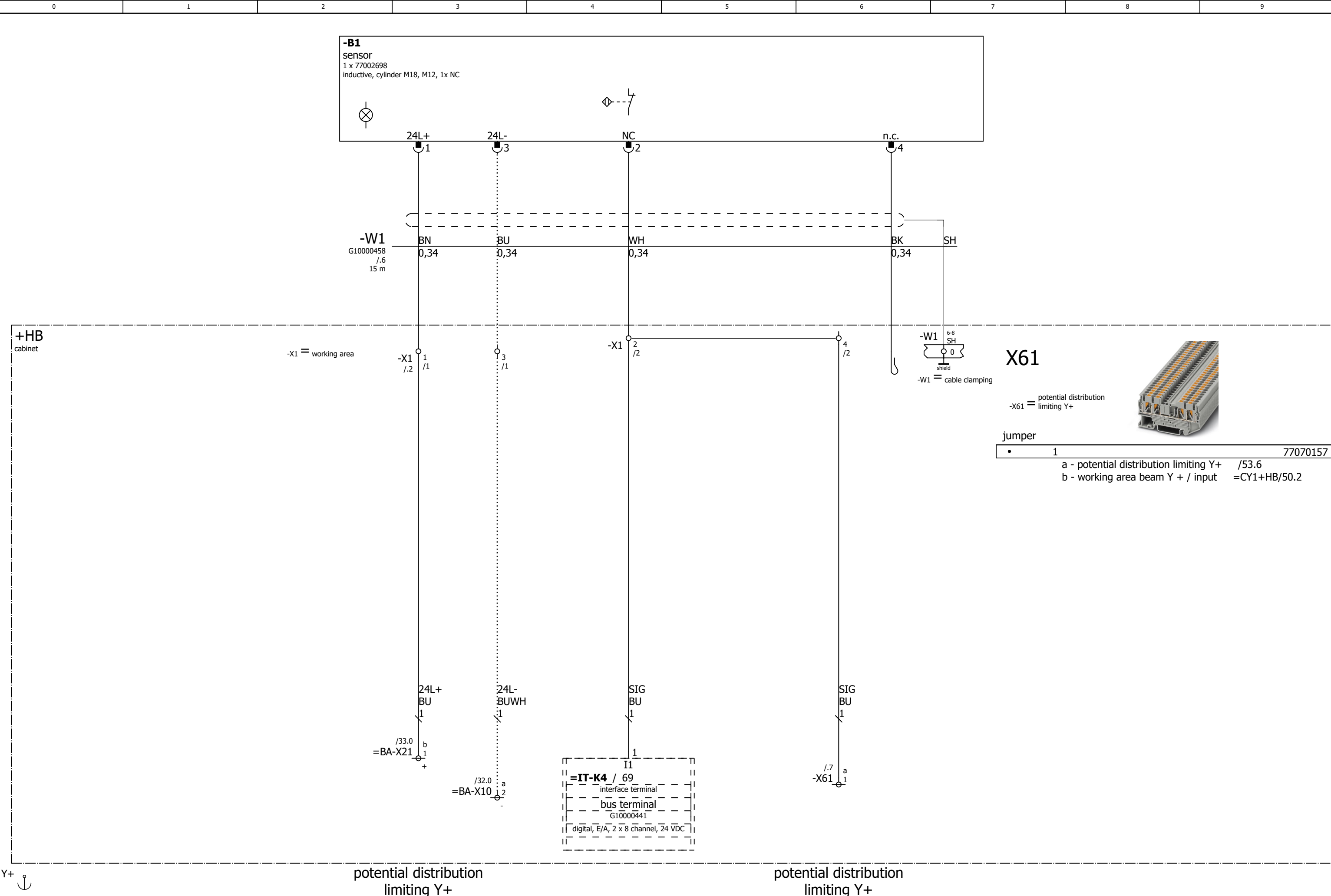
-X1 =
G10000246
M 8, 3-pole, male pin, 0°

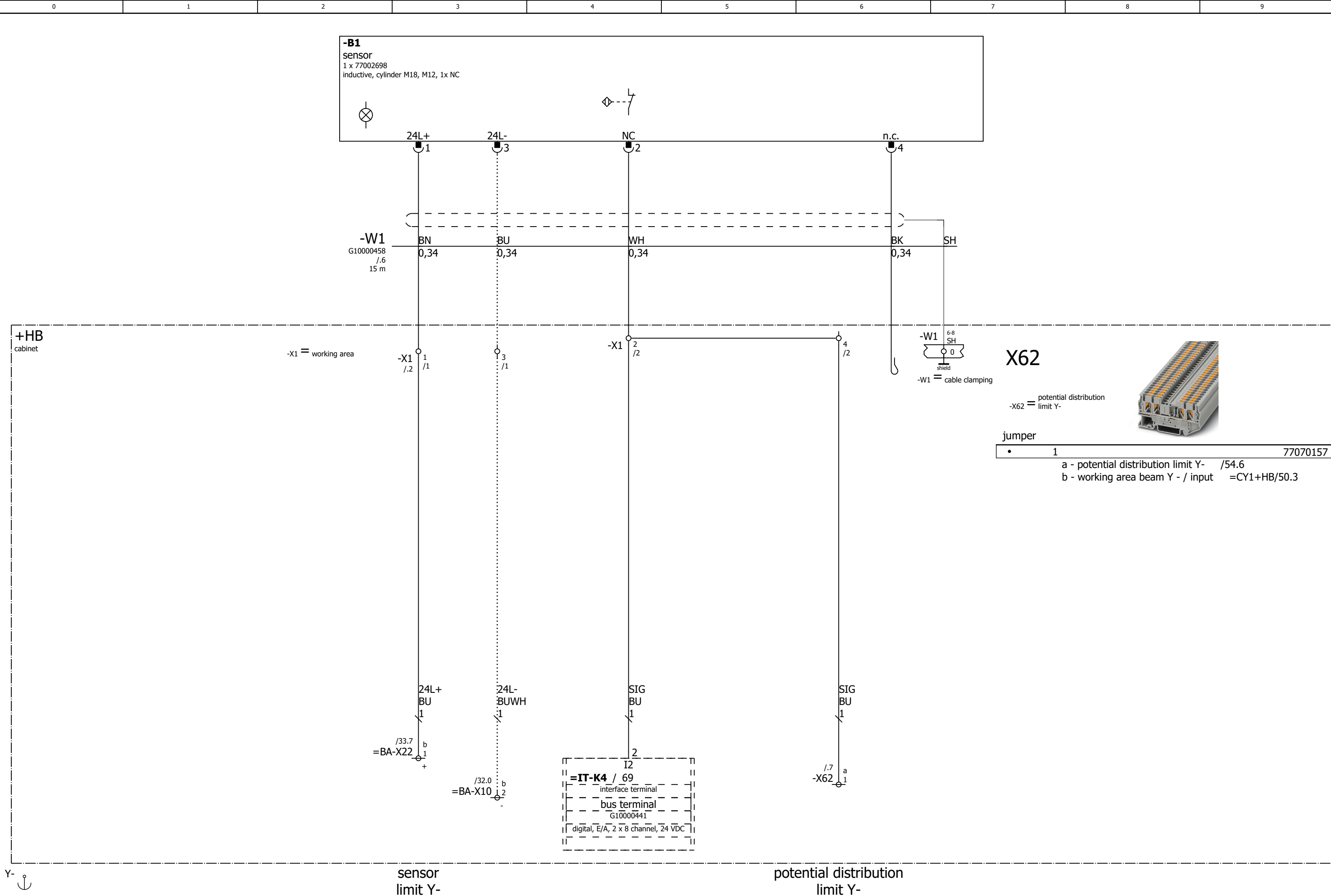
1

4



collision
nozzle





X33



-X33 = potential controller enable

jumper

•	1	77070157
---	---	----------

- a - potential Drive enable / 1/24L+ /56.3
- b - controller enable signal / input =CXXA/46.6
- c - controller enable signal / input =CY1/49.6

X34

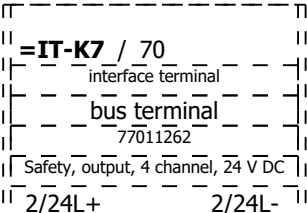
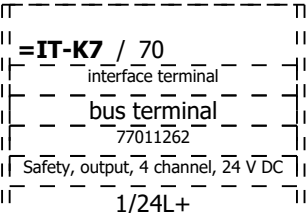


-X34 = potential Brake activate

jumper

•	1	77070157
---	---	----------

- a - potential Brake activate /56.6
- b - controller power supply DC / Up+ =CXXA/46.3
- c - controller power supply DC / Up+ =CY1/49.3



SAFETY
YE
1

SAFETY
YE
1
A1

SAFETY
YE
1
A2

=BU-X29
/44.7
T6,3A
2
a

24L+
BU
2,5

11
14 12

24L+
BU
2,5

/,6
-X34
1
+

potential
Drive enable

potential
Brake activate



72a	
72b	
A1a	Safety interface external /58.1
A1b	
A2a	Safety interface external /58.1
A2b	

G10000109

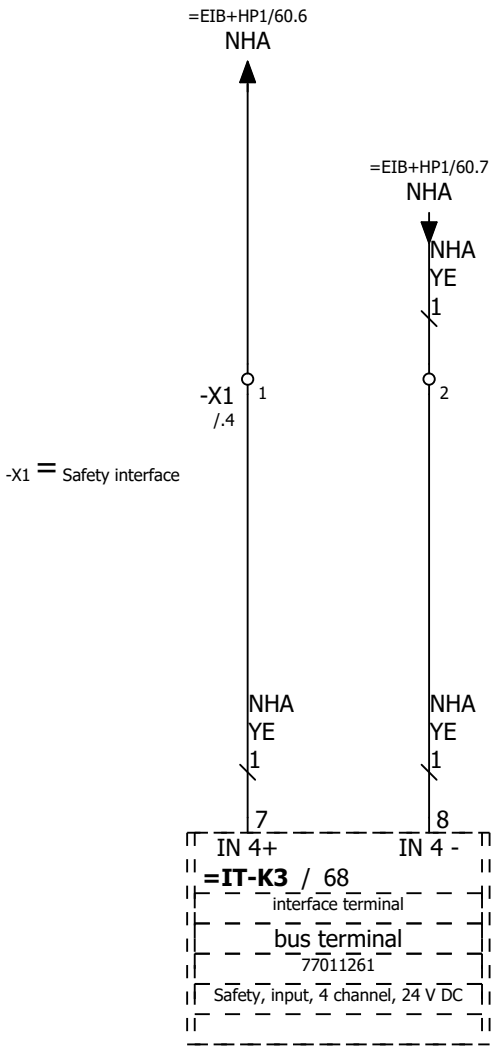
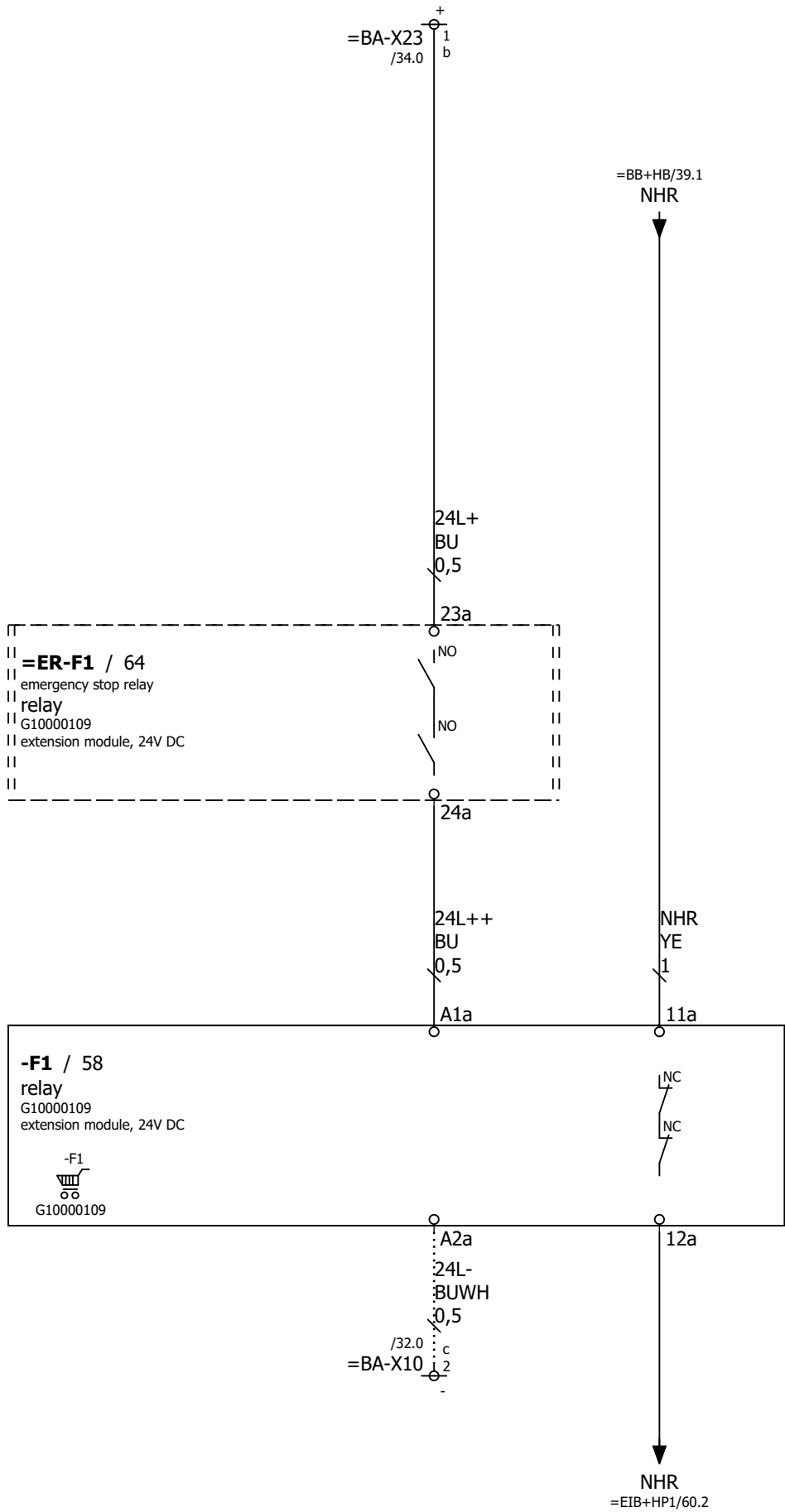
relay

-F1

extension module, 24V DC

Logic

11a	Safety interface return /58.2
11b	
12a	Safety interface return /58.2
12b	
23a	
23b	
24a	
24b	
33a	
33b	
34a	
34b	
43a	
43b	
44a	
44b	
53a	
53b	
54a	
54b	
63a	
63b	
64a	
64b	
71a	
71b	





72a	
72b	
A1a	Safety interface external /60.1
A1b	
A2a	Safety interface external /60.1
A2b	

G10000109

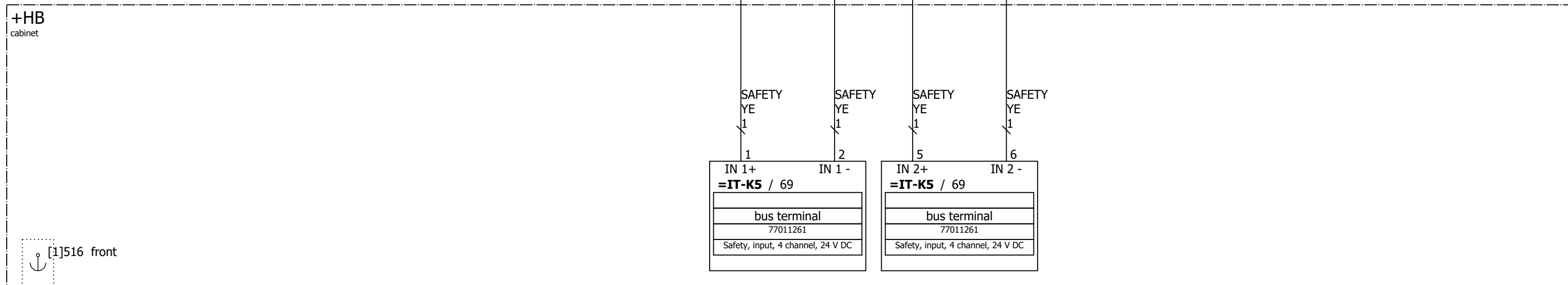
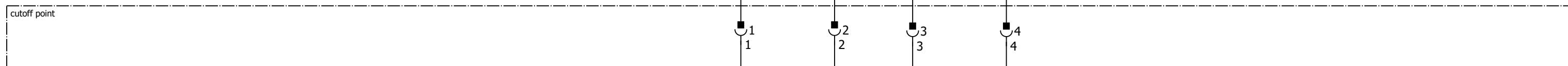
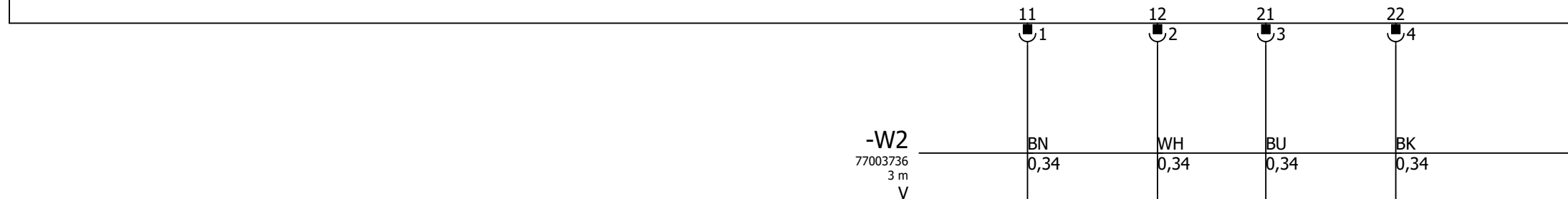
relay

-F1

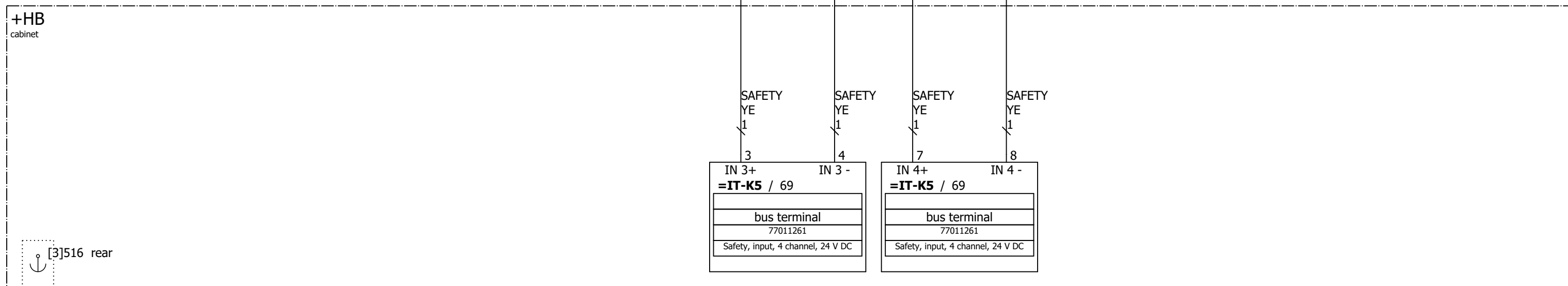
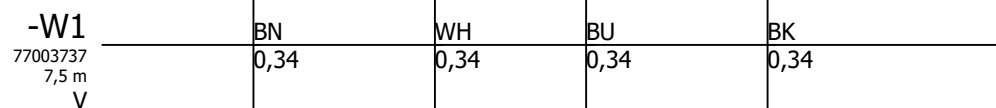
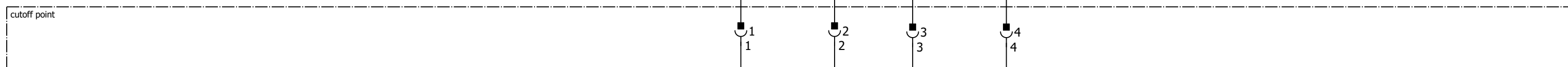
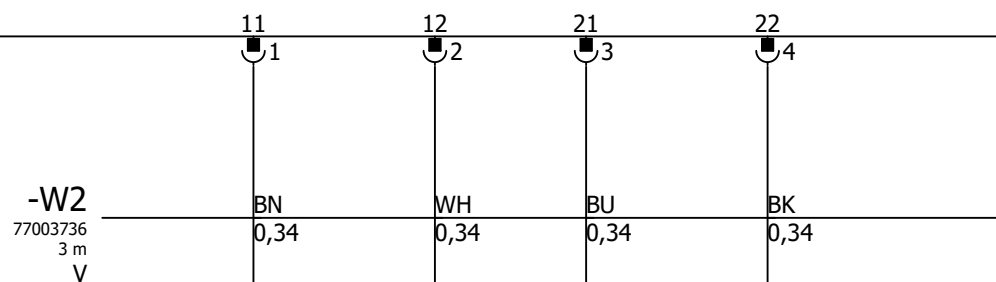
extension module, 24V DC

Logic

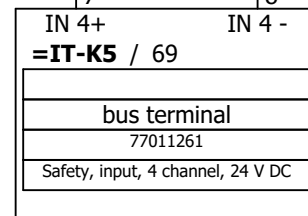
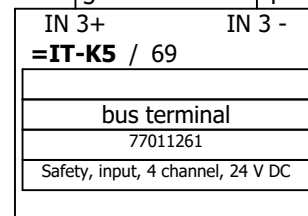
11a	Safety interface return /60.2
11b	
12a	Safety interface return /60.2
12b	
23a	plasma power source switch-off =PA1+HJ1/87.8
23b	
24a	plasma power source switch-off =PA1+HJ1/87.8
24b	
33a	
33b	
34a	
34b	
43a	
43b	
44a	
44b	
53a	
53b	
54a	
54b	
63a	
63b	
64a	
64b	
71a	
71b	



Reißleine
vorne



SAFETY YE	SAFETY YE	SAFETY YE	SAFETY YE
1	1	1	1
3	4	7	8



Reißleine
hinten



72a	
72b	
A1a	safety-related control voltage /64.0
A1b	
A2a	safety-related control voltage /64.1
A2b	

G10000109

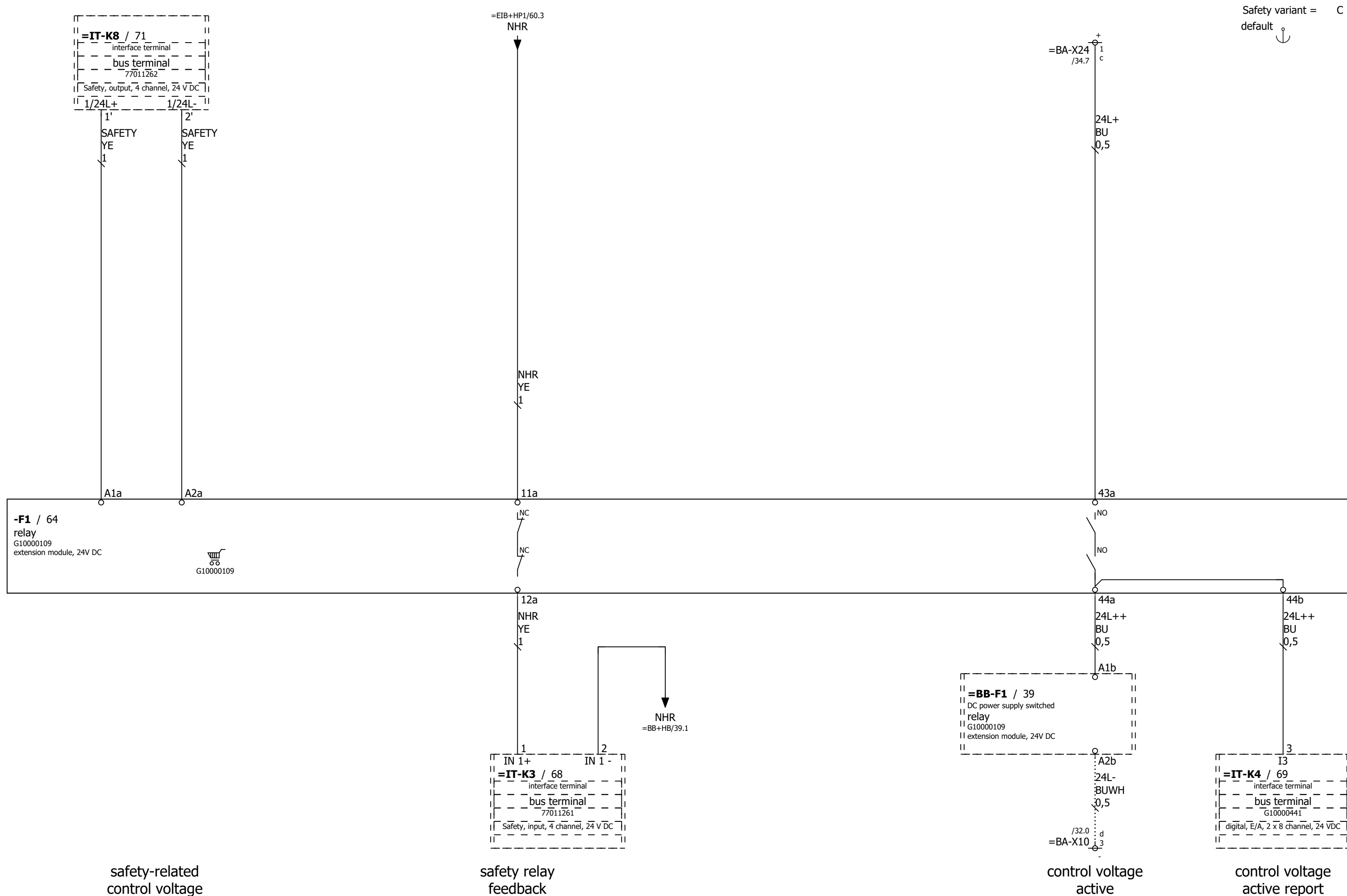
relay

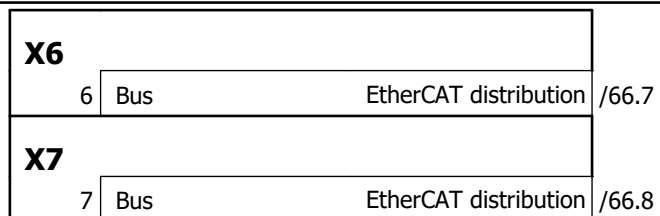
-F1

extension module, 24V DC

Logic

11a	safety relay feedback	/64.3
11b		
12a	safety relay feedback	/64.3
12b		
23a	Safety interface external	=EI/58.1
23b		
24a	Safety interface external	=EI/58.1
24b		
33a	Safety interface external	=EIB+HP1/60.1
33b		
34a	Safety interface external	=EIB+HP1/60.2
34b		
43a	control voltage active	/64.8
43b		
44a	control voltage active	/64.8
44b	control voltage active report	/64.9
53a		
53b		
54a		
54b		
63a		
63b		
64a		
64b		
71a		
71b		





-K1 EtherCAT, star hub, 8 ports

Power IN		
1	24L+s	/66.1
2	24L+p	EP module supply /66.2
3	24L-s	= /66.3
4	24L-p	= /66.3

	Power OUT
1	24L+s
2	24L+p
3	24L-s
4	24L-p

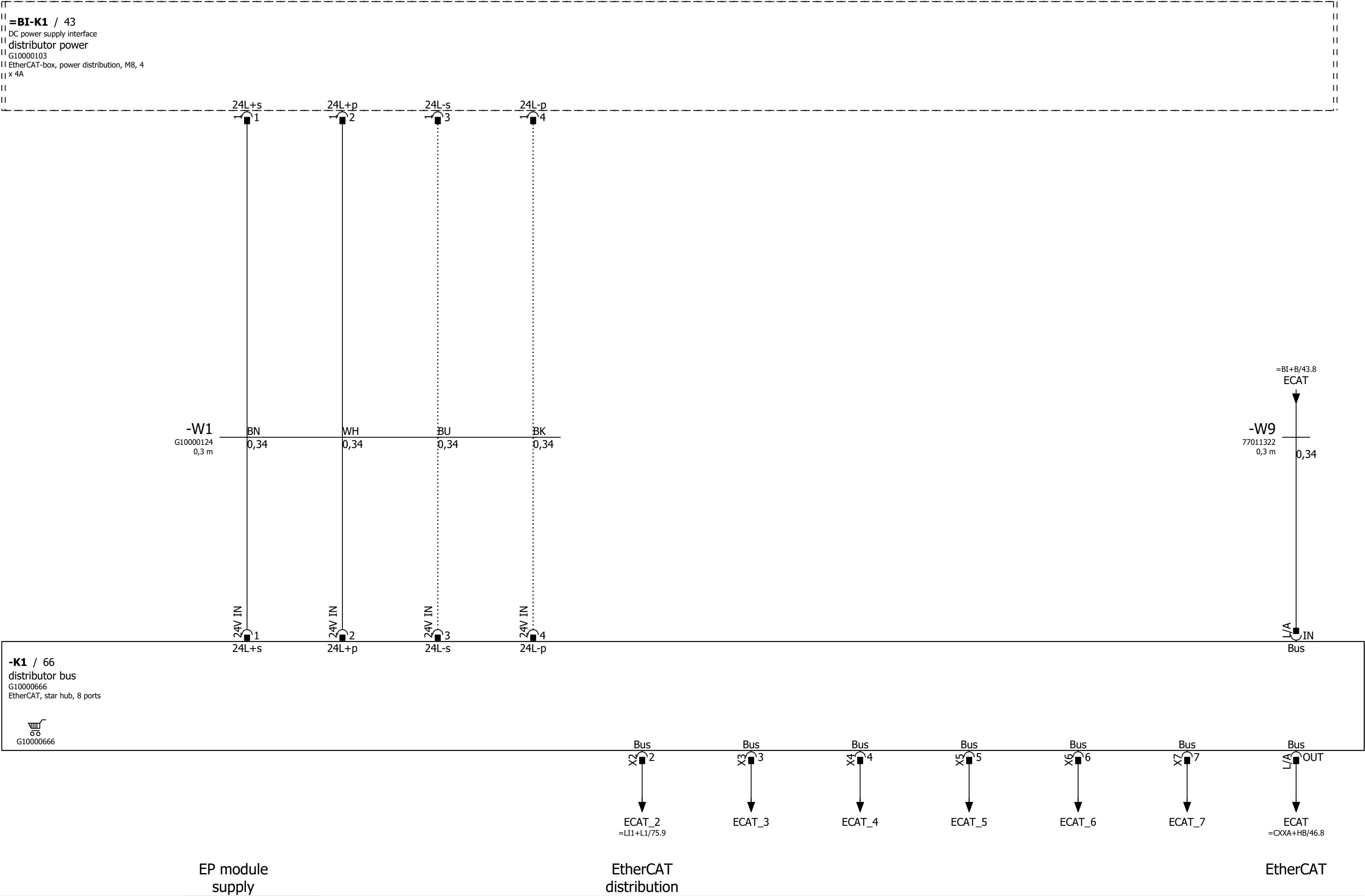
	Bus	
IN	Bus	EtherCAT /66.9
OUT	Bus	= /66.9

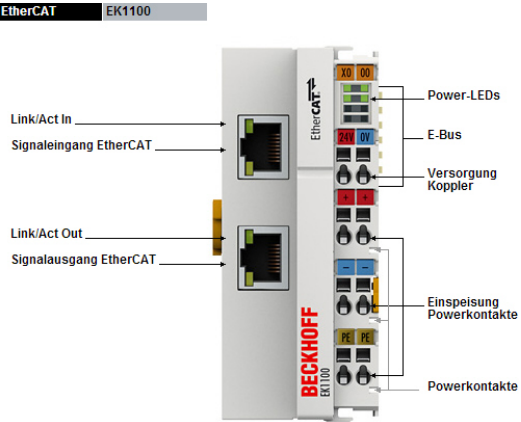
X2	
2	Bus EtherCAT distribution /66.4

X3	
3	Bus EtherCAT distribution /66.5

X4			
4	Bus	EtherCAT distribution	/66.6

X5			
5	Bus	EtherCAT distribution	/66.7





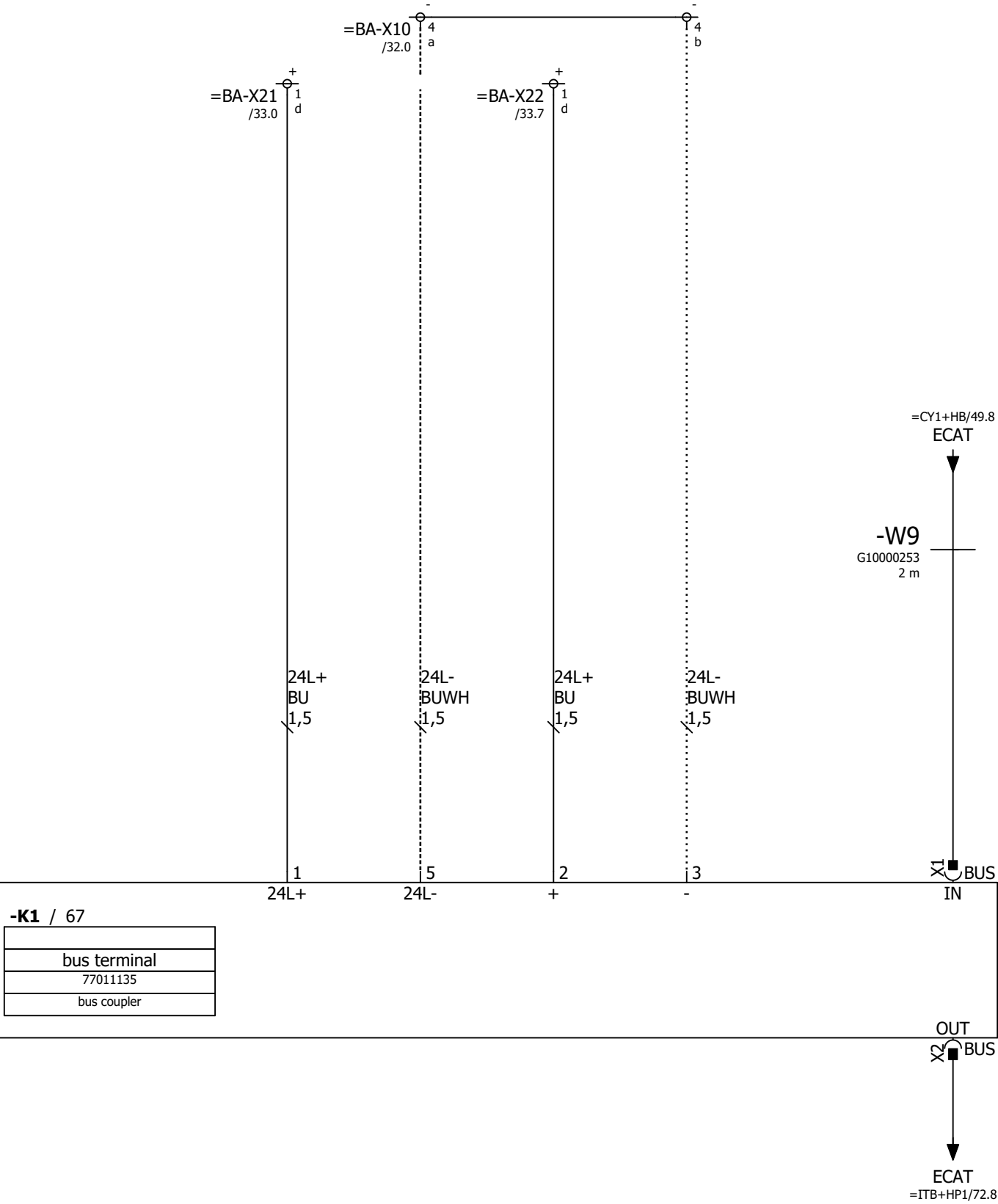
77011135
bus terminal

-K1 bus coupler

supply		
1	24L+	EtherCAT coupler operating supply /67.5
power contacts		
2	+	EtherCAT coupler powerbus /67.6
3	-	= /67.7
4	PE	
supply		
5	24L-	EtherCAT coupler operating supply /67.6
power contacts		
6	+	
7	-	
8	PE	

X1		
Bus		
BUS	IN	EtherCAT coupler Bus /67.8
X2		
BUS	OUT	EtherCAT coupler Bus /67.8

EtherCAT coupler
-K1
77011135



-K1 / 67	
bus terminal	
77011135	
bus coupler	

EtherCAT coupler operating supply
EtherCAT coupler powerbus
EtherCAT coupler Bus

Digital-EingangEL1904

Input 1

Input 3

Diagnose 1

Diagnose 3

Input 2

Input 4

Diagnose 2

Diagnose 4

Eingang 1+

Eingang 1-
Powerkontakt +24 V

Eingang 3+
Powerkontakt 0 V

Eingang 3-

Eingang 2+

Eingang 2-

Eingang 4+

Eingang 4-

0708

1112

1314

1516

1718

EL1904
BECKHOFF

77011261

bus terminal

-K2

Safety, input, 4 channel, 24 V DC

IN 01

1IN 1+IN 1 -

IN 03

3IN 3+IN 3 -

IN 02

5IN 2+IN 2 -

IN 04

7IN 4+IN 4 -

emergency stop

emergency stop

emergency stop

Safety interface internal

=EA+G/55.0

=EA+G/55.1

=EA+G/55.1

=EA+G/55.2

ON

1

2

3

4

5

6

7

8

Digital-EingangEL1904

Input 1

Input 3

Diagnose 1

Diagnose 3

Input 2

Input 4

Diagnose 2

Diagnose 4

Eingang 1+

Eingang 1-
Powerkontakt +24 V

Eingang 3+
Powerkontakt 0 V

Eingang 3-

Eingang 2+

Eingang 2-

Eingang 4+

Eingang 4-

0708

1112

1314

1516

1718

EL1904
BECKHOFF

77011261

bus terminal

-K3

Safety, input, 4 channel, 24 V DC

IN 01

1IN 1+IN 1 -

IN 03

3IN 3+IN 3 -

IN 02

5IN 2+IN 2 -

IN 04

7IN 4+IN 4 -

safety relay feedback

Safety interface internal

=ER/64.3

=ER/64.4

=EI/58.5

ON

1

2

3

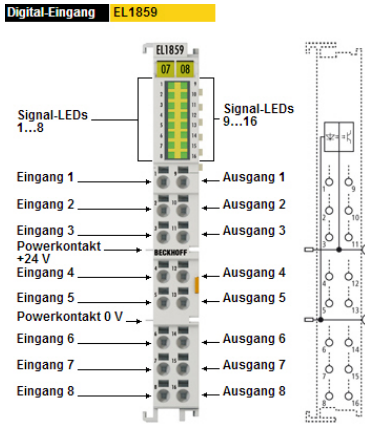
4

5

6

7

8



	O5
13	O5
	O6
14	O6
	O7
15	O7
	O8
16	O8

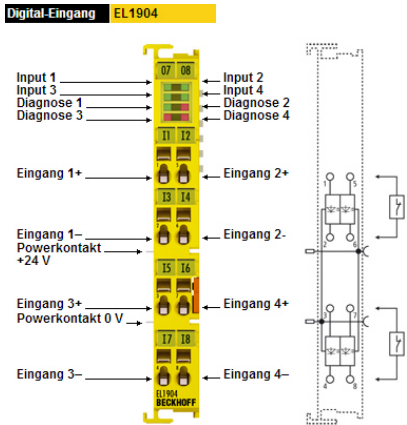
G10000441

bus terminal

-K4

digital, E/A, 2 x 8 channel, 24 VDC

	I1	
1	I1	potential distribution limiting Y+ =DL1+L1/53.4
	I2	
2	I2	sensor limit Y- =DL1B+L1/54.4
	I3	
3	I3	control voltage active report =ER/64.9
	I4	
4	I4	control voltage turn on =OP+E/85.6
	I5	
5	I5	machine start =OP+E/85.7
	I6	
6	I6	machine Stop =OP+E/85.8
	I7	
7	I7	
	I8	
8	I8	
	O1	
9	O1	machine in start =OP+E/86.3
	O2	
10	O2	machine in Stop =OP+E/86.2
	O3	
11	O3	=OP+E/86.1
	O4	
12	O4	



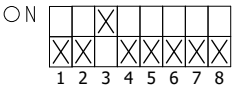
77011261

bus terminal

-K5

Safety, input, 4 channel, 24 VDC

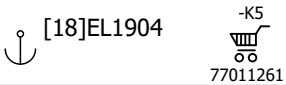
	IN 01	
1	IN 1+	Reißleine vorne =ELC+E/61.4
2	IN 1 -	= =ELC+E/61.5
	IN 03	
3	IN 3+	Reißleine hinten =ELD+E/62.4
4	IN 3 -	= =ELD+E/62.5
	IN 02	
5	IN 2+	Reißleine vorne =ELC+E/61.5
6	IN 2 -	= =ELC+E/61.6
	IN 04	
7	IN 4+	Reißleine hinten =ELD+E/62.5
8	IN 4 -	= =ELD+E/62.6



[3]EL1859



G10000441

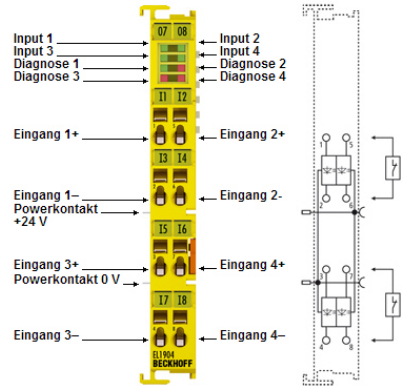


[18]EL1904



77011261

Digital-Eingang EL1904



77011261

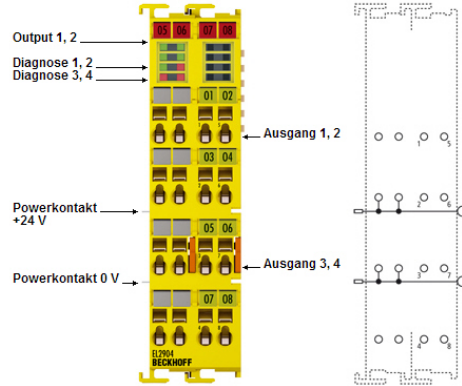
bus terminal

-K6 Safety, input, 4 channel, 24 V DC

IN 01		
1	IN 1+ machine reset	=OP+E/85.3
2	IN 1 -	=OP+E/85.4
IN 03		
3	IN 3+ panel Safety channel	=OP+E/85.2
4	IN 3 -	=OP+E/85.2
IN 02		
5	IN 2+ panel Safety channel	=OP+E/85.1
6	IN 2 -	=OP+E/85.1
IN 04		
7	IN 4+	
8	IN 4 -	



Digital-Ausgang EL2904



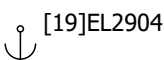
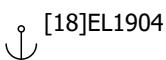
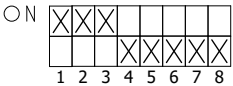
77011262

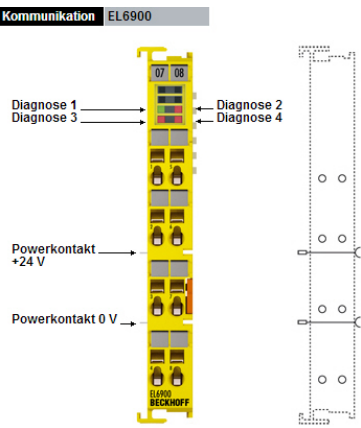
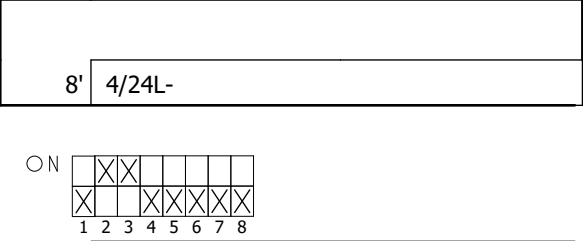
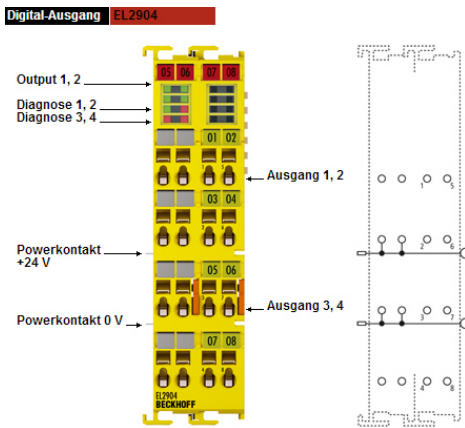
bus terminal

-K7 Safety, output, 4 channel, 24 V DC

Out 01		
1'	1/24L+ potential Drive enable	=ED/56.3
Power		
2	24L+	
Out 01		
2'	1/24L-	
Power		
3	24L-	
Out 02		
3'	2/24L+ potential Brake activate	=ED/56.5
Out 02		
4'	2/24L- potential Brake activate	=ED/56.5
Out 03		
5'	3/24L+	
Power		
6	24L+	
Out 03		
6'	3/24L-	
Power		
7	24L-	
Out 04		
7'	4/24L+	

8' 4/24L-





77011262

bus terminal

-K8

Safety, output, 4 channel, 24 V DC

Out 01		
1'	1/24L+	safety-related control voltage =ER/64.0
Power		
2	24L+	
Out 01		
2'	1/24L-	safety-related control voltage =ER/64.1
Power		
3	24L-	
Out 02		
3'	2/24L+	
4'	2/24L-	
Out 03		
5'	3/24L+	
Power		
6	24L+	
Out 03		
6'	3/24L-	
Power		
7	24L-	
Out 04		
7'	4/24L+	

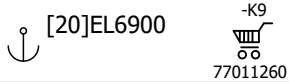
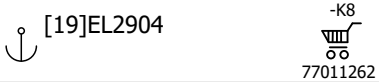
77011260

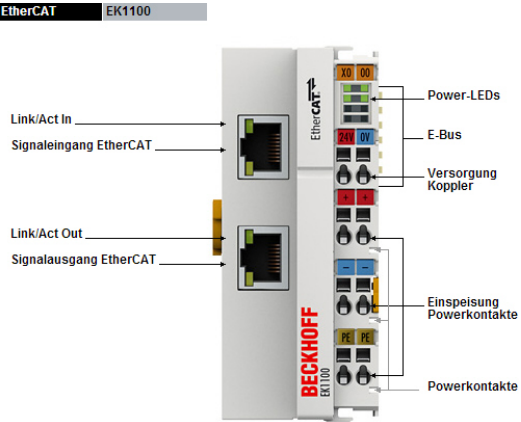
bus terminal

-K9

Safety, TwinSAFE-logic-terminal

1	n.c.
2	n.c.
3	n.c.
4	n.c.
5	n.c.
6	n.c.
7	n.c.
8	n.c.



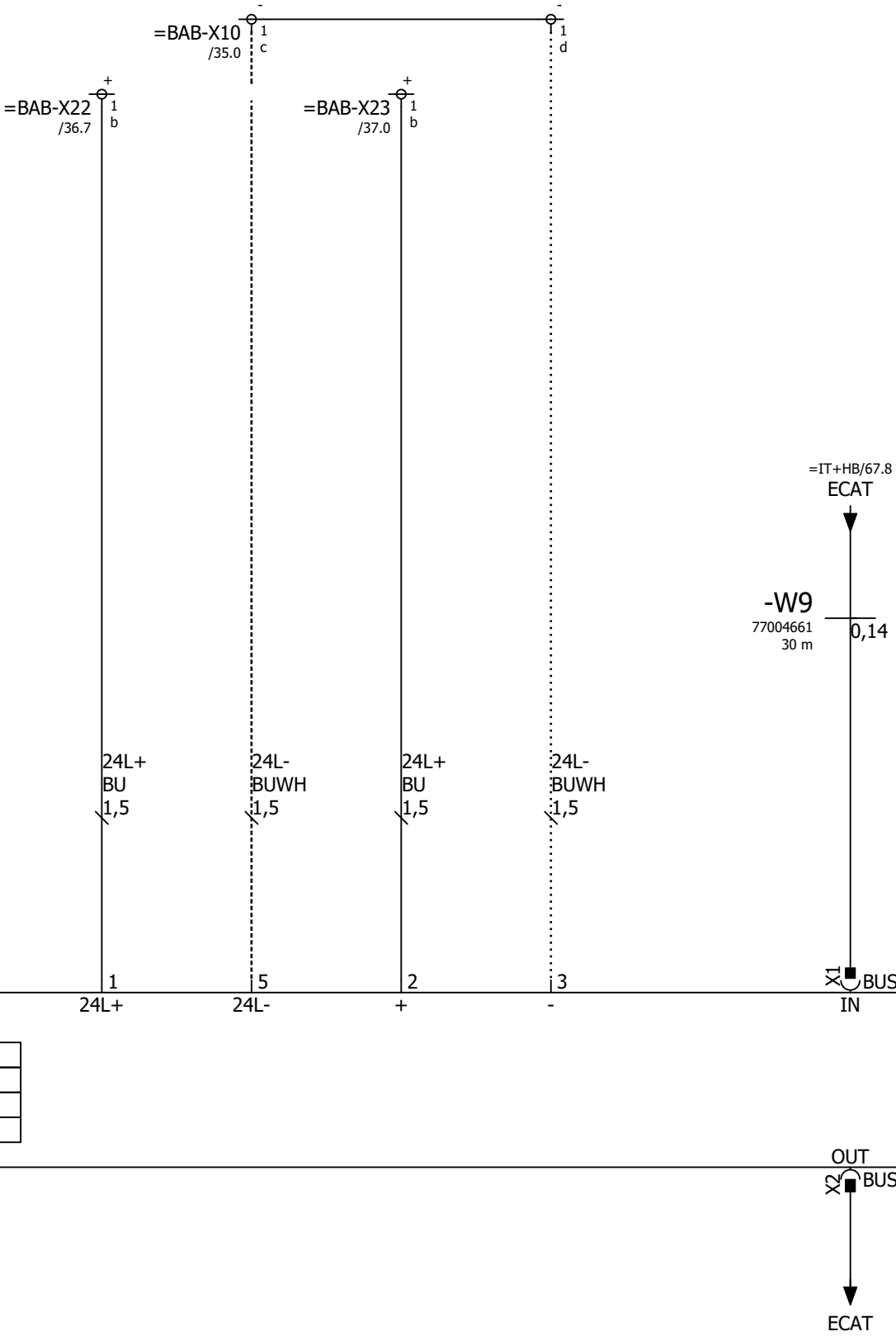


77011135
bus terminal

-K1 bus coupler

supply		
1	24L+	EtherCAT coupler operating supply /72.5
power contacts		
2	+	EtherCAT coupler powerbus /72.6
3	-	= /72.7
4	PE	
supply		
5	24L-	EtherCAT coupler operating supply /72.6
power contacts		
6	+	
7	-	
8	PE	

X1		
Bus		
BUS	IN	EtherCAT coupler Bus /72.8
X2		
BUS	OUT	EtherCAT coupler Bus /72.8



-K1 / 72
bus terminal
77011135
bus coupler

EtherCAT coupler

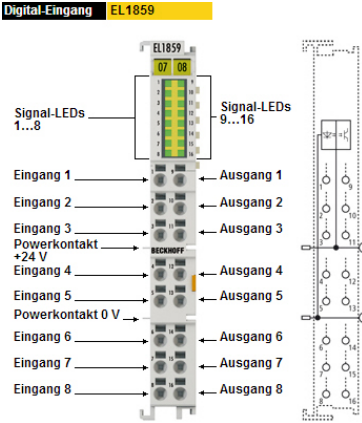


77011135

EtherCAT coupler
operating supply

EtherCAT coupler
powerbus

EtherCAT coupler
Bus



G10000441

bus terminal

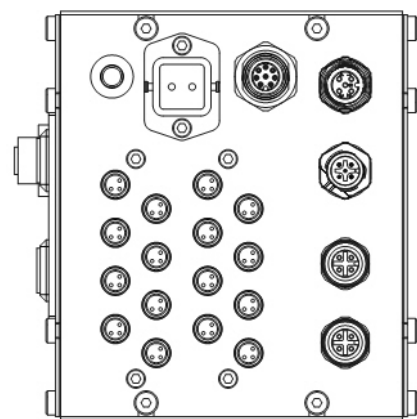
digital, E/A, 2 x 8 channel, 24 VDC

-K2

I1	
1	I1 exhaustion active =SP+P/97.3

O3
11 O3
O4
12 O4
O5
13 O5
O6
14 O6
O7
15 O7
O8
16 O8

I2
2 I2
I3
3 I3
I4
4 I4
I5
5 I5
I6
6 I6
I7
7 I7
I8
8 I8
O1
9 O1
O2
10 O2



72249810

lifter

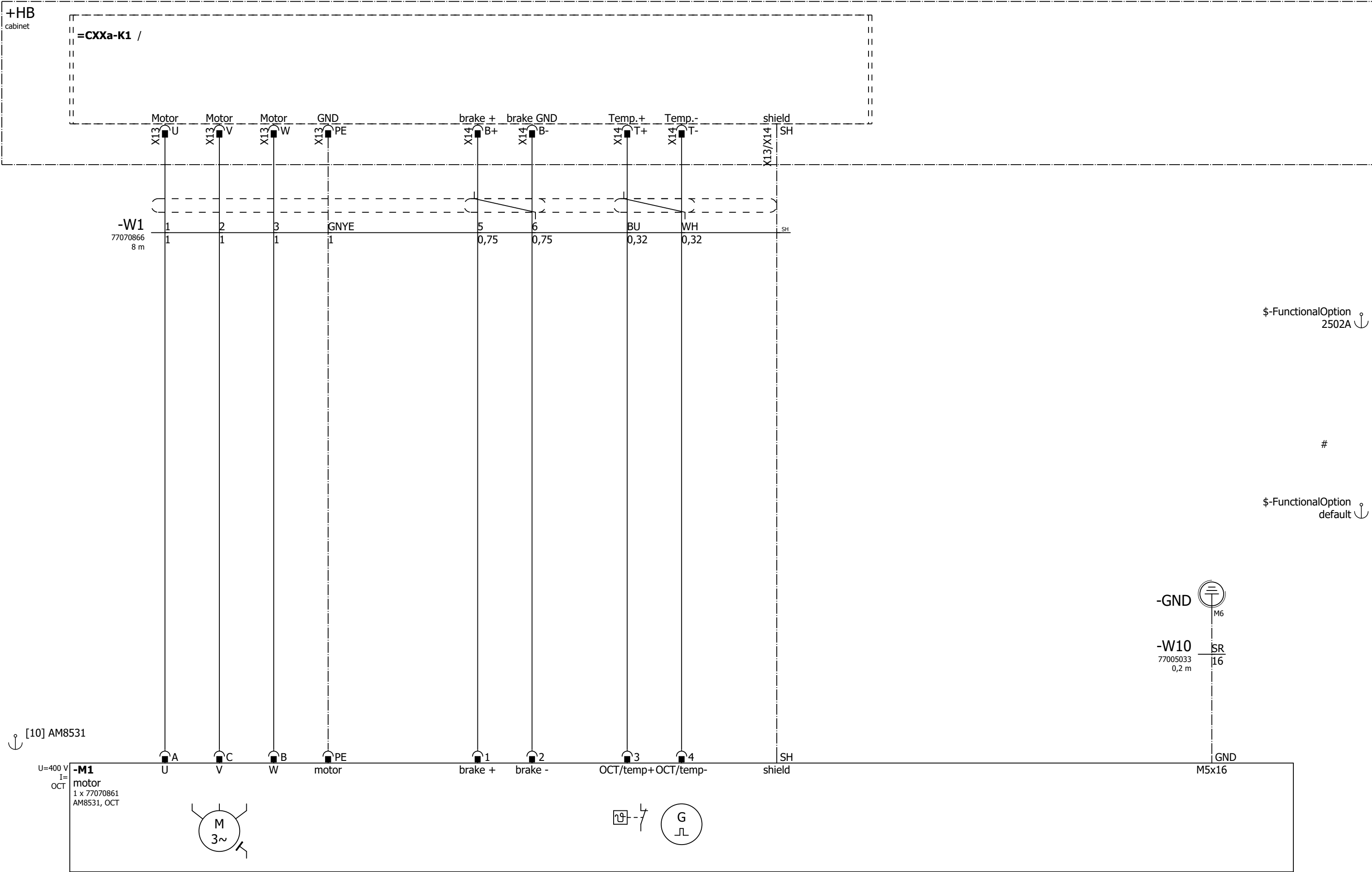
-K1 SmartLifter 200, 2 x 4 DI, 2 x 4 DO

J5			
4	PE	SmartLifter AC supply	/75.1
SH		SmartLifter periphery	/75.3
J7_I8			
	I8		
1	24L+	magnet collision torch	=DC1/51.2
3	24L-		=DC1/51.2
4	I8		=DC1/51.2
J7_I9			
	I9		
1	24L+	collision nozzle	=DC1B/52.2
3	24L-		=DC1B/52.2
4	I9		=DC1B/52.2
J7_I10			
	I10		
1	24L+		
3	24L-		
4	I10		
J7_I11			
	I11		
1	24L+		
3	24L-		
4	I11		
J7_I12			
	I12		
1	24L+		
3	24L-		
4	I12		
J7_I13			
	I13		
1	24L+		
3	24L-		
4	I13		
J7_I14			
	I14		
1	24L+		
3	24L-		
4	I14		

J7_I15			
	I15		
1	24L+		
3	24L-		
4	I15		
J7_O8			
	O8		
1	24L+		=UP1/100.2
3	24L-	laser diode	=UP1/100.2
4	O8		=UP1/100.2
J7_O9			
	O9		
1	24L+		
3	24L-		
4	O9		
J7_O10			
	O10		
1	24L+		
3	24L-		
4	O10		
J7_O11			
	O11		
1	24L+		
3	24L-		
4	O11		
J7_O12			
	O12		
1	24L+		
3	24L-		
4	O12		
J7_O13			
	O13		
1	24L+		
3	24L-		
4	O13		
J7_O14			
	O14		
1	24L+		
3	24L-		
4	O14		

J7_O15		
	O15	
1	24L+	
3	24L-	
4	O15	

J1			
	EtherCAT		
Bus	IN	SmartLifter EtherCAT	/75.9
J2			
Bus	OUT	SmartLifter EtherCAT	/75.9
J3			
	PwrDC		
1	Us	SmartLifter DC operating voltage	/75.5
2	Up	SmartLifter DC control voltage	/75.7
3	GND	SmartLifter DC operating voltage	/75.5
4	n.c		= /75.6
5	n.c		/75.6
SH			/75.8
J4			
	PwrALFA		
1	24L		
2	nc		
3	24N		
4	nc		
5	nc		
SH			
J5			
	PwrDrive		
1	36Lp	SmartLifter AC power supply	/75.3
2	36N		= /75.2
3	36Ls		/75.3



\$-FunctionalOption 2502A

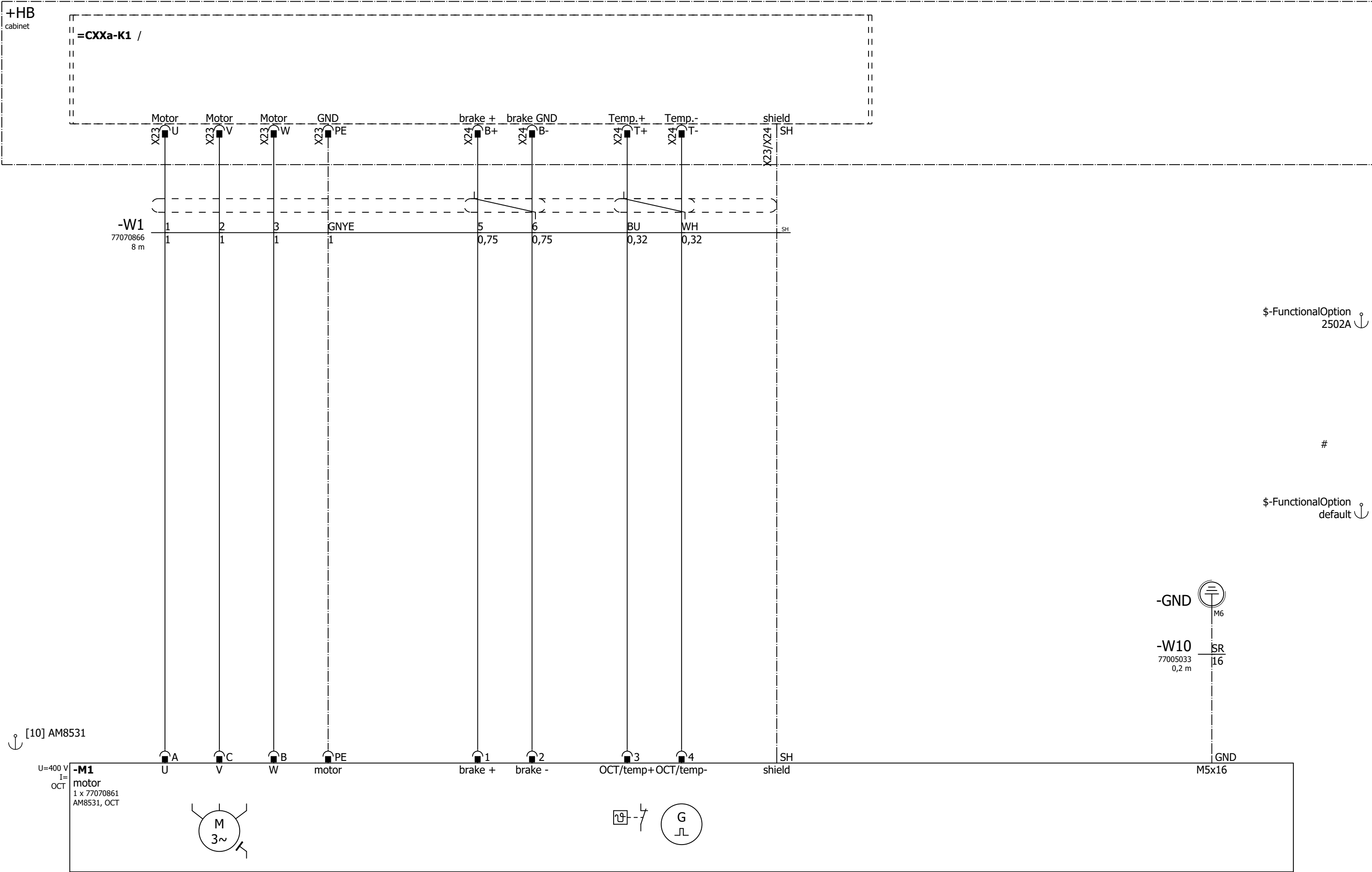
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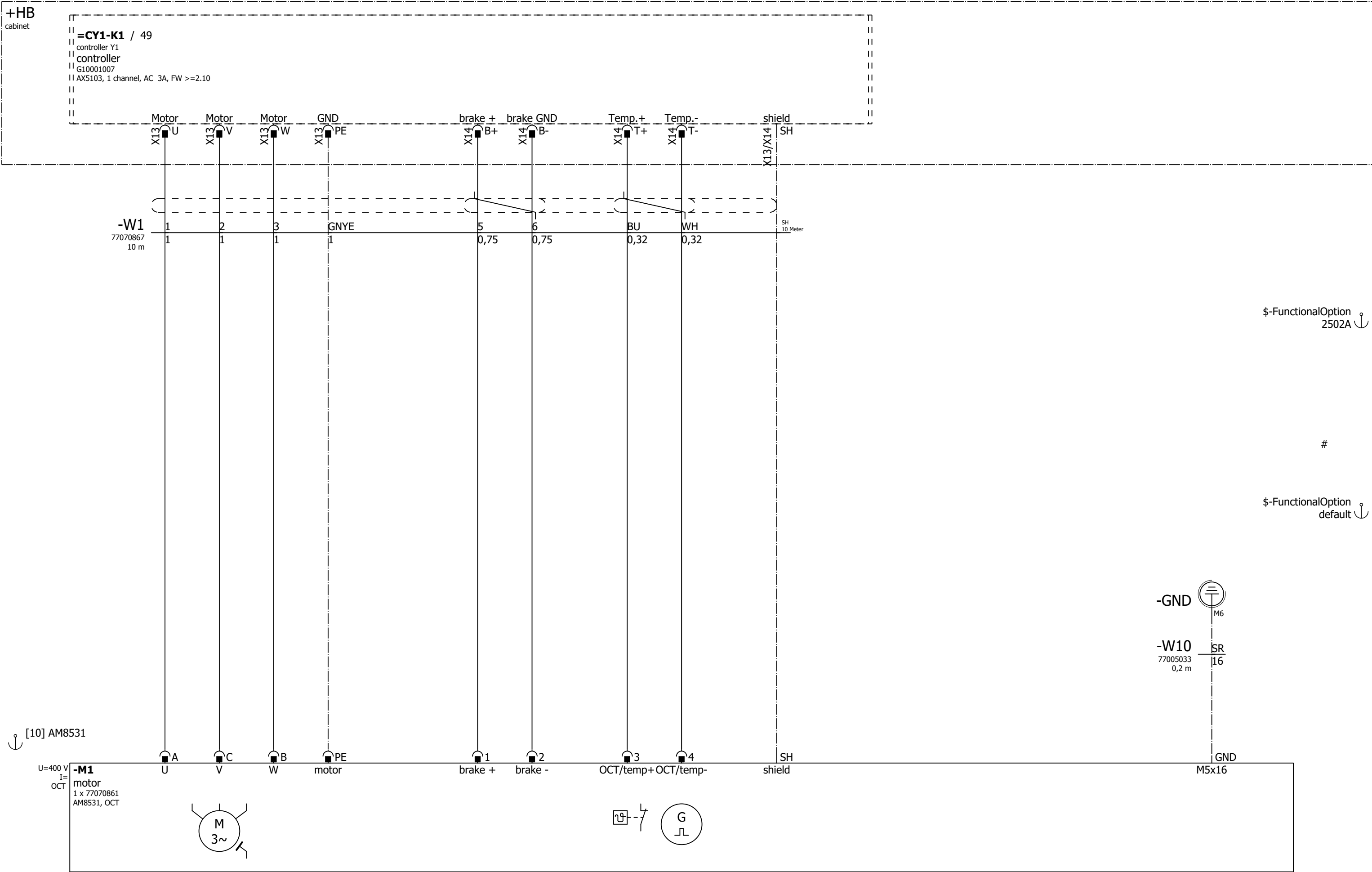
\$-FunctionalOption default

-GND M6

-W10 77005033 0,2 m SR 16

GND M5x16

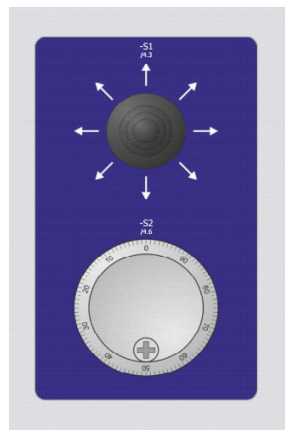




\$-FunctionalOption 2502A

#

\$-FunctionalOption default

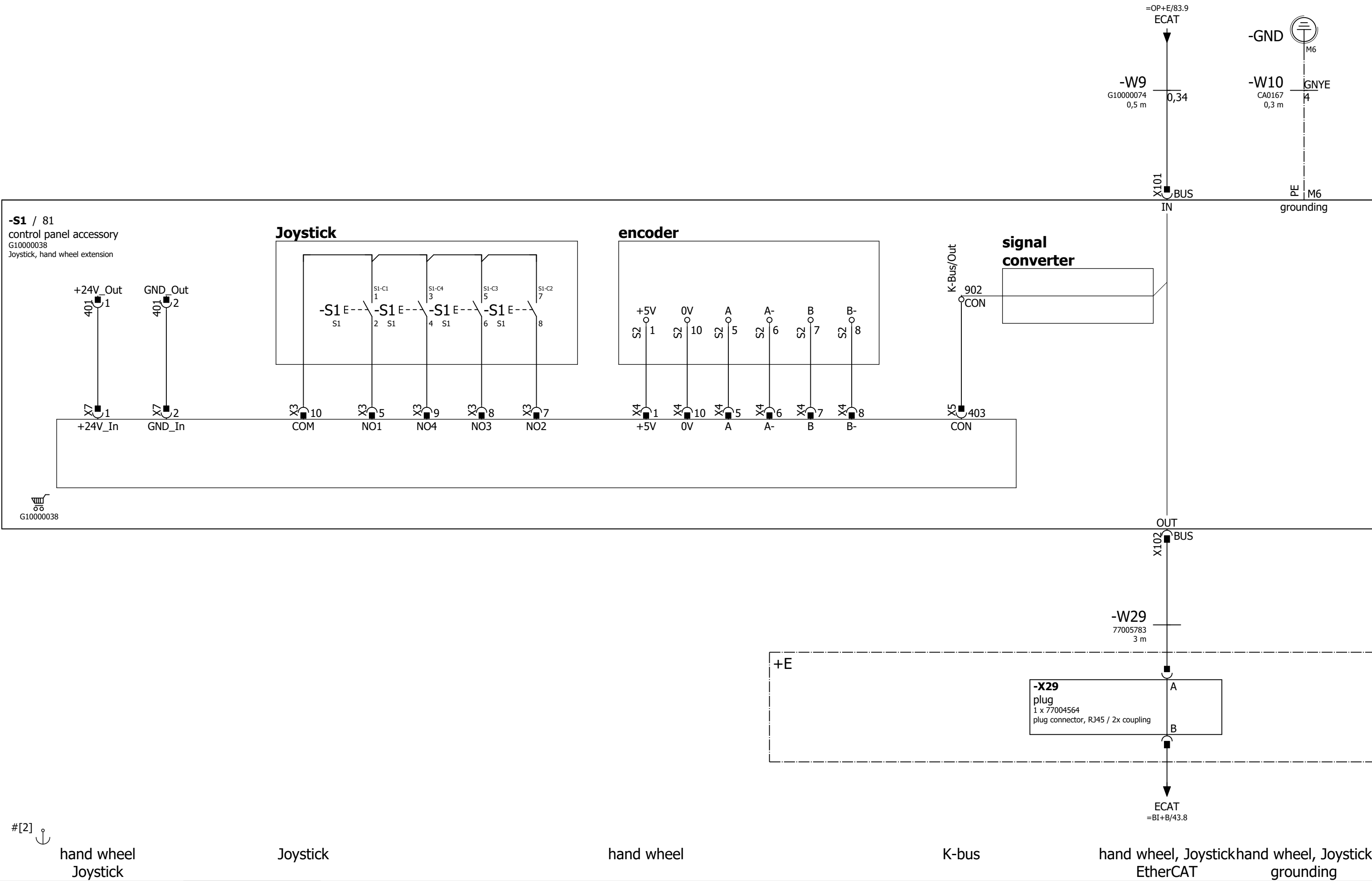


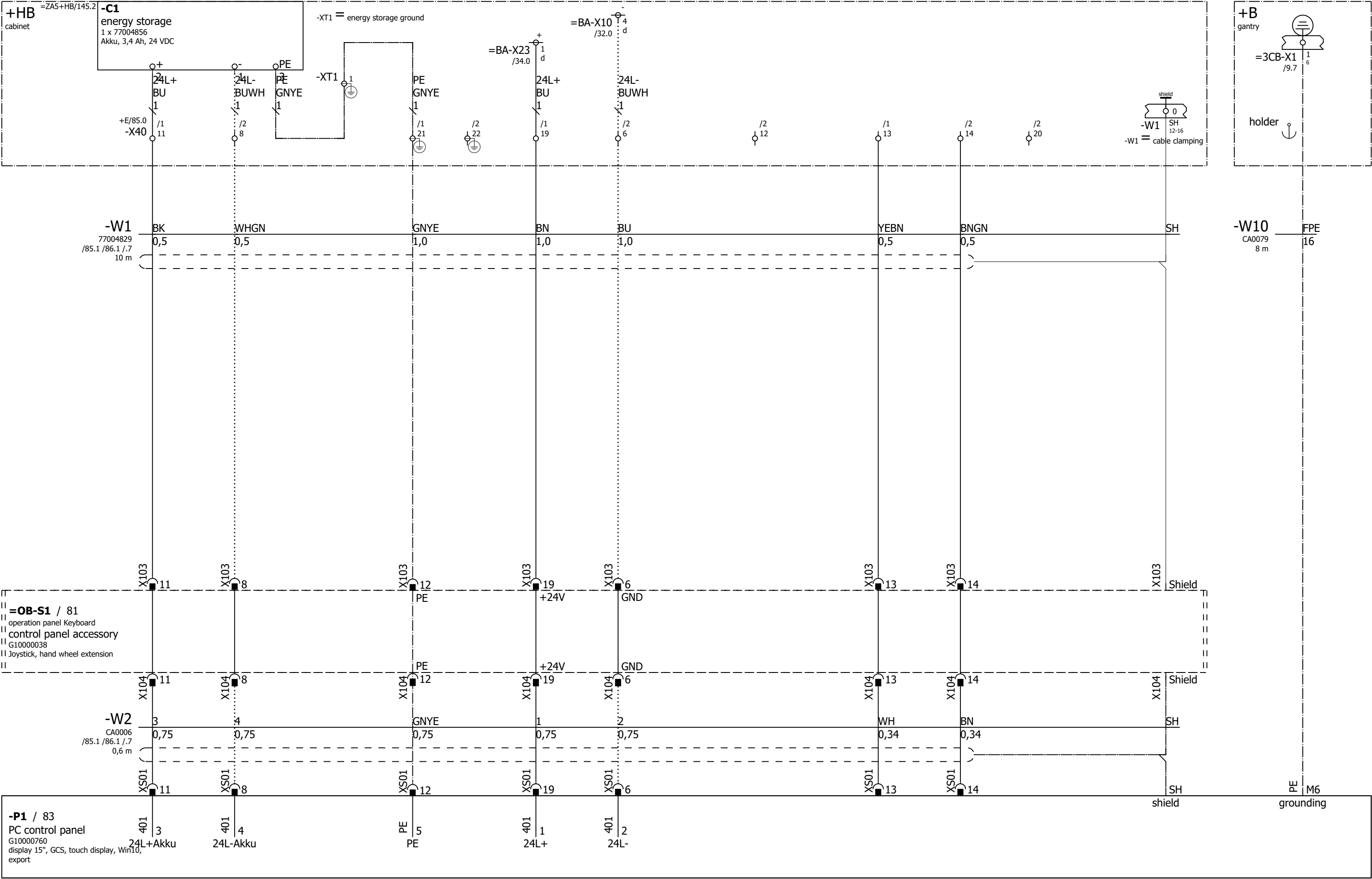
G10000038
control panel accessory
-S1 Joystick, hand wheel extension

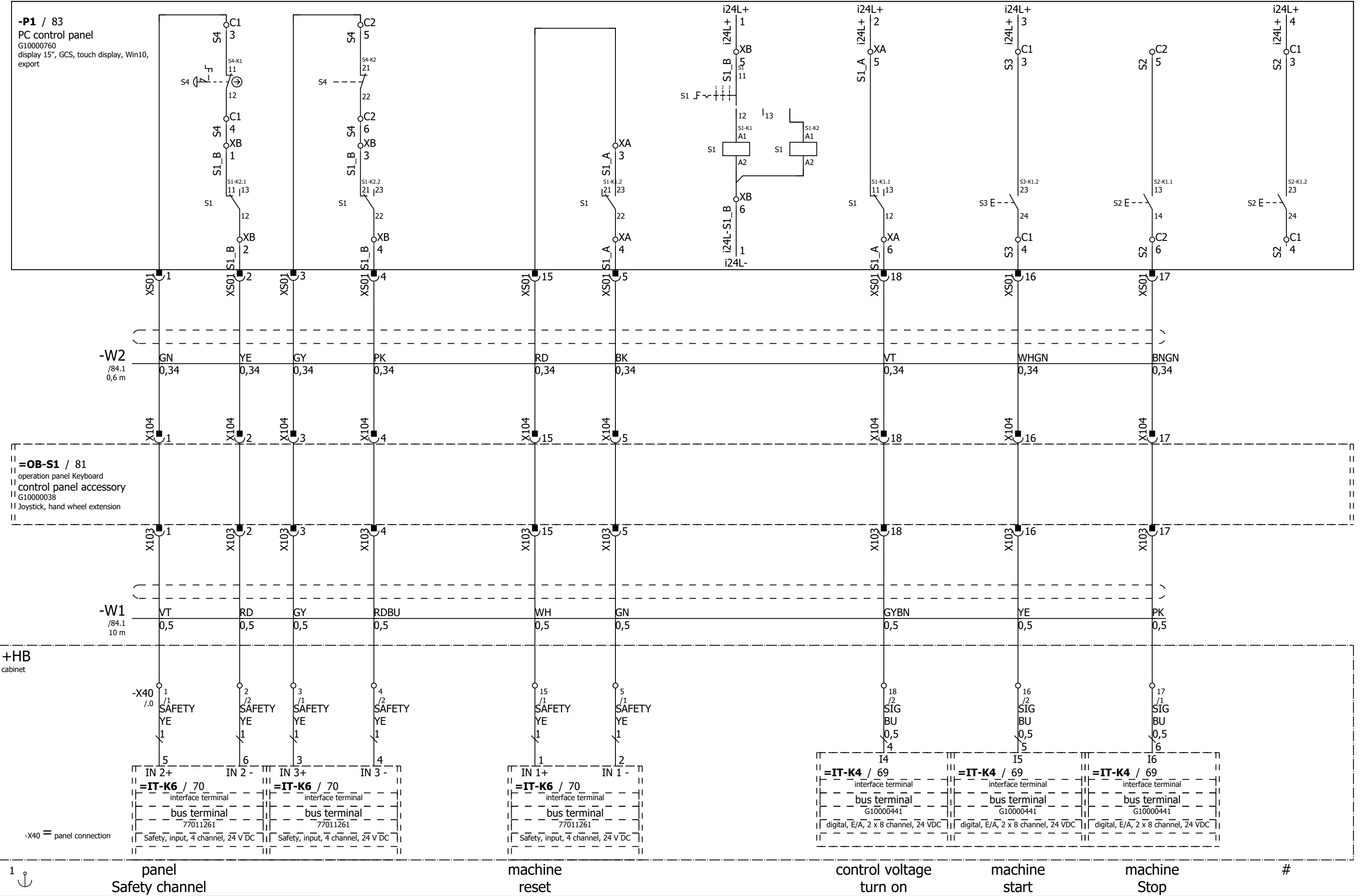
400			
Versorgung			
1	+24V_In		
2	GND_In		
401			
1	+24V_Out	hand wheel Joystick	/81.0
2	GND_Out		/81.1
K-Bus/Out			
K-Bus			
902	CON	K-bus	/81.7
PE			
grounding			
M6	grounding	hand wheel, Joystick grounding	/81.9
S2			
Encoder			
1	+5V	hand wheel	/81.4
5	A		/81.5
6	A-		/81.5
7	B		/81.5
8	B-		/81.6
10	0V		/81.5
X01			
Verteiler			
1	+24V		
2	+24V		
3	PE		

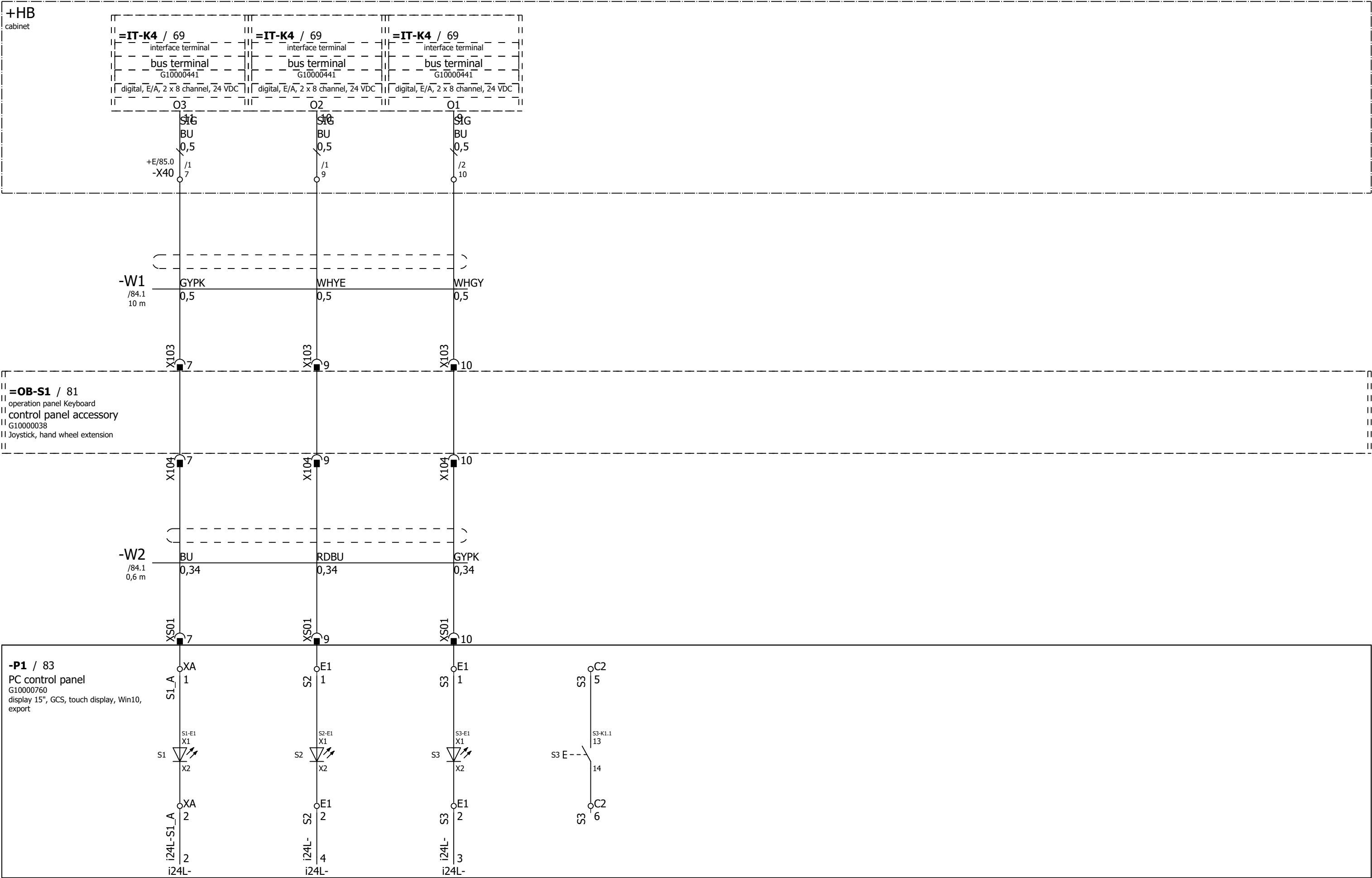
X01			
4	GND		
5	GND		
X3			
Joystick			
5	NO1		/81.2
7	NO2		/81.3
8	NO3		/81.3
9	NO4		/81.3
10	COM	Joystick	/81.2
X4			
Encoder			
1	+5V	hand wheel	/81.4
5	A		/81.5
6	A-		/81.5
7	B		/81.5
8	B-		/81.6
10	0V		/81.5
X5			
K-Bus			
403	CON	K-bus	/81.7
X7			
Versorgung			
1	+24V_In	hand wheel Joystick	/81.0
2	GND_In		/81.1
X101			
Bus			
BUS	IN	hand wheel, Joystick EtherCAT	/81.8
X102			
BUS			
BUS	OUT	hand wheel, Joystick EtherCAT	/81.8
X103			
Interface-in			
1		panel Safety channel	=OP/85.1
2		=	=OP/85.1
3		=	=OP/85.2
4		=	=OP/85.2
5			=OP/85.4
6	GND	operation panel supply	=OP/84.4
7			=OP/86.1
8		operation panel battery pack	=OP/84.1

X103			
9		machine in Stop	=OP/86.2
10		machine in start	=OP/86.3
11		operation panel battery pack	=OP/84.1
12	PE	operation panel supply	=OP/84.3
13			=OP/84.6
14			=OP/84.7
15		machine reset	=OP/85.3
16		machine start	=OP/85.7
17		machine Stop	=OP/85.8
18		control voltage turn on	=OP/85.6
19	+24V	operation panel supply	=OP/84.3
Shield			=OP/84.8
X104			
Interface-out			
1		panel Safety channel	=OP/85.1
2		=	=OP/85.1
3		=	=OP/85.2
4		=	=OP/85.2
5			=OP/85.4
6	GND	operation panel supply	=OP/84.4
7			=OP/86.1
8		operation panel battery pack	=OP/84.1
9		machine in Stop	=OP/86.2
10		machine in start	=OP/86.3
11		operation panel battery pack	=OP/84.1
12	PE		=OP/84.3
13			=OP/84.6
14			=OP/84.7
15		machine reset	=OP/85.3
16		machine start	=OP/85.7
17		machine Stop	=OP/85.8
18		control voltage turn on	=OP/85.6
19	+24V	operation panel supply	=OP/84.3
Shield			=OP/84.8









control voltage
selected

machine
in Stop

machine
in start

			Toolbox_31	Option: GCs+HWJ {R.29}		Operation interface	module	operation panel	Page 86
			Toolbox version	customer: Blocotelha		operating elements			
			OP6 - ABI - 25.30.00 - CC: EU	commission: 289001273					
Modification	Date	Name	Toolbox module designation	Ed. by. Thomas.Lehle	01.04.2021		sector 3C2E location	+E	=OP



72424271

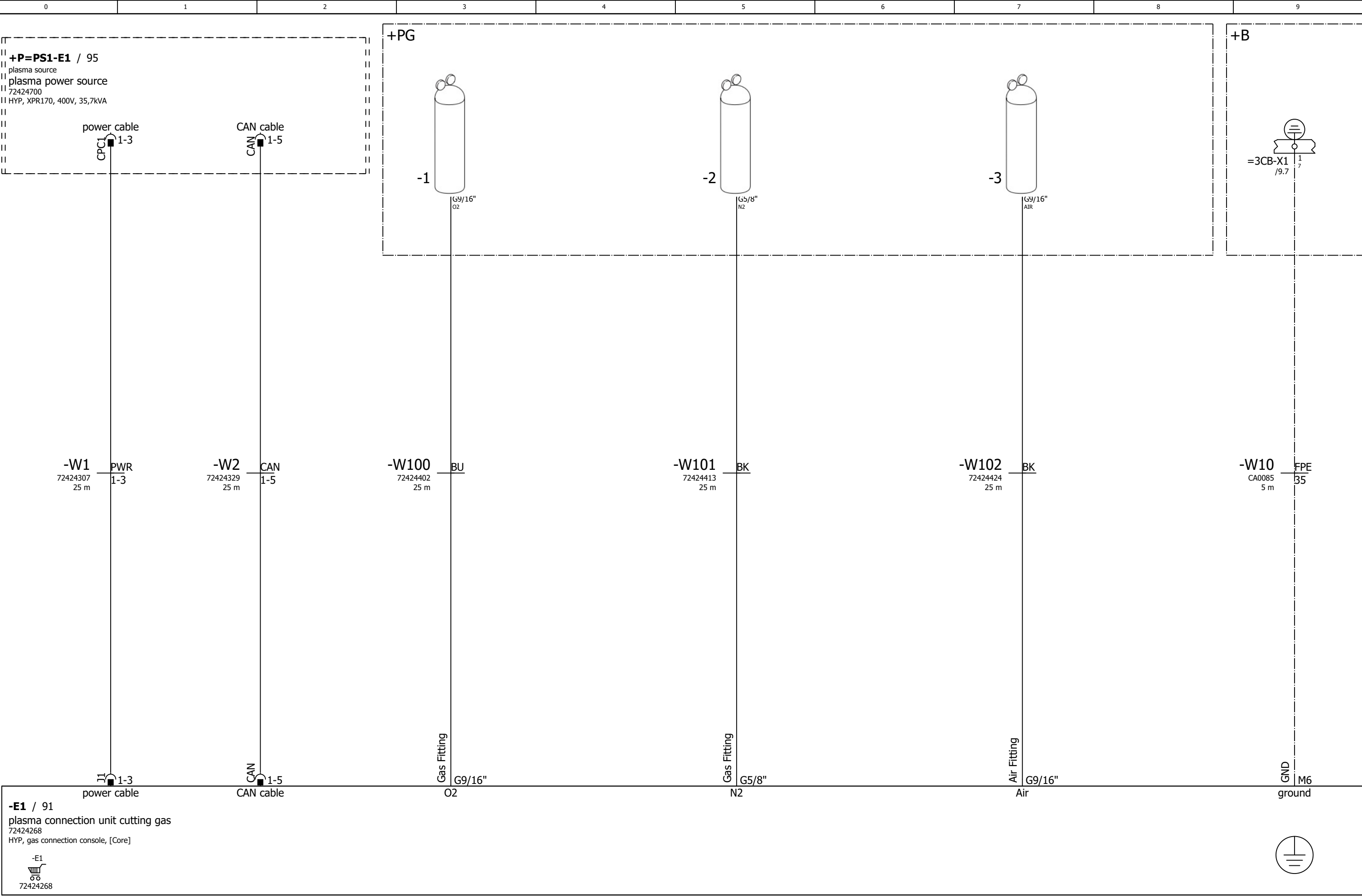
plasma connection unit torch

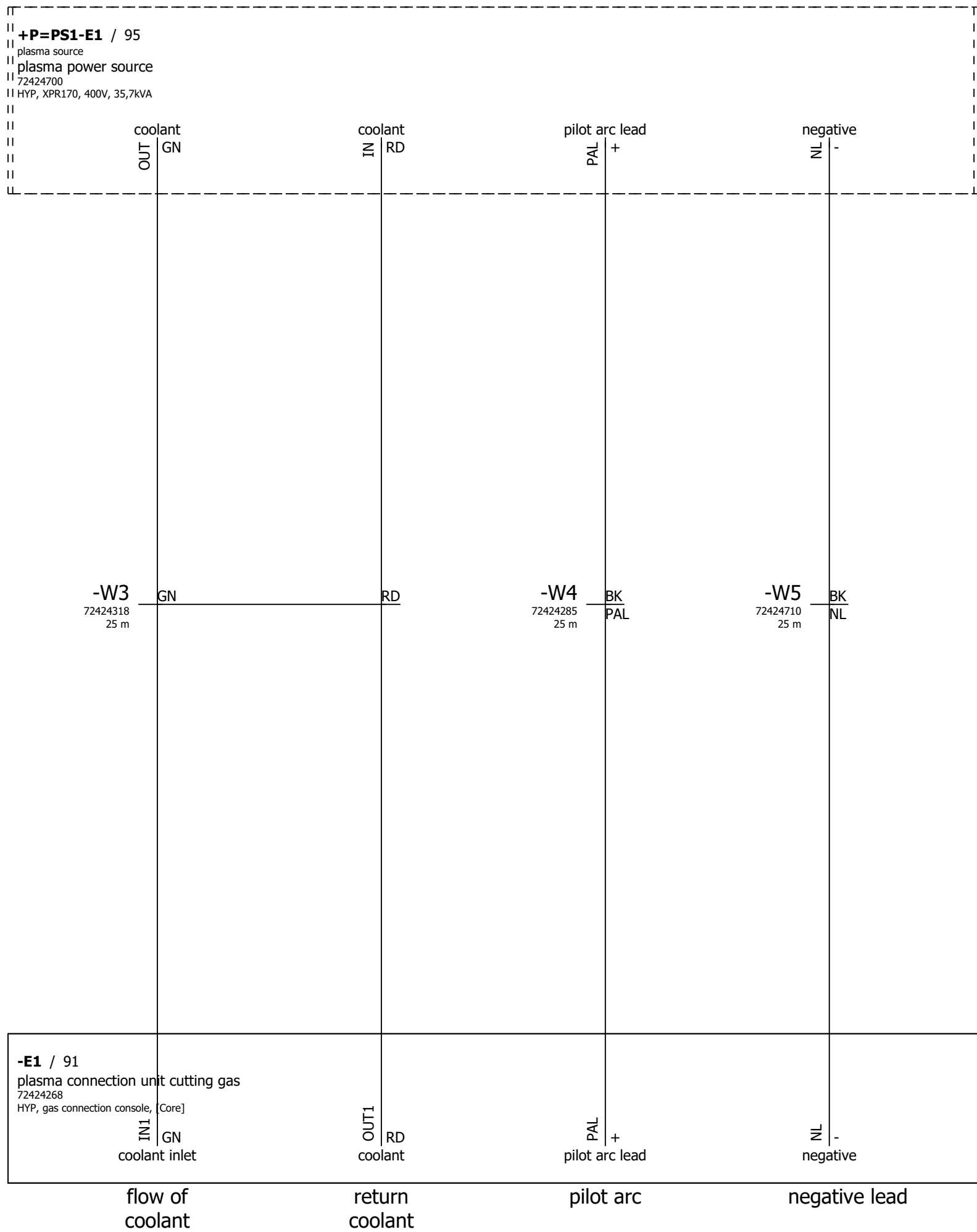
-E1

HYP, XPR300, EasyConnect

CAN			
gas console (lead 1) [Core]			
1-5	CAN cable	CAN bus	/89.1
gas console (lead 1) [VWI/Opti			
1-5	CAN cable		
Coupler			
Torch			
BK	plasmagas A	plasma gas 1	=PT1/96.1
BU	shield gas	gas secondary	=PT1/96.3
GN	coolant outlet	flow of coolant	=PT1/96.4
RD	coolant inlet	return coolant	=PT1/96.5
YE	plasmagas B	plasma gas 2	=PT1/96.2
YE	pilot arc lead	pilot arc	=PT1/96.6
YE	ohmic lead	line resistor	=PT1/96.7
GND			
grounding			
M6	ground	torch connection console grounding	/89.9
IN			
gas console (lead 2) [Core]			
GN	coolant inlet	flow of coolant	/89.6
gas console (lead 2) [VWI/Opti			
GN	coolant inlet		
Inlet Air			
gas console (lead 1) [Core]			
G1/8"NPT	Air	gas air	/89.2

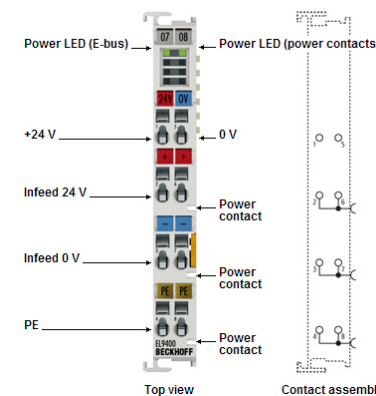
Inlet Air			
gas console (lead 1) [VWI/Opti			
G1/8"NPT	Air		
Inlet Gas			
gas console (lead 1) [Core]			
G1/4"NPT.RHBF	N2	gas N2	/89.3
G1/8"NPT.RHFA	O2	gas O2	/89.4
gas console (lead 1) [VWI/Opti			
G1/4"NPT	AR		
G1/4"NPT.RHBF	N2		
G1/8"NPT.RHFA	O2		
G1/8"NPTLHFA	H2 MIX		
Inlet Water			
gas console (lead 2) [VWI/Opti			
.	shield water		
J1			
gas console (lead 1) [Core]			
1-3	power cable		/89.0
gas console (lead 1) [VWI/Opti			
1-3	power cable		
J2			
Torch			
1-4	plasma valve cable	plasma valve control system	=PT1/96.0
OUT			
gas console (lead 2) [Core]			
RD	coolant outlet	return coolant	/89.7
gas console (lead 2) [VWI/Opti			
RD	coolant outlet		
PAL			
gas console (lead 2) [Core]			
-	pilot arc lead	pilot arc	/89.8
gas console (lead 2) [VWI/Opti			
-	pilot arc lead		



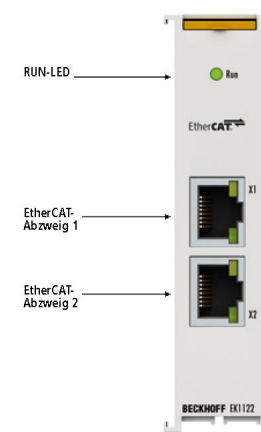
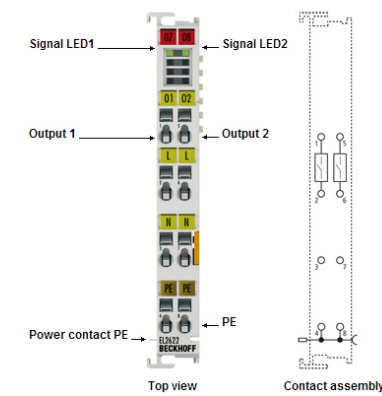


#03	25.03.2021	Thomas.Lehle	Toolbox_31	Option:	L1-HYP XPR170core+SwBox {R.30}		type core		module	plasma gas console	Page 92
02	12.03.2021	Adeline.Wegang	Toolbox version	customer:	Blocotelha		cooling				
01	28.01.2021	Adeline.Wegang	PG5 - ABI - 25.25.00 - CC: EU	commission:	289001273						
Modification	Date	Name	Toolbox module designation	Ed. by:	Thomas.Lehle 01.04.2021			sector	4A1B location	+E	

System Terminals EL9400, EL9410



Digital Output EL26X2



77011215

bus terminal

-K1 potential supply terminal, E-bus refreshment

power contacts		
2	24L+ plasma power source supply 24VDC	+P/95.1
3	24L- =	+P/95.2
4	PE	
6	24L+ plasma power source supply 24VDC	+P/95.1
7	24L- =	+P/95.2
8	PE	
supply		
1	24L+ plasma power source supply 24VDC	+P/95.1
5	24L- =	+P/95.2

77011138

bus terminal

-K2 digital, relay, 2x NO , 230 V AC/30 V DC

#		
3	nc	
K1		
1	NO plasma power source turn on	+P/95.3
2	COM =	+P/95.4
K2		
5	NO	
6	COM	
PE		
4	PE	
8	PE	
nc		
7	nc	

G10000781

bus terminal

-K3 EtherCAT port junction module, 2x RJ45, 100MBit/s

X1		
Bus		
OUT	Bus plasma power source EtherCAT	+P/95.6
X2		
OUT	Bus	



72424700

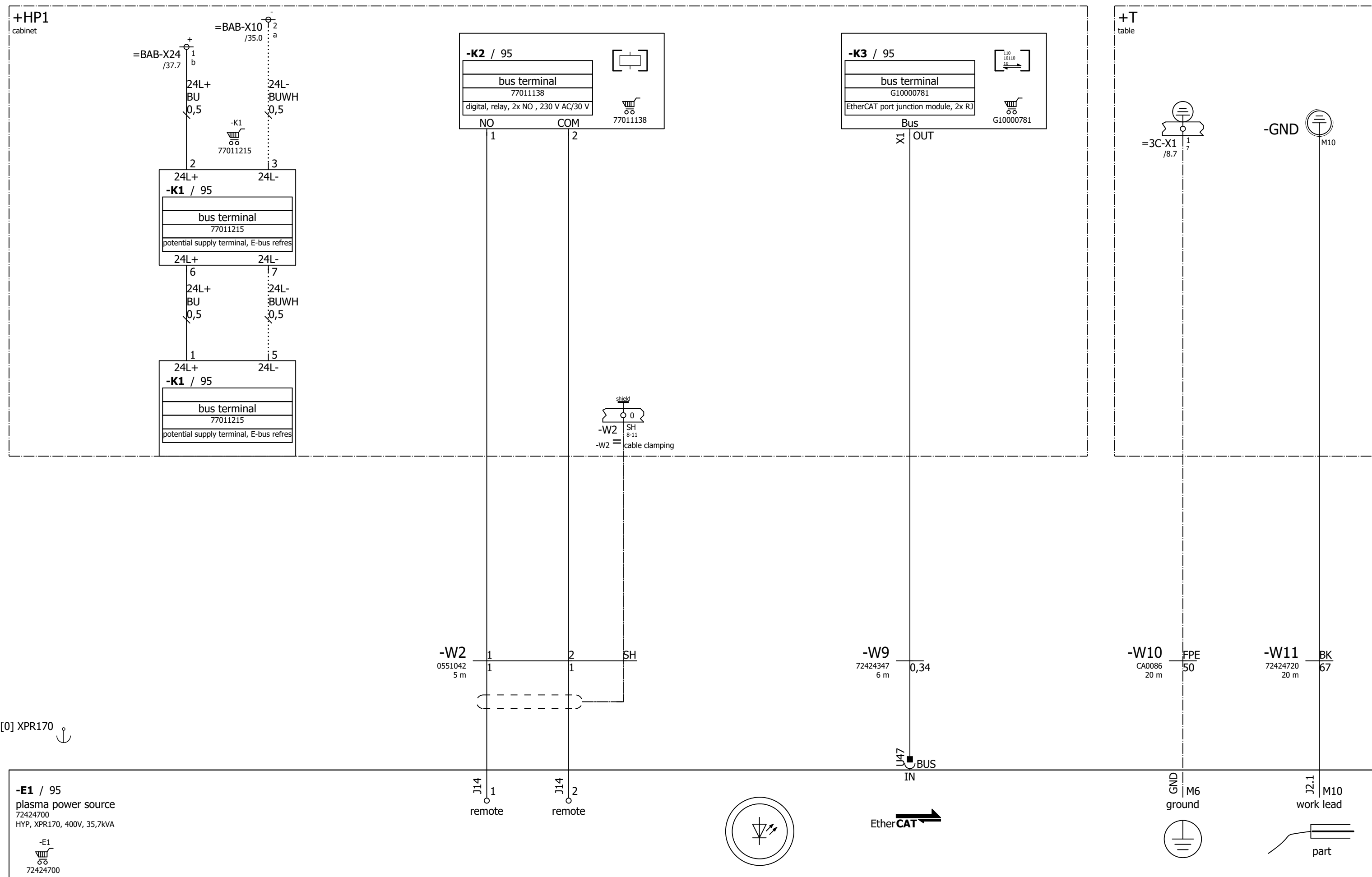
plasma power source

-E1HYP, XPR170, 400V, 35,7kVA

CAN			
Gas Connect Console			
1-5	CAN cable	CAN bus	=PG1+E/91.2
CPC1			
1-3	power cable	gas console supply	=PG1+E/91.0
GND			
Table			
M6	ground	plasma power source functional ground	/95.8
IN			
Gas Connect Console			
RD	coolant return	return coolant	=PG1+E/92.2
J2.1			
Table			
M10	work lead	grounding workpiece	/95.9
J12			
RS422			
2	Tx-		
3	Rx-		
4	Tx+		
5	GND		
7	Rx+		
J14			
Plasma start			
3	plasma start +		
4	plasma start -		

J14			
Remote ON			
1	remote ON/OFF plasma power source turn on	/95.3	
2	remote ON/OFF -	=	/95.4
hold			
7	hold +		
8	hold -		
motion			
5	motion +		
6	motion -		
shield pierce gas			
9	shield pierce gas +		
10	shield pierce gas -		
supply CNC			
11	F +24V CNC		
12	F PWRNGND		
J19			
Error			
1	error +		
2	error -		
Shield OHMIC			
7	shield ohmic contact +		
8	shield ohmic contact -		
auto pierce detect			
5	auto pierce detect +		
6	auto pierce detect -		
ready to start			
3	not ready for start +		
4	not ready for start -		
NL			
Gas Connect Console			
-	negative lead	=PG1+E/92.4	
OUT			
GN	coolant supply	flow of coolant	=PG1+E/92.0
PAL			
+	pilot arc lead	pilot arc	=PG1+E/92.3
TB1			
mains			
PE	PE	plasma power source power supply	=PA1+HJ1/87.1
U	L1	=	=PA1+HJ1/87.2
V	L2	=	=PA1+HJ1/87.3

TB1			
W	L3	plasma power source power supply	=PA1+HJ1/87.3
U47			
Bus			
BUS	IN	plasma power source EtherCAT	/95.6
U49			
BUS	OUT		



[0] XPR170

[21] XPR 170

plasma power source
supply 24VDC

plasma power source
turn on

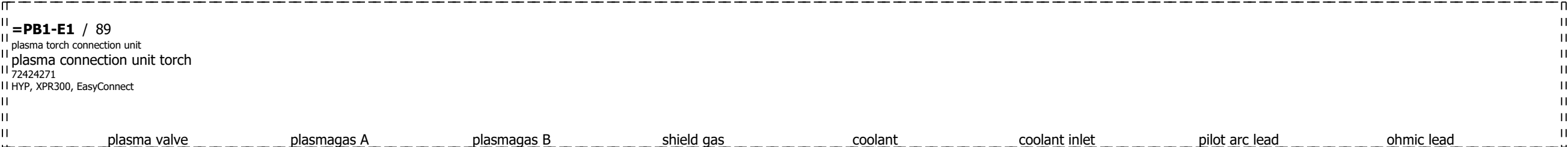
power source
state

plasma power source
EtherCAT

plasma power source
functional ground

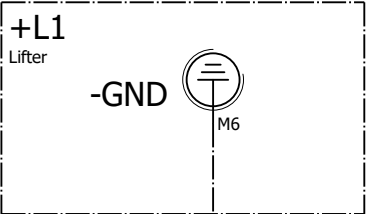
grounding
workpiece

#03	25.03.2021	Thomas.Lehle	Toolbox_31	Option:	L1-HYP XPR170core+SwBox {R.30}		type XPR 170		module plasma source		Page 95
02	28.01.2021	Adeline.Wegang	Toolbox version	customer:	Blocotelha		EtherCAT, grounding				
01	28.01.2021	Adeline.Wegang	PS19 - ABI - 30.30.00 - CC: EU	commission:	289001273						
Modification	Date	Name	Toolbox module designation	Ed. by:	Thomas.Lehle 01.04.2021				sector 3D7P location	+P =PS1	



-W1
72424391
2 m

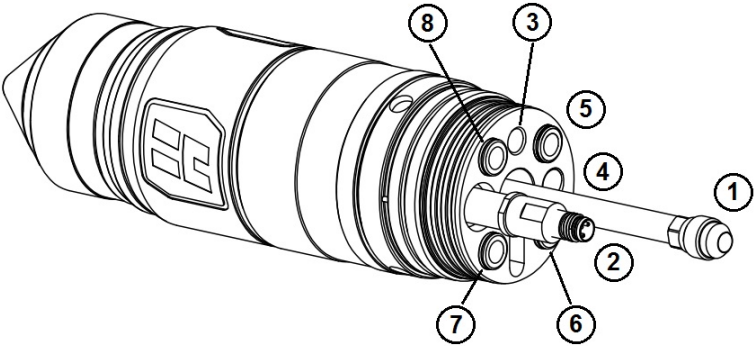
-W2
CA0067
2 m



Protective Sleeve

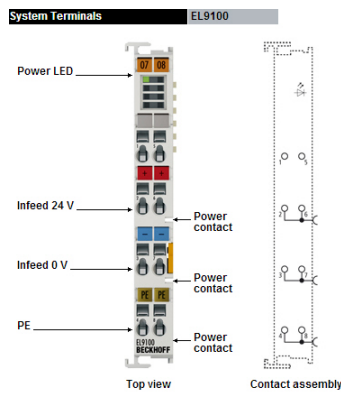
-E1
plasma valve
plasma torch add-on
1 x 72424272
HYP, torch pipe, mounting sleeve, standard length

GND
FS6,3
ground

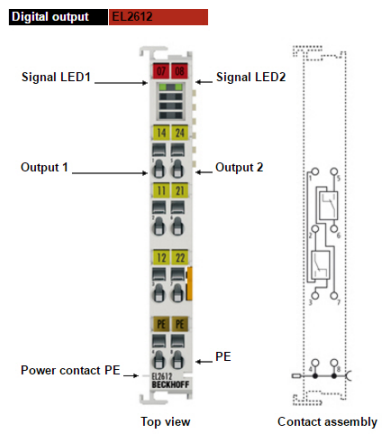


- 1 = coolant return
- 2 = plasma valve
- 3 = pilot arc
- 4 = ohmic line
- 5 = coolant feed-in
- 6 = secondary gas
- 7 = plasma gas 2
- 8 = plasma gas 1





EL9100 | Potential supply terminal, 24 V DC



77011141

bus terminal

-K1 potential supply terminal, 24 V DC

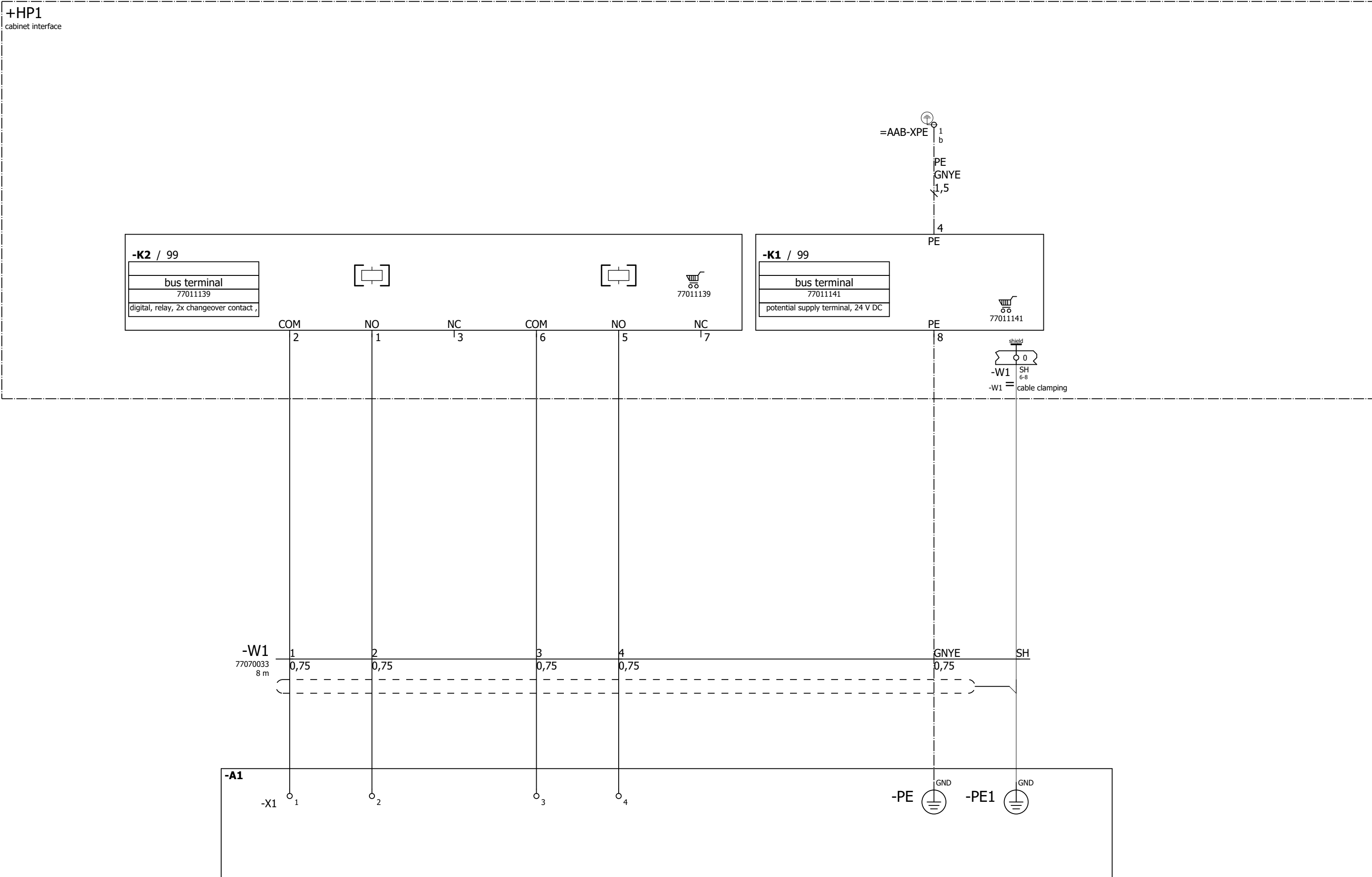
PE		
4	PE	filter fan grounding +P/99.6
8	PE	= +P/99.6
n.c.		
1	n.c	
5	n.c	
power contacts		
2	24L+	
3	24L-	
supply 24L+		
6	24L+	
supply 24L-		
7	24L-	

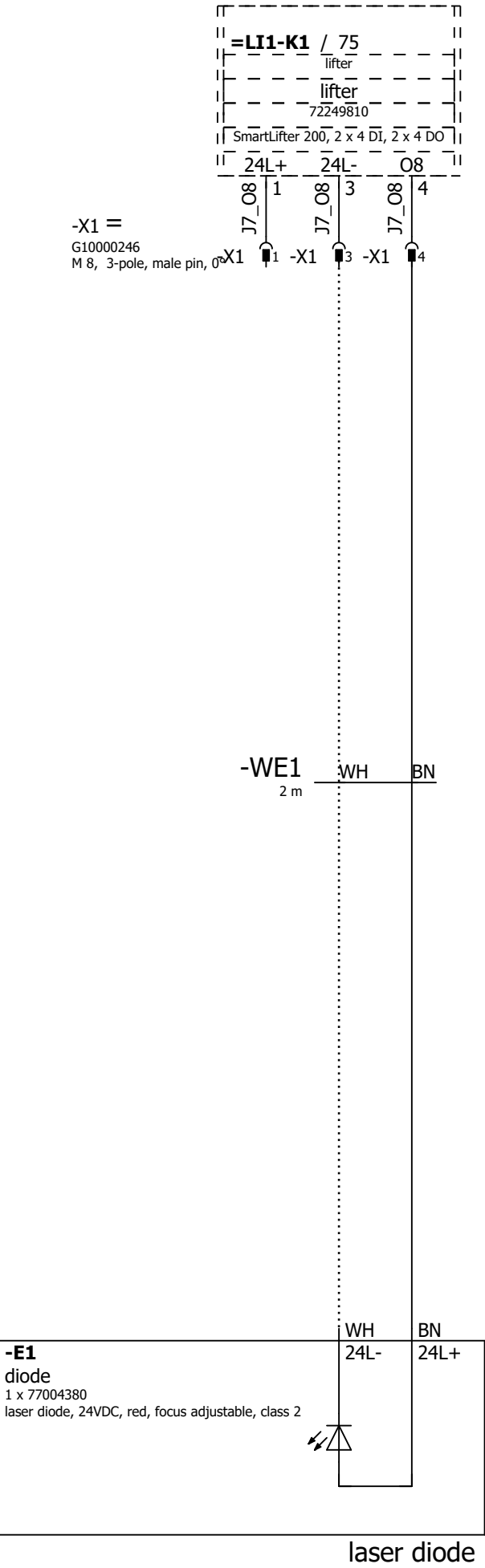
77011139

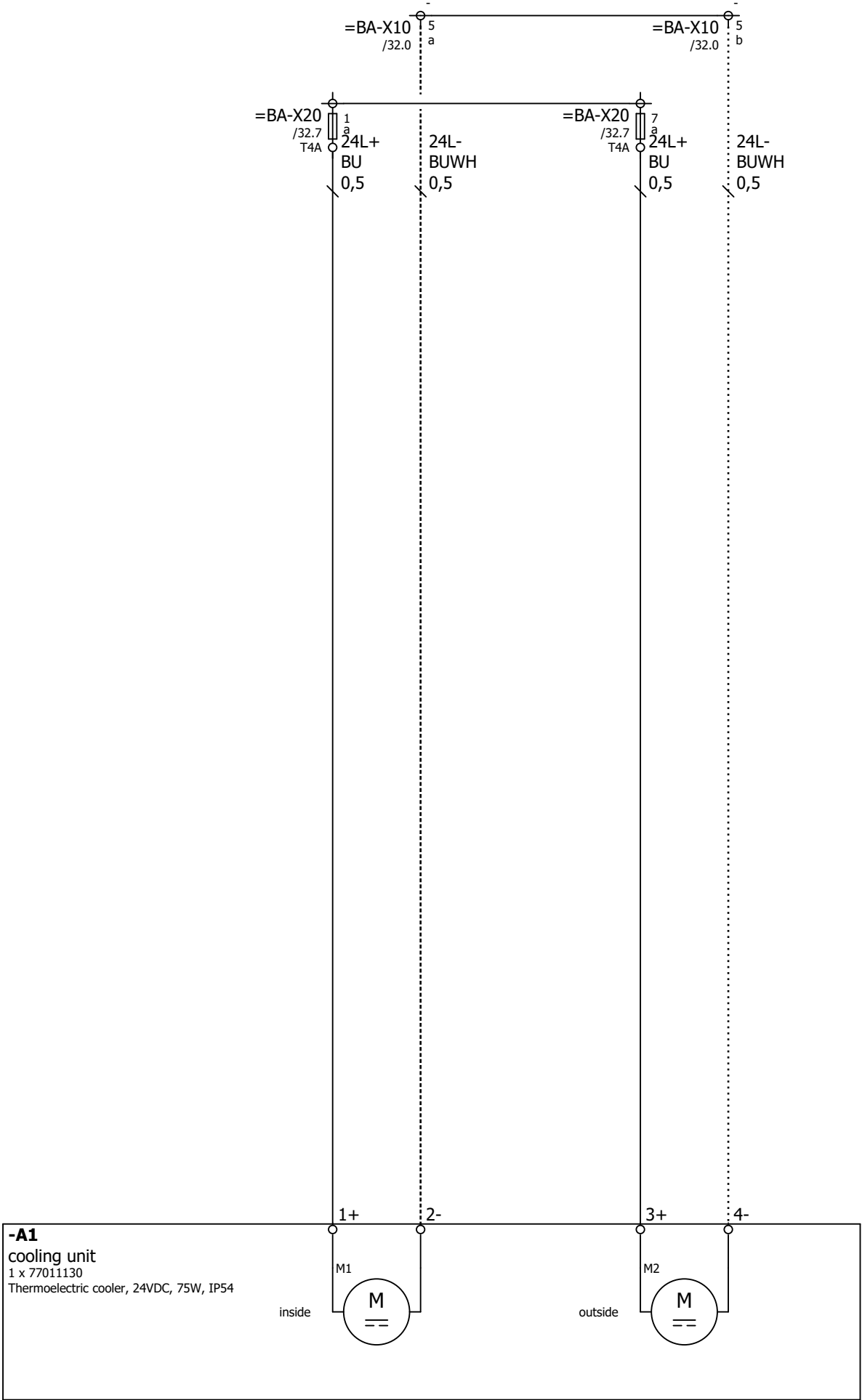
bus terminal

-K2 digital, relay, 2x changeover contact , 125 V AC 2A

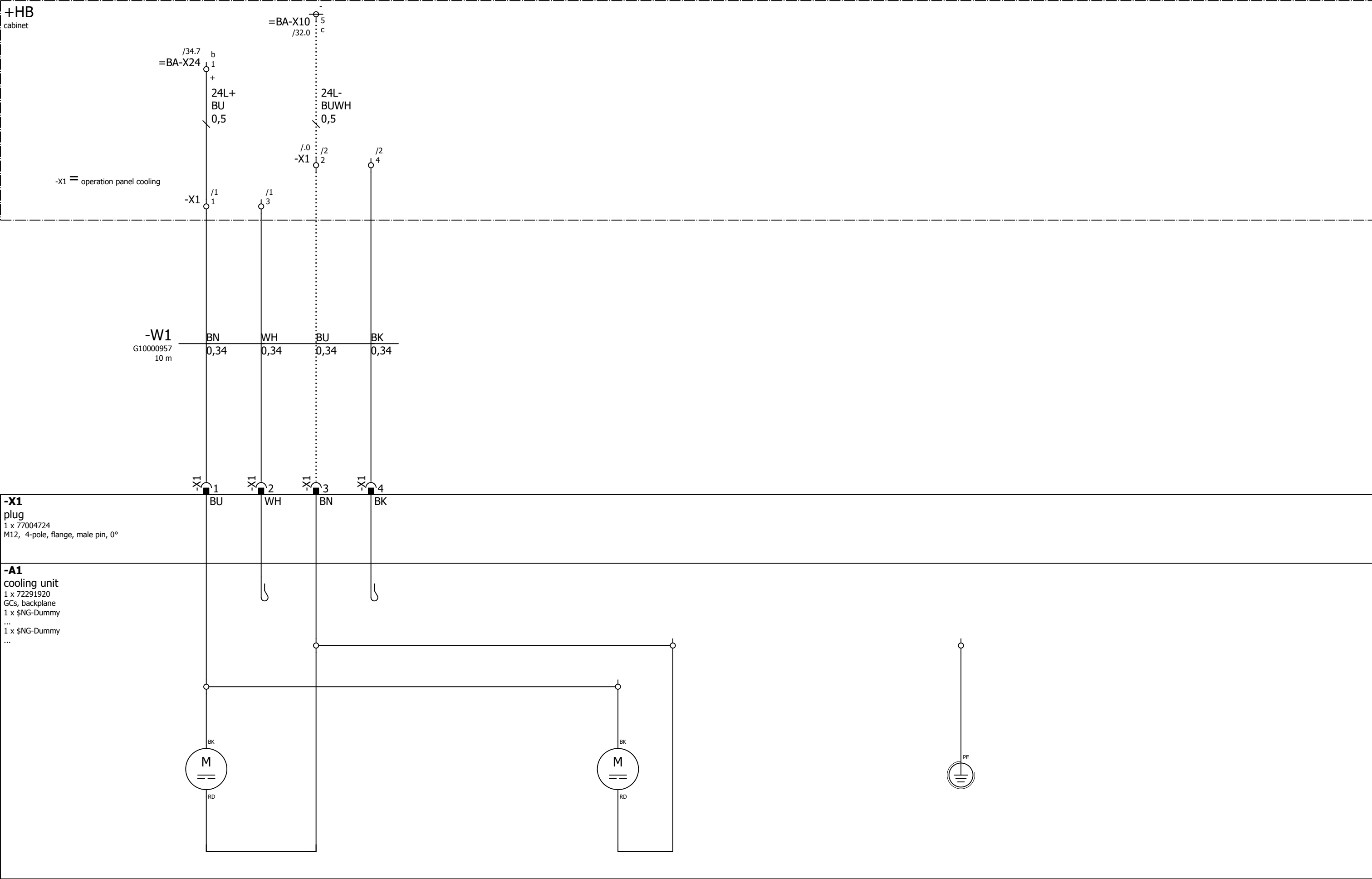
K1		
2	COM	filter fan turn on +P/99.2
7	NC	filter fan enabling +P/99.5
K1		
1	NO	filter fan turn on +P/99.2
3	NC	= +P/99.3
K2		
5	NO	filter fan enabling +P/99.4
6	COM	= +P/99.3
PE		
4	PE	
8	PE	







operating voltage
power supply 24VDC



operation panel cooling

operation panel cooling

operation panel cooling grounding box

CA0078

+HB=3B-W10

MCS* Cable Production

7 m - 16 mm²

△

M10 - female

+B=3CB-X1

c

Page: 5

cable // FPE, ring, OE-M10, 16mm²

OE -

+HB=3B-X0FPE

a

grounding cabinet //

Cable Plan Formular: =Z6B++111+A/110

77005032

+HB=3B-WH1

0,1 m - 16 mm²



M6 - female

+HB=3B-PE1

Page: 5

ground strap // M6-M6, 0,1 m

M6 - female

+HB=3B-PEM

grounding cabinet //

77005034

+HB=3B-WH2

0,3 m - 16 mm²



M6 - female

+HB=3B-PE2

Page: 5

ground strap // M6-M6, 0,3 m

M6 - female

+HB=3B-PEM

grounding cabinet //

77005035

+HB=3B-WH3

0,6 m - 16 mm²



M6 - female

+HB=3B-PE3

Page: 5

ground strap // M6-M6, 0,6 m

M6 - female

+HB=3B-PEM

grounding cabinet //

CA0078

+HB=3BB-W10

MCS* Cable Production

7 m - 16 mm²

△

M10 - female

+B=3CB-X1

d

Page: 6

cable // FPE, ring, OE-M10, 16mm²

OE -

+HB=3BB-X0FPE

a

grounding cabinet //

Cable Plan Formular: =Z6B++111+A/111

77005032

+HB=3BB-WH1

0,1 m - 16 mm²



M6 - female

+HB=3BB-PE1

Page: 6

ground strap // M6-M6, 0,1 m

M6 - female

+HB=3BB-PEM

grounding cabinet //

77005034

+HB=3BB-WH2

0,3 m - 16 mm²



M6 - female

+HB=3BB-PE2

Page: 6

ground strap // M6-M6, 0,3 m

M6 - female

+HB=3BB-PEM

grounding cabinet //

77005035

+HB=3BB-WH3

0,6 m - 16 mm²



M6 - female

+HB=3BB-PE3

Page: 6

ground strap // M6-M6, 0,6 m

M6 - female

+HB=3BB-PEM

grounding cabinet //

CA0078

+HP1=3BC-W10

MCS* Cable Production

18 m - 16 mm²

△

OE -

+HP1=3BC-X0FPE

a

Page: 7

cable // FPE, ring, OE-M10, 16mm²

M10 - female

+T=3C-X1

e

grounding cabinet //

Cable Plan Formular: =Z6B++111+A/112

77005032

+HP1=3BC-WH1

0,1 m - 16 mm²



M6 - female

+HP1=3BC-PE1

Page: 7

ground strap // M6-M6, 0,1 m

M6 - female

+HP1=3BC-PEM

grounding cabinet //

77005034

+HP1=3BC-WH2

0,3 m - 16 mm²



M6 - female

+HP1=3BC-PE2

Page: 7

ground strap // M6-M6, 0,3 m

M6 - female

+HP1=3BC-PEM

grounding cabinet //

77005035

+HP1=3BC-WH3

0,6 m - 16 mm²



M6 - female

+HP1=3BC-PE3

Page: 7

ground strap // M6-M6, 0,6 m

M6 - female

+HP1=3BC-PEM

grounding cabinet //

			Toolbox_31	Option:		MESSER		module	Page 103
			Toolbox version	customer:	Blocotelha		layout plan		
			automated	commission:	289001273				
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle01.04.2021				
							sector 111	location +A	=Z6A

CA0065

+T=3C-W1

MCS* Cable Production

14 m - 70 mm²

M10 - female

+T=3C-X1

a

Page: 8

cable // grounding, PE, ring, OE-M10, 70mm²

+PZ=CUSTOMERGND

OE -

grounding bar // grounding bar

Cable Plan Formular: =Z6B++111+A/113

CA0066

+T=3C-W2

MCS* Cable Production

14 m - 70 mm²

M10 - female

+T=3C-X1

b

Page: 8

cable // FPE, ring, OE-M10, 70mm²

+PZ=CUSTOMERGND

OE -

grounding bar // grounding bar

Cable Plan Formular: =Z6B++111+A/114

CA0061

+T=3C-W12

MCS* Cable Production

0,2 m - 70 mm²

△

M10 - female

+T=3C-X1

c

Page: 8

cable // FPE,, ring, M10-M10, 70mm²

+T=3C-GND

M10 - female

grounding bar // grounding bar

Cable Plan Formular: =Z6B++111+A/115

CA0074

+B=3CB-W1

MCS* Cable Production

0,2 m - 35 mm²

△

M10 - female

+B=3CB-X1

a

Page: 9

cable // FPE, ring, M8-M10, 35mm²

+B=3CB-GND

M8 - female

grounding bar //

Cable Plan Formular: =Z6B++111+A/116

CA0049

+B=3CB-W2

MCS* Cable Production

15 m - 50 mm²

△

M10 - female

+B=3CB-X1

b

Page: 9

cable // FPE, ring, M10-M10, 50mm²

M10 - female

+T=3C-X1

d

grounding bar //

Cable Plan Formular: =Z6B++111+A/117

77005032

+E=3D-W1

MCS* Cable Production

0,1 m - 16 mm²

M6 - female

+E=3D-GND

Page: 10

ground strap // M6-M6, 0,1 m

+E=3D-GND

M6 - female

grounding machine // grounding machine

77005032

+G=3D-W1

MCS* Cable Production

0,1 m - 16 mm²

M6 - female

+G=3D-GND

Page: 10

ground strap // M6-M6, 0,1 m

+G=3D-GND

M6 - female

grounding machine // grounding machine

77005035

+L1=3E1-W1

MCS* Cable Production

0,6 m - 16 mm²

M6 - female

+L1=3E1-GND2-1

Page: 11

ground strap // M6-M6, 0,6 m

+L1=3E1-GND2-2

M6 - female

grounding elements // grounding Location

CA0079

+L1=3E1-W10

MCS* Cable Production

8 m - 16 mm²

△

M10 - female

+B=3CB-X1

e

Page: 11

cable // FPE, ring, M6-M10, 16mm²

+L1=3E1-GND

M6 - female

grounding elements // grounding Location

Cable Plan Formular: =Z6B++111+A/118

G10000911

+HB=AA-W0

MCS* Cable Production

25 m - 4 mm²

△

OE -

+HB=AA-X0

Page: 19

cable sold by meter // PVC, 5x 4, PE, CY

+PZ=AA-X1

OE -

AC power supply cabinet // AC operating voltage

Cable Plan Formular: =Z6B++111+A/119

G10000797

+HP1=AAB-W1

MCS* Cable Production

30 m - 2 mm²

△

OE -

+HB=AAB-W1

Page: 22

cable sold by meter // PVC, 3x 2,5, PE, CY

+HP1=AAB-X0

OE -

AC power supply cabinet // feed-in

Cable Plan Formular: =Z6B++111+A/120

CA0092

+B=BI-W1

MCS* Cable Production

6 m - 2 mm²

△

OE -

+HB=BI-X1

Page: 43

cable // power line 7/8", 5-core

+B=BI-K1 X1

7/8 - female

DC power supply interface // EP power - distributor

Cable Plan Formular: =Z6B++111+A/121

77005809
+B=BI-W9

10 m - 0 mm²



RJ45 - male
+E=OB-X29

Page: 43

M8 - male
+B=BI-K1 L/A

cable network // EtherCAT, M8-RJ45, 10m

DC power supply interface // EP power - distributor

77005809
+HB=CXXA-W9

10 m - 0 mm²



Page: 46

cable network // EtherCAT, M8-RJ45, 10m

controller X,Xa // AX5206 <axis>

77005033
+HB=CXXA-WH

0,2 m - 16 mm²



Page: 46

ground strap // M6-M6, 0,2 m

controller X,Xa // AX5206 <axis>

G10000209
+HB=CY1-W9

0,25 m - 0 mm²



RJ45 - male
+HB=CY1-K1 X04

Page: 49

RJ45 - male
+HB=CXXa-K1 X05

cable network // EtherCAT, RJ45-RJ45, 0,25m

controller Y1 // AX5103 <axis>

Industrial Ethernet/EtherCAT patch cable

77005033
+HB=CY1-WH

0,2 m - 16 mm²



M6 - female
+HB=CY1-GND

Page: 49

M6 - female
+HB=CY1-K1 GND

ground strap // M6-M6, 0,2 m

controller Y1 // AX5103 <axis>

+L1=DC1-W1

2 m

Page: 51

potted line end

detection collision // torch magnetic holder

+L1=DC1B-WB1

2 m

Page: 52

+L1=DC1B-B1

potted line end

+L1=DC1B-X1

detection collision // torch dig switch tactile

Cable Plan Formular: =Z6B++111+A/122

G10000458
+L1=DL1-W1

15 m - 0 mm²



OE -
+HB=DL1-W1

Page: 53

M12 - female
+L1=DL1-B1

cable // OE-M12, 5-pole, socket, 15m

detection limiting // limiting Y+

Cable Plan Formular: =Z6B++111+A/123

G10000458
+L1=DL1B-W1

15 m - 0 mm²



OE -
+HB=DL1B-W1

Page: 54

M12 - female
+L1=DL1B-B1

cable // OE-M12, 5-pole, socket, 15m

detection limiting // limit Y-

Cable Plan Formular: =Z6B++111+A/124

CA0004
+G=EA-W1

MCS* Cable Production

3 m - 0 mm²

OE -
+HB=EA-W1

Page: 55

M12 - female
+HB=IT-K2

cable // OE-M12, 5-core

emergency stop additions // emergency stop

Cable Plan Formular: =Z6B++111+A/125

77002538
+HP1=EIB-W1

30 m - 0 mm²



OE -
+HB=EIB-W1

Page: 60

OE -
+HP1=EIB-X1

cable sold by meter // PVC, 7x 0,5,PE, CY

emergency stop interface // cabinet external

Cable Plan Formular: =Z6B++111+A/126

77003737
+E=ELC-W1

7,5 m - 0,34 mm²



M12 - female
+E=ELC-X1

Page: 61

+HB=IT-K5

cable // M12-M12, 4-pole, 7,5m

Not-Halt Reißleine // Reißleine vorne

			Toolbox_31	Option:	
			Toolbox version	customer:	Blocotelha
			automated	commission:	289001273
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle 01.04.2021

77003736
+E=ELC-W2

3 m - 0,34 mm²

M12 - female
+E=ELC-B1

Page: 61

M12 - male
+E=ELC-X2

cable //

Not-Halt Reißleine // Reißleine vorne

77003737
+E=ELD-W1

7,5 m - 0,34 mm²

M12 - female
+E=ELD-X1

Page: 62

M12 - male
+HB=IT-K5

cable // M12-M12, 4-pole, 7,5m

Not-Halt Reißleine // Reißleine hinten

77003736
+E=ELD-W2

3 m - 0,34 mm²

M12 - female
+E=ELD-B1

Page: 62

M12 - male
+E=ELD-X2

cable //

Not-Halt Reißleine // Reißleine hinten

G10000124
+B=IB-W1

0,3 m - 0 mm²

M8 - male
+B=BI-K1 1

Page: 66

M8 - female
+B=IB-K1 24V IN

cable // power line, M8-M8, 0,3m

interface box // Fieldbus star hub

77011322
+B=IB-W9

0,3 m - 0 mm²

M8 - male
+B=IB-K1 L/A

Page: 66

M8 - male
+B=BI-K1 L/A

cable network // EtherCAT, M8-M8, 0,3m

interface box // Fieldbus star hub

G10000253
+HB=IT-W9

2 m - 0 mm²

RJ45 - male
+HB=IT-K1 X1

Page: 67

RJ45 - male
+HB=CY1-K1 X05

cable network // EtherCAT, RJ45-RJ45, 2m

interface terminal // bus coupler

77004661
+HP1=ITB-W9

30 m - 0 mm²

77005037 // cable protection conduit M20 divisible

RJ45 - male
+HP1=ITB-K1 X1

Page: 72

RJ45 - male
+HB=IT-K1 X2

cable network // EtherCAT, RJ45-RJ45, 30 m

interface terminal // bus coupler

CA0000
+L1=LI1-W1

MCS* Cable Production

10 m - 1 mm²

OE -
+HB=LI1-W1

Page: 75

M22 - female
+L1=LI1-K1 J5

cable // OE-M22, 4-core

lifter // SL 200

Cable Plan Formular: =Z6B++111+A/127

G10000458
+L1=LI1-W2

15 m - 0 mm²

OE -
+HB=LI1-W2

Page: 75

M12 - female
+L1=LI1-K1 J3

cable // OE-M12, 5-pole, socket, 15m

lifter // SL 200

Cable Plan Formular: =Z6B++111+A/128

G10000243
+L1=LI1-W9

10 m - 0 mm²

M8 - male
+B=IB-K1 X2

Page: 75

M12 - male
+L1=LI1-K1 J1

cable network // EtherCAT, M8-M12, 10m

lifter // SL 200

77070866
+E=MX-W1

8 m - 1 mm²

M23 - female
+E=MX-M1

Page: 76

X13+X14 - female
+HB=CXXa-K1 X13

cable motor // motor, AMxxxx, OCT, 1,0+0,75+0,32, 8m

motor X // AM8531 OCT X

77005033
+E=MX-W10

0,2 m - 16 mm²

M6 - female
+E=MX-GND

Page: 76

M6 - female
+E=MX-M1

ground strap // M6-M6, 0,2 m

motor X // AM8531 OCT X

			Toolbox_31	Option:	
			Toolbox version	customer:	Blocotelha
			automated	commission:	289001273
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle 01.04.2021

Diagram illustrating a cable assembly configuration. The main cable is labeled **77005037** and **+P=NC-W1.1**. The cable length is **30 m - 20 mm²**. The assembly includes two **HM20 - female** connectors and a **+E=OP-1** cable accessory. The cable accessory is labeled **cable accessory // cable protection conduit M20 divisible**. The assembly is connected to a **network connectivity // network**.

The diagram illustrates the connection between the operation panel and the operation interface. The operation panel is represented by a box on the left, and the operation interface is represented by a box on the right. A cable, labeled '8 m - 16 mm²', connects the two. The cable is shown with a cross-section of three conductors (two red, one blue) and a green/yellow ground wire. The connection points are labeled 'M10 - female' on the operation panel side and 'M6 - female' on the operation interface side. The cable is also labeled '+E=OP-W10' and '+E=OP-P1 PE'. The diagram is titled 'CA0079' and 'MCS* Cable Production'.

			Toolbox_31	Option: customer: commission: Ed. by.				module		=Z6A	Page 107	
			Toolbox version		Blocotelha							
			automated		289001273							
Modification	Date	Name	evaluation has been generated		Thomas.Lehle 01.04.2021			layout plan	sector 111 location			+A

77005783
+E=OP-W17

3 m - 0 mm²



RJ45 - male
+E=OP-X17

Page: 83
cable network // EtherCAT, RJPP-RJ45, 3m

RJPP - male
+E=OP-P1 X106

operation panel // Operation interface

77005783
+E=OP-W18

3 m - 0 mm²



RJ45 - male
+E=OP-X18

Page: 83
cable network // EtherCAT, RJPP-RJ45, 3m

RJPP - male
+E=OP-P1 X105

operation panel // Operation interface

77002882
+HJ1=PA1-W0

5 m - 35 mm²



OE -
+HJ1=PA1-XPE

Page: 87
cable sold by meter // PVC, 4x35, PE, CY

OE -
+P=PS1-E1 TB1

plasma power switch box // XPR 170

Cable Plan Formular: =Z6B++111+A/133

77002882
+HJ1=PA1-W00

8 m - 35 mm²



OE -
+HJ1=PA1-XPE

Page: 87
cable sold by meter // PVC, 4x35, PE, CY

OE -
+PZ=PA1-F0

plasma power switch box // XPR 170

Cable Plan Formular: =Z6B++111+A/134

0571423
+HJ1=PA1-W1

9 m - 1 mm²



OE -
+HJ1=PA1-X1

Page: 87
cable sold by meter // PVC, 5x 1,5, PE, CY

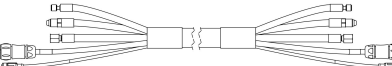
OE -
+HP1=PA1-X1

plasma power switch box // XPR 170

Cable Plan Formular: =Z6B++111+A/135

72424337
+L1=PB1-W1

7,5 m - 0 mm²



\$XPR300SGB - female
+E=PG1-E1 J2

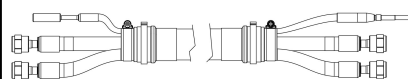
Page: 89
plasma connection // HYP, Set gas console(Core), 3x connection, 7,5m

\$XPR300SGB - female
+L1=PB1-E1 J1

plasma torch connection unit // type EasyConnect

72424334
+L1=PB1-W2

7,5 m - 0 mm²



\$XPR300SGB - female
+E=PG1-E1 OUT2

Page: 89
plasma connection // HYP, Set gas console(Core), 3x connection, 7,5m

\$XPR300SGB - female
+L1=PB1-E1 IN

plasma torch connection unit // type EasyConnect

CA0085
+L1=PB1-W10

MCS* Cable Production

19 m - 35 mm²



M10 - female
+T=3C-X1

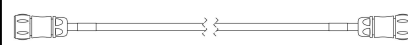
Page: 89
cable // FPE, ring, M6-M10, 35mm²

M6 - female
+L1=PB1-E1 GND

plasma torch connection unit // type EasyConnect

72424307
+E=PG1-W1

25 m - 0 mm²



Page: 91
plasma connection // HYP, power cord, gas console, 25m

plasma gas console // type core

72424329
+E=PG1-W2

25 m - 0 mm²



Page: 91
plasma connection // HYP, control cable, D-Sub, 25m

plasma gas console // type core

72424318
+E=PG1-W3

25 m - 0 mm²



Page: 92
hose gas // hose, Set, coolant, 25m

plasma gas console // type core

72424285
+E=PG1-W4

25 m - 0 mm²



Page: 92
plasma connection // HYP, pilot arc line, 25m

plasma gas console // type core

			Toolbox_31	Option:	
			Toolbox version	customer:	Blocotelha
			automated	commission:	289001273
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle 01.04.2021



layout plan

module

sector 111 location

+A

=Z6A

Page
108

72424710 25 m - 50 mm² ☐

+E=PG1-W5

Page: 92

plasma connection // HYP, negative lead, strain relief, 10m

plasma gas console // type core

CA0085 MCS* Cable Production 5 m - 35 mm² ☐

+E=PG1-W10

M10 - female Page: 91 M6 - female

+B=3CB-X1 cable // FPE, ring, M6-M10, 35mm² **+E=PG1-E1 GND**

plasma gas console // type core

0551042 5 m - 1 mm² ☐

+P=PS1-W2

OE - Page: 95 OE -

+HP1=PS1-W2 cable sold by meter // PVC, 3x 1, PE, CY **+P=PS1-E1 J14**

plasma source // type XPR 170

Cable Plan Formular: =Z6B++111+A/136

72424347 6 m - 0 mm² ☐

+P=PS1-W9

RJ45 - male Page: 95 RJ45 - male

+HP1=PS1-K3 X1 plasma connection // HYP, EtherCAT, RJ45-RJ45, 6m **+P=PS1-E1 U47**

plasma source // type XPR 170

CA0086 MCS* Cable Production 20 m - 50 mm²

+P=PS1-W10

M10 - female Page: 95 M6 - female

+T=3C-X1 cable // FPE, ring, M6-M10, 50mm² **+P=PS1-E1 GND**

plasma source // type XPR 170

Cable Plan Formular: =Z6B++111+A/137

72424720 20 m - 50 mm² ☐

+P=PS1-W11

XPR170WL - female Page: 95 M10 - female

+P=PS1-E1 J2.1 plasma // workpiece cable, 50mm², 15m **+T=PS1-GND**

plasma source // type XPR 170

72424391 2 m - 0 mm² ☐

+L1=PT1-W1

\$XPR300TL - female Page: 96 \$XPR300TL - female

+L1=PB1-E1 J2 plasma torch // HYP, machine torch, hose package, 2m **+L1=PT1-E1 Torch**

plasma torch // type XPR 300 torch

CA0067 MCS* Cable Production 2 m - 4 mm² ☐

+L1=PT1-W2

FS6,3 - female Page: 96 M6 - female

+L1=PT1-E1 GND cable // M6-POC, 1-core, ring-socket **+L1=PT1-GND**

plasma torch // type XPR 300 torch

Cable Plan Formular: =Z6B++111+A/138

77070094 8 m - 0 mm² ☐

+P=SP-W1

OE - Page: 97 OE -

+HP1=SP-W1 cable sold by meter // PVC, 3x 0,5, PE, CY **+P=SP-X1**

sensor pressure // differential pressure monitor

Cable Plan Formular: =Z6B++111+A/139

77070033 8 m - 0 mm² ☐

+P=UE-W1

OE - Page: 99 OE -

+HP1=UE-W1 cable sold by meter // PVC, 5x 0,75, PE, CY **+P=UE-PE1**

unit exhaustion // external control

Cable Plan Formular: =Z6B++111+A/140

----- 2 m

+L1=UP1-WE1

+L1=UP1-X1 Page: 100

J7 08:3 potted line end **+L1=UP1-E1**

unit pointer // laser diode

Cable Plan Formular: =Z6B++111+A/141

G10000957 10 m - 0 mm² ☐

+E=UTB-W1

OE - Page: 102 M12 - female

+HB=UTB-X1 cable // OE-M12, 5-polig, Buchse, 10m **+E=UTB-X1 -X1**

unit temperature // climatisation

Cable Plan Formular: =Z6B++111+A/142

			Toolbox_31	Option:	
			Toolbox version	customer:	Blocotelha
			automated	commission:	289001273
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle 01.04.2021



layout plan

module

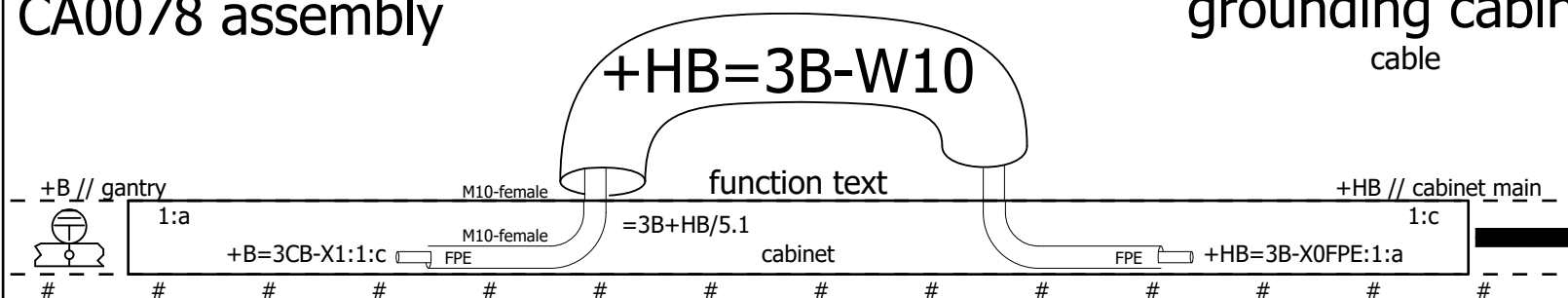
sector 111 location

+A

=Z6A

grounding cabinet
cable

length: 7 m

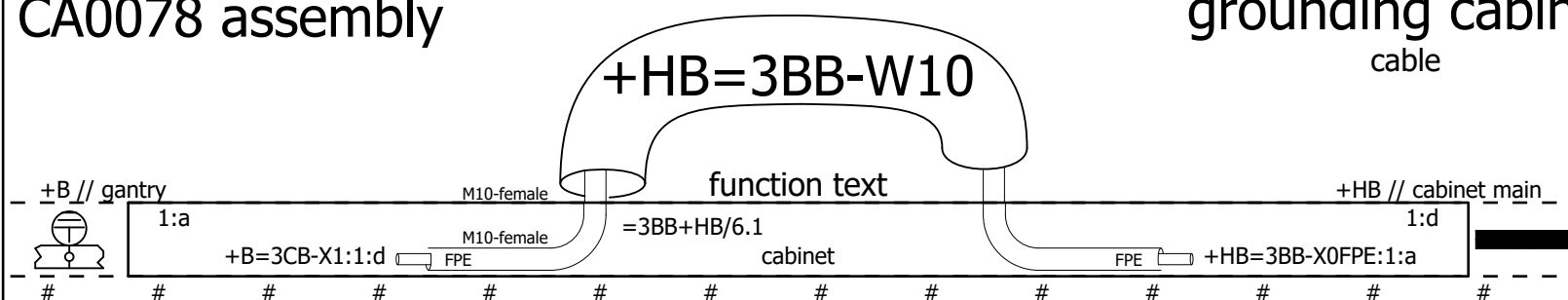


Quantity	Designation	Part number	A
	cable sold by meter PVCT, 1x16	77070130	
1	cable accessory cable lug, M10, 16mm ²	556027	

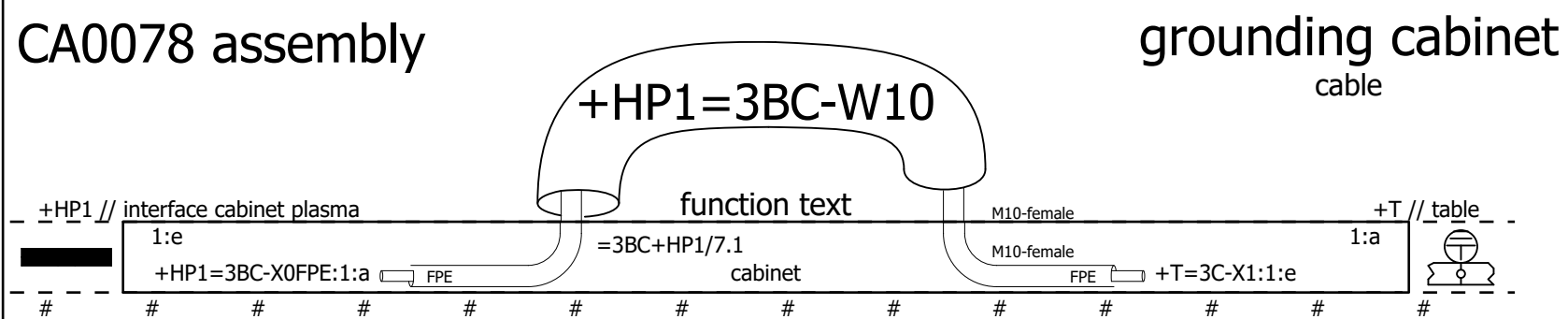
			Toolbox_31	Option: customer: commission: Ed. by:				module	sector 111 location +A =Z6B	Page 110
			Toolbox version		Blocotelha					
			automated		289001273					
Modification	Date	Name	evaluation has been generated		Thomas.Lehle 01.04.2021		connection plan			

grounding cabinet
cable

length: 7 m



Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130	A
1	cable accessory cable lug, M10, 16mm ²	556027	



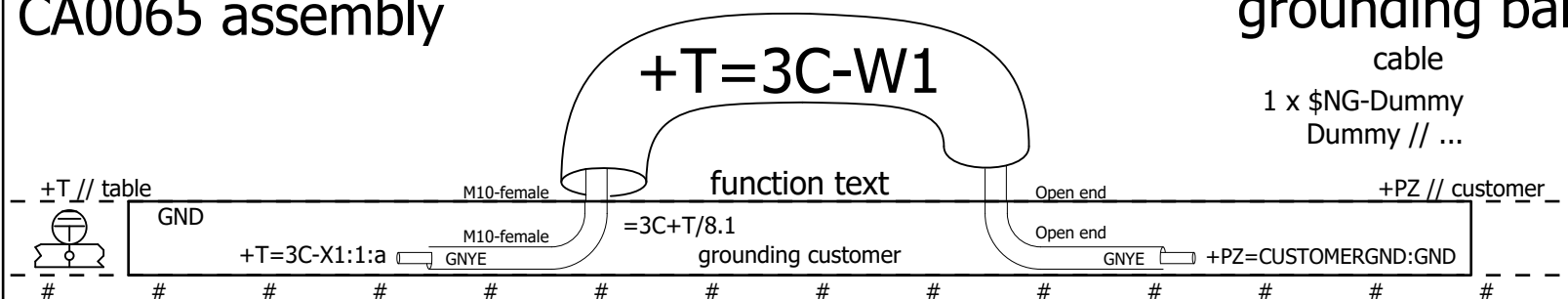
description: FPE, ring, OE-M10, 16mm²
number of cores:
cross section 16 mm²
length: 18 m

Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130	A
1	cable accessory cable lug, M10, 16mm ²	556027	

grounding bar

1 x \$NG-Dummy
Dummy // ...

length: 14 m

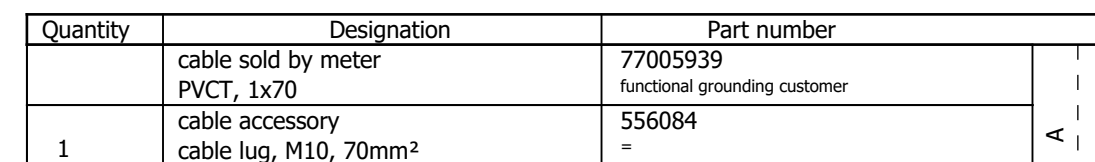


Quantity	Designation	Part number	
	cable sold by meter PVC, 1x70, PE	0551080 grounding customer	A
1	cable accessory cable lug, M10, 70mm ²	556084 =	

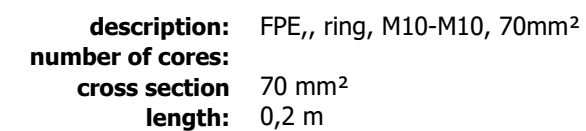
grounding bar

cable

description:	FPE, ring, OE-M10, 70mm ²
number of cores:	
cross section	70 mm ²
length:	14 m

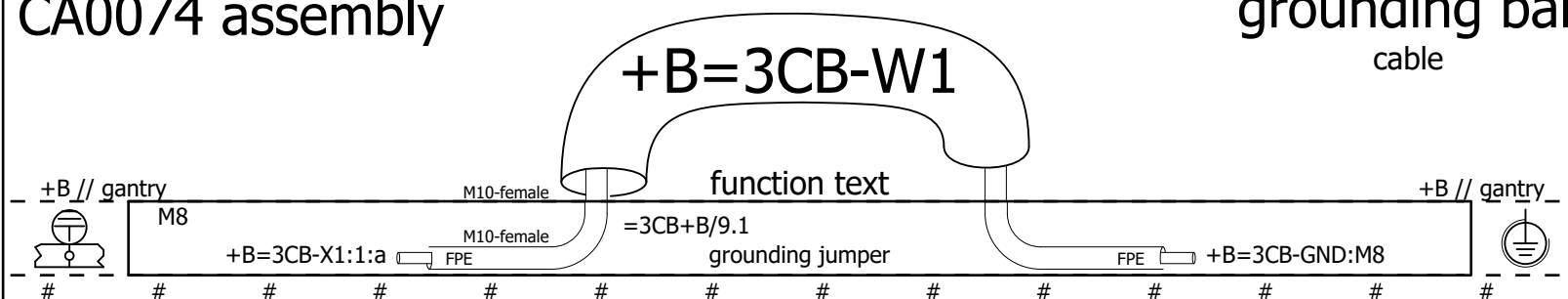


grounding bar
cable



Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x70	77005939 grounding table	A
1	cable accessory cable lug, M10, 70mm ²	556084 =	
1	cable accessory cable lug, M10, 70mm ²	556084 =	

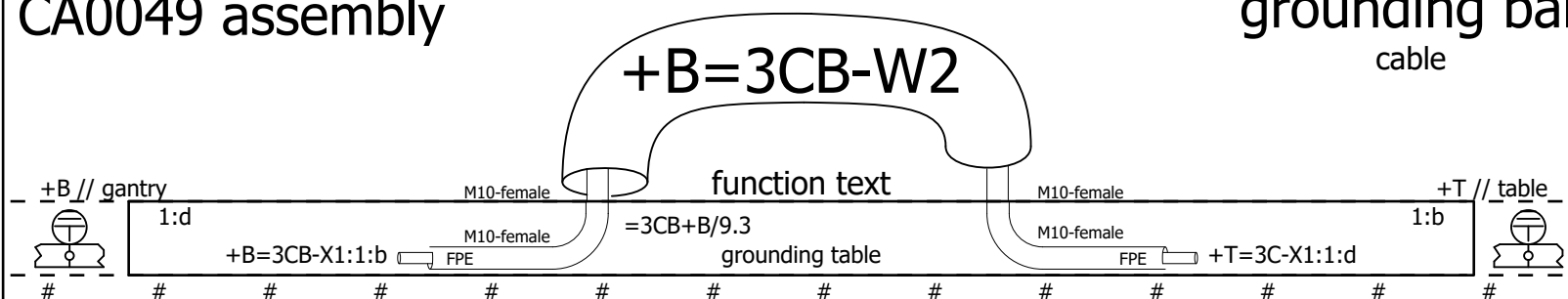
grounding bar
cable



description:	FPE, ring, M8-M10, 35mm ²
number of cores:	
cross section	35 mm ²
length:	0,2 m

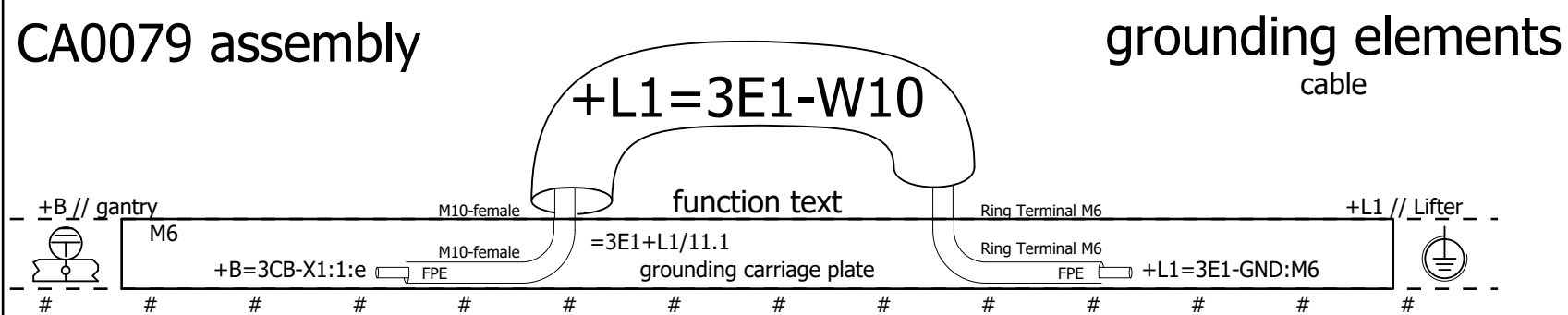
Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x35	77005938 grounding jumper	A
Stück 1	cable accessory cable lug, M8, 35mm²	556080 =	
Stück 1	cable accessory cable lug, M10, 35mm²	G10000648 =	

grounding bar
cable



description:	FPE, ring, M10-M10, 50mm ²
number of cores:	
cross section	50 mm ²
length:	15 m

Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x50	G10000792 grounding table	A
1	cable accessory cable lug, M10, 50mm ²	556083 =	
1	cable accessory cable lug, M10, 50mm ²	556083 =	



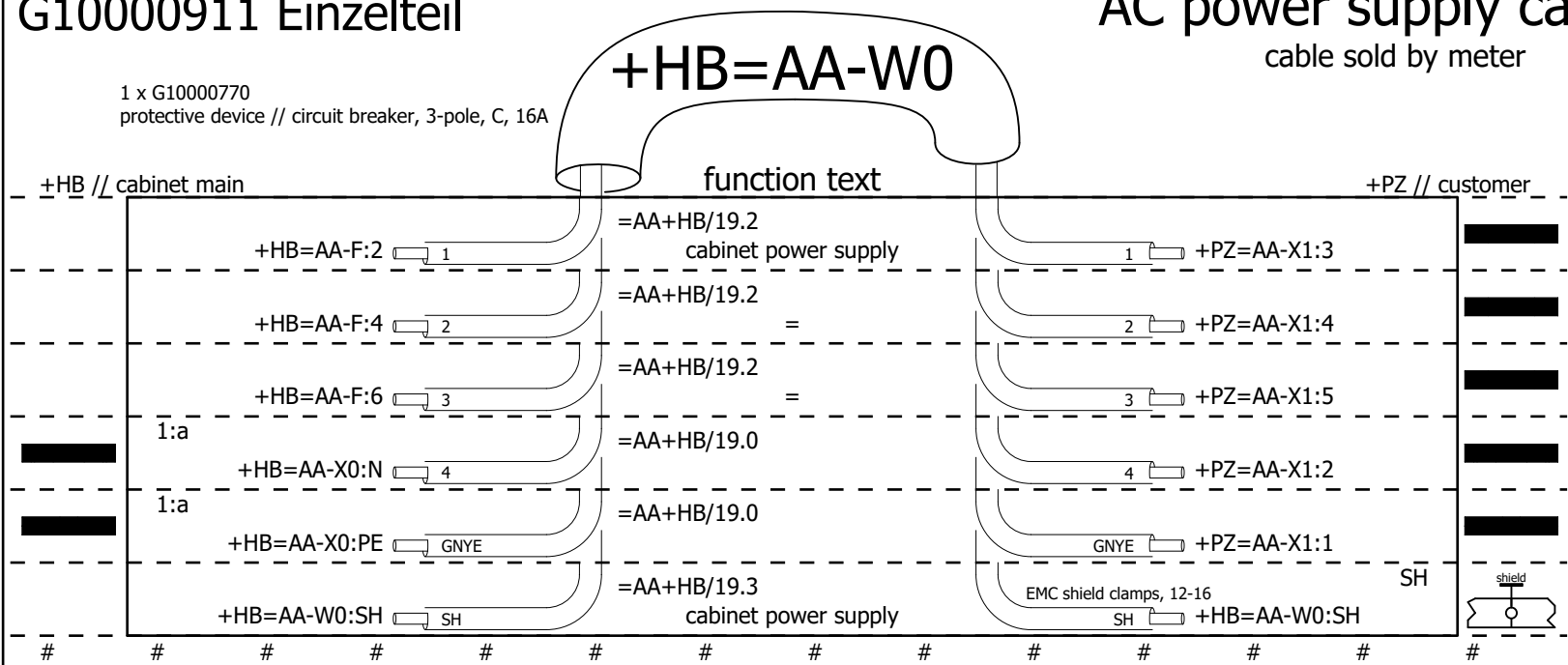
description: FPE, ring, M6-M10, 16mm²
number of cores:
cross section 16 mm²
length: 8 m

Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130	A B
		grounding carriage plate	
1	cable accessory cable lug, M6, 16mm ²	556018 =	
1	cable accessory cable lug, M10, 16mm ²	556027 =	

G10000911 Einzelteil

AC power supply cabinet
cable sold by meter

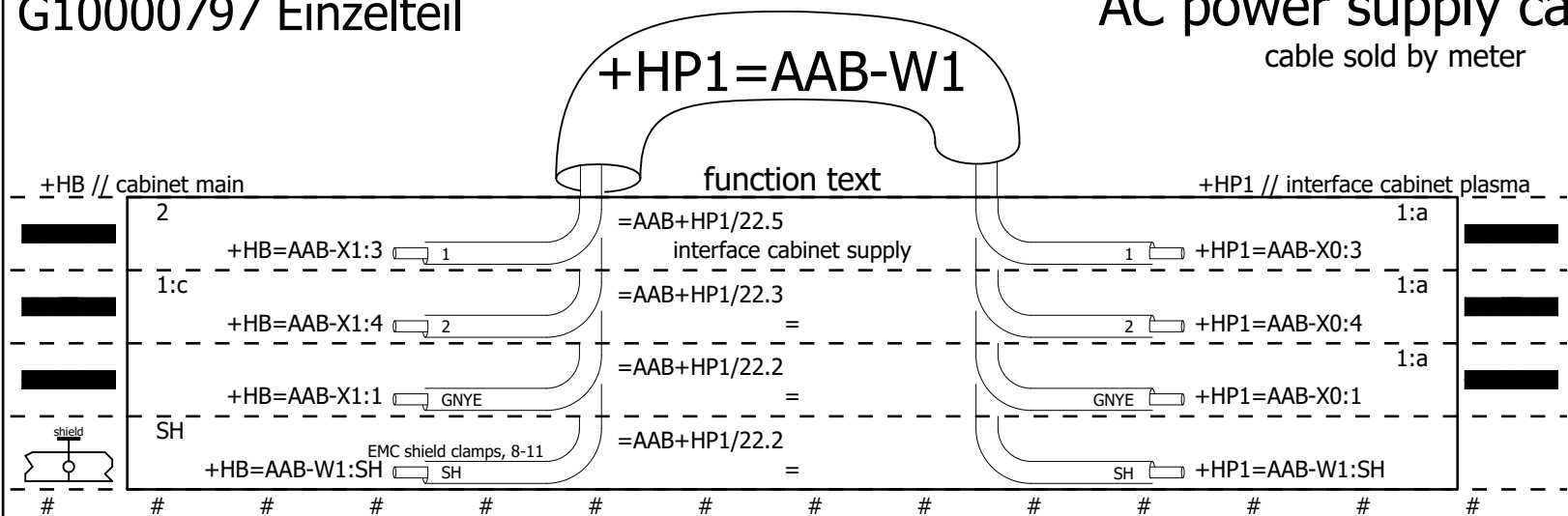
description: PVC, 5x 4, PE, CY
number of cores: 5
cross section: 4 mm²
length: 25 m



G10000797 Einzelteil

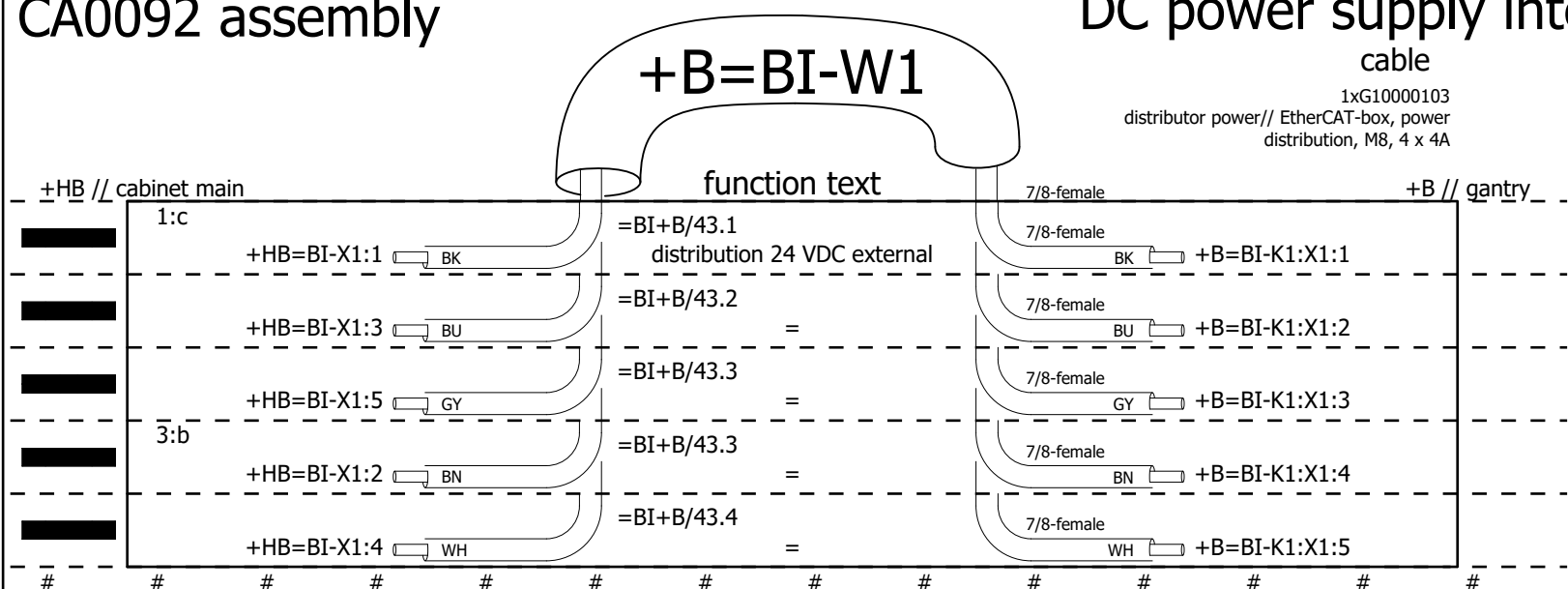
AC power supply cabinet
cable sold by meter

description: PVC, 3x 2,5, PE, CY
number of cores: 3
cross section 2 mm²
length: 30 m
Add-on: cable accessory: G10000423
Schirmklammer, Zugentlastung, Hutschienen, 8-11



DC power supply interface

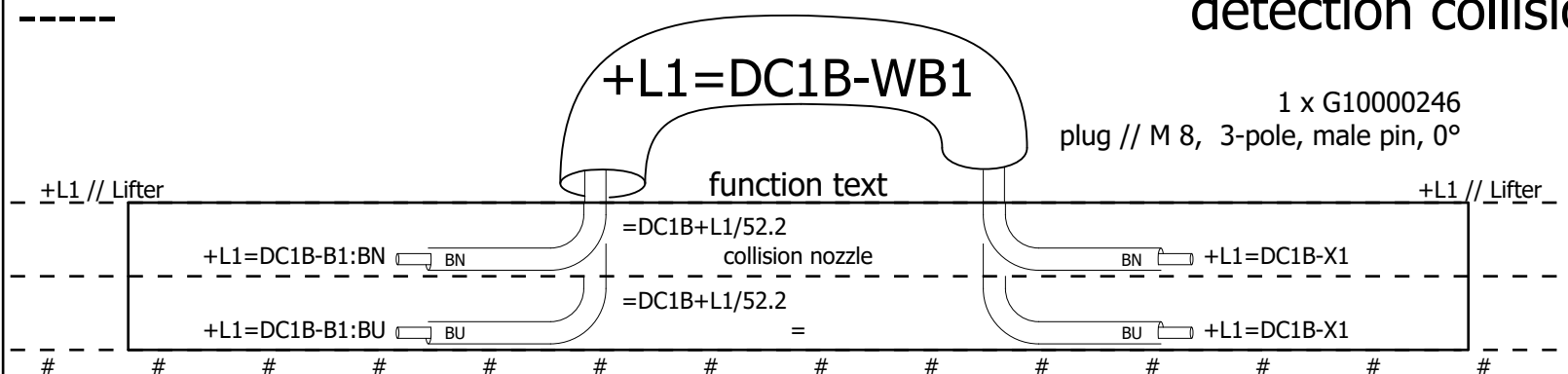
length: 6 m



Quantity	Designation	Part number
1	cable OE-7/8", socket, 5 x 2,5, 10m	77011391 distribution 24 VDC external

			Toolbox_31	Option: customer: commission:				module		Page 121	
			Toolbox version		Blocotelha						
			automated		289001273						
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle	01.04.2021	connection plan	sector 111	location +A	=Z6B	

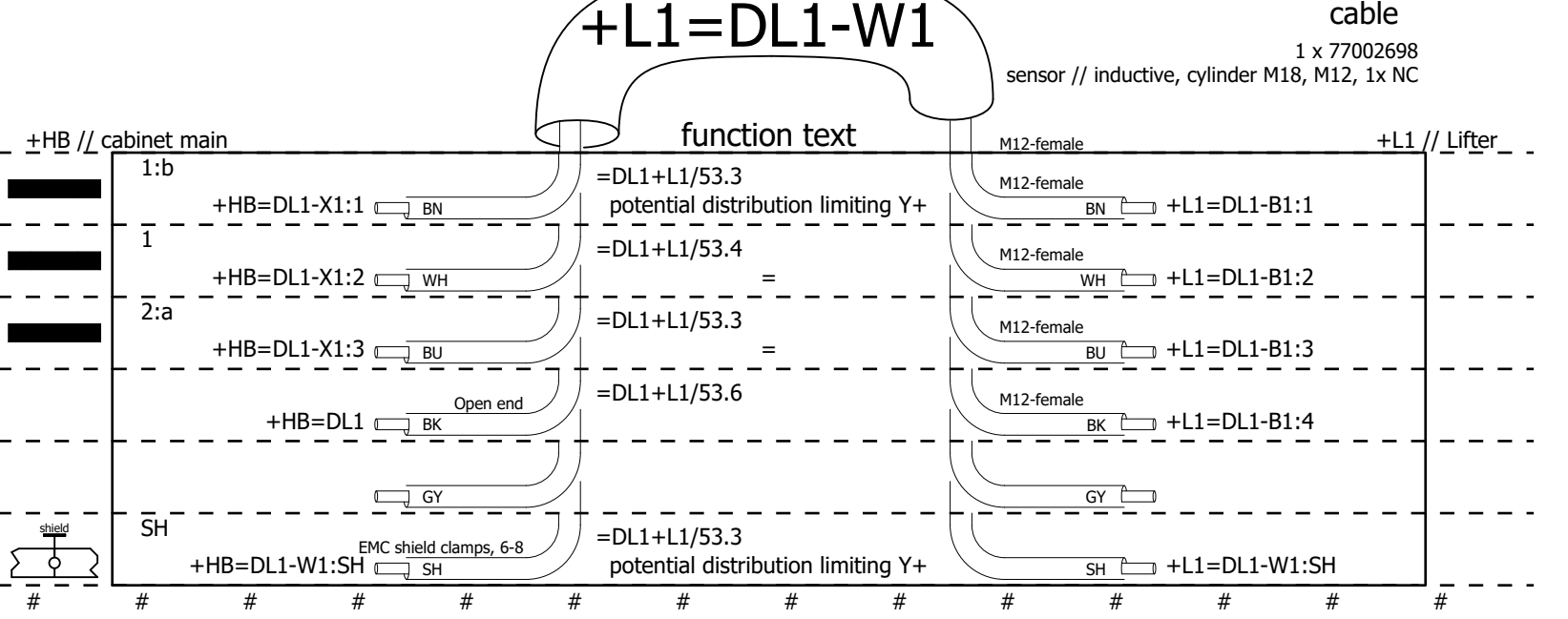
description:
number of cores:
cross section
length: 2 m



			Toolbox_31	Option: customer: commission:				module	sector 111 location +A =Z6B	Page 122
			Toolbox version		Blocotelha					
			automated		289001273					
Modification	Date	Name	evaluation has been generated	Ed. by:	Thomas.Lehle 01.04.2021		connection plan			

G10000458 Einzelteil

detection limiting



description: OE-M12, 5-pole, socket, 15m
number of cores: 5
cross section: 0 mm²
length: 15 m



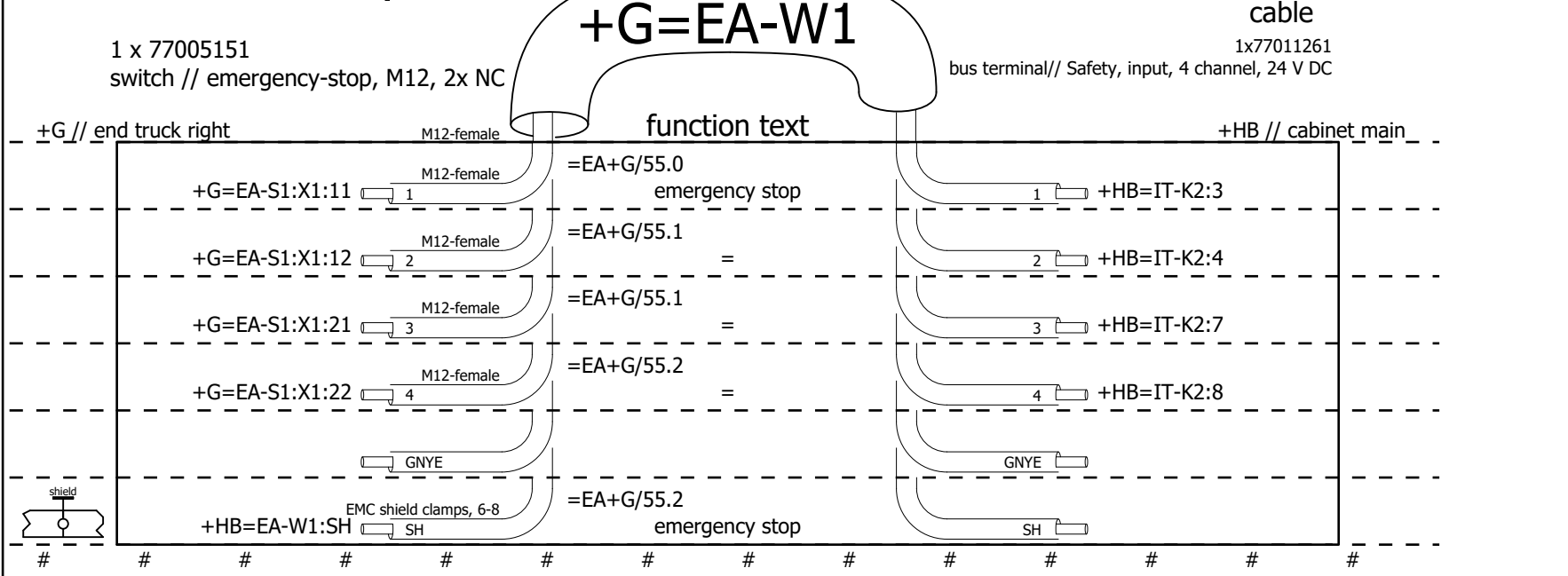
detection limiting

cable
1 x 77002698



CA0004 assembly

emergency stop additions

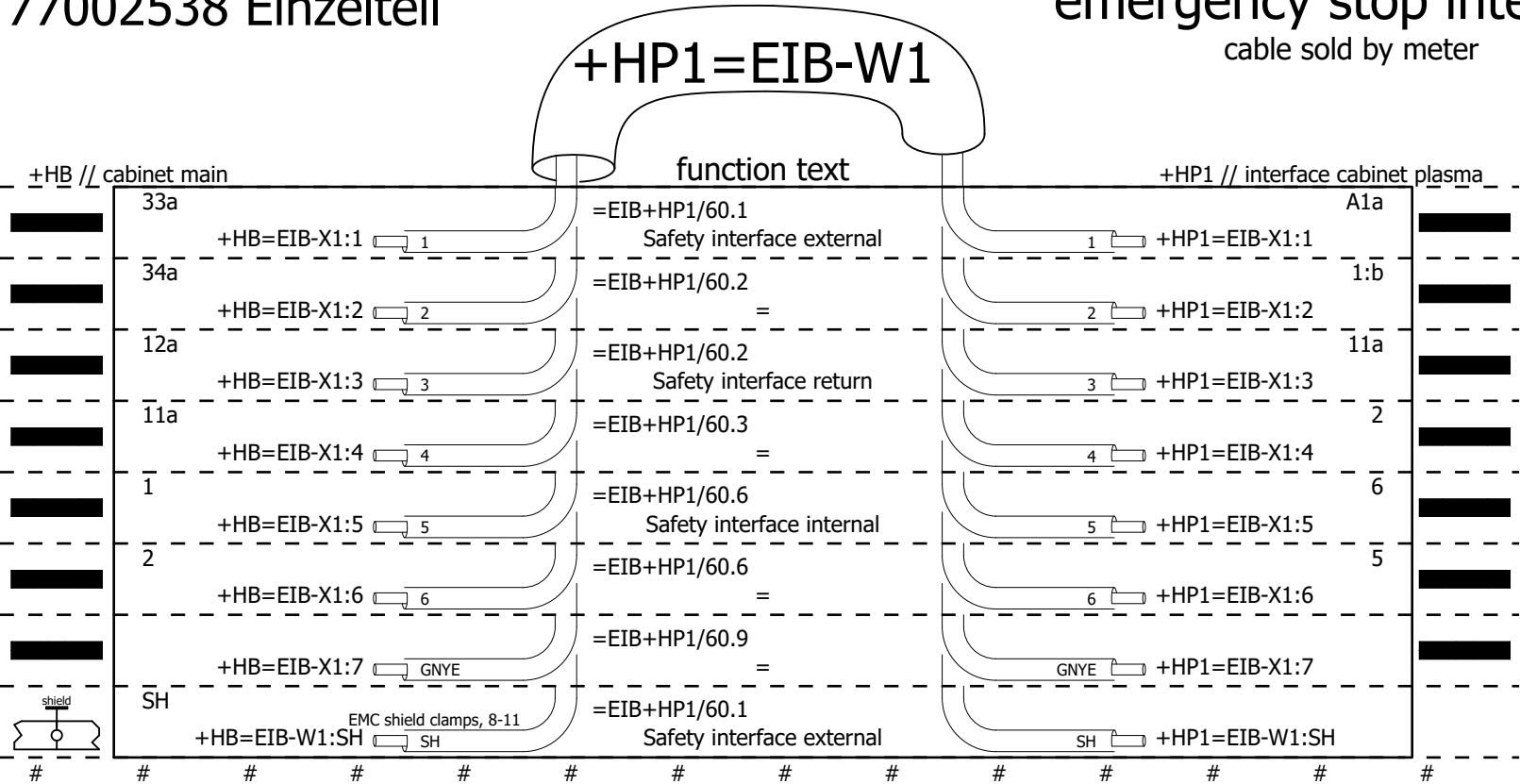


77002538 Einzelteil

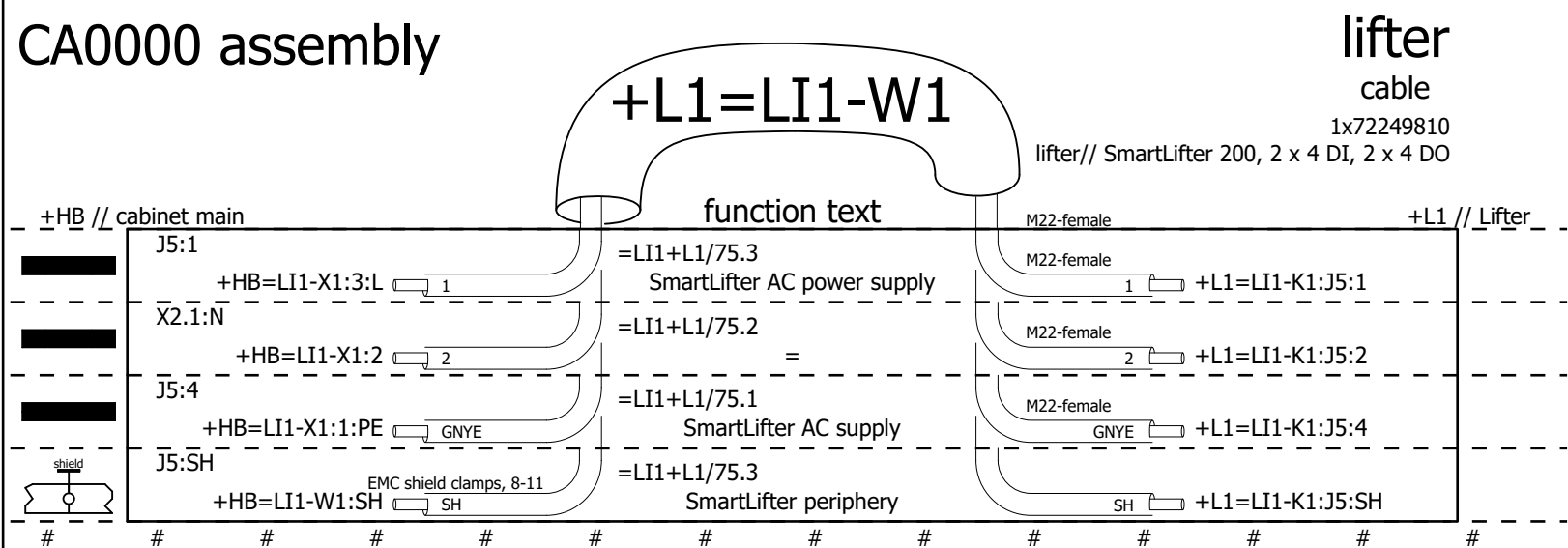
emergency stop interface

cable sold by meter

description: PVC, 7x 0,5,PE, CY
number of cores: 7
cross section: 0 mm²
length: 30 m



CA0000 assembly



description: OE-M22, 4-core
number of cores:
cross section 1 mm²
length: 10 m

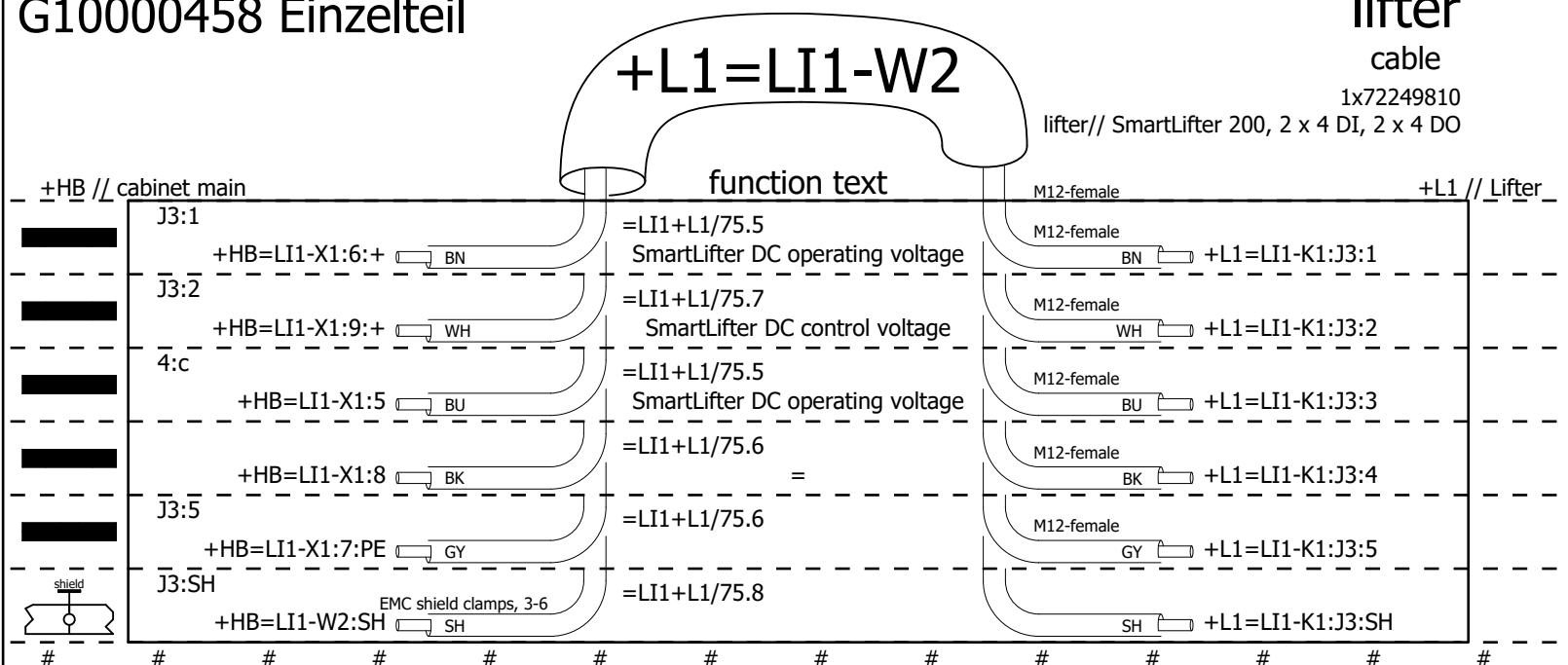
Quantity	Designation	Part number	
	cable sold by meter PVC, 3x 1, PE, CY	0551042	A
Stück 1	plug M22, 4-pole, socket, 0°	77070191	

cable

1x72249810

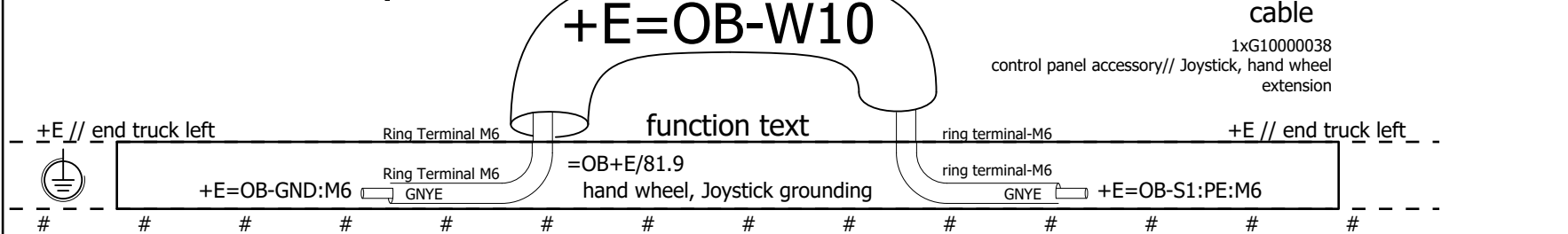
lifter// SmartLifter 200, 2 x 4 DI, 2 x 4 DO

description:	OE-M12, 5-pole, socket, 15m
number of cores:	5
cross section	0 mm ²
length:	15 m



CA0167 assembly

operation panel Keyboard



description: grounding, PE, ring, M6- M6, 4mm²

number of cores:

cross section 4 mm²

length: 0,3 m

Quantity	Designation	Part number	
	cable sold by meter PVC, 1x 4	570130	A B
		hand wheel, Joystick grounding	
1	cable accessory cable lug, M6, 16mm ²	556018 =	
1	cable accessory cable lug, M6, 16mm ²	556018 =	

description:	OE-M23, 19-pole, socket (plug female), 10m
number of cores:	16+3
cross section	0 mm ²
length:	10 m



operation panel

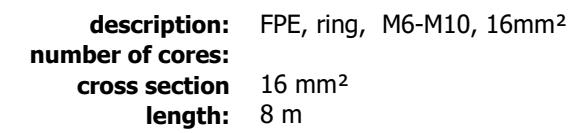
1xG10000760
PC control panel// display 15", GCS, touch display,
Win10, export

description:	M23-M23, 16+5-core
number of cores:	
cross section	0 mm ²
length:	0,6 m

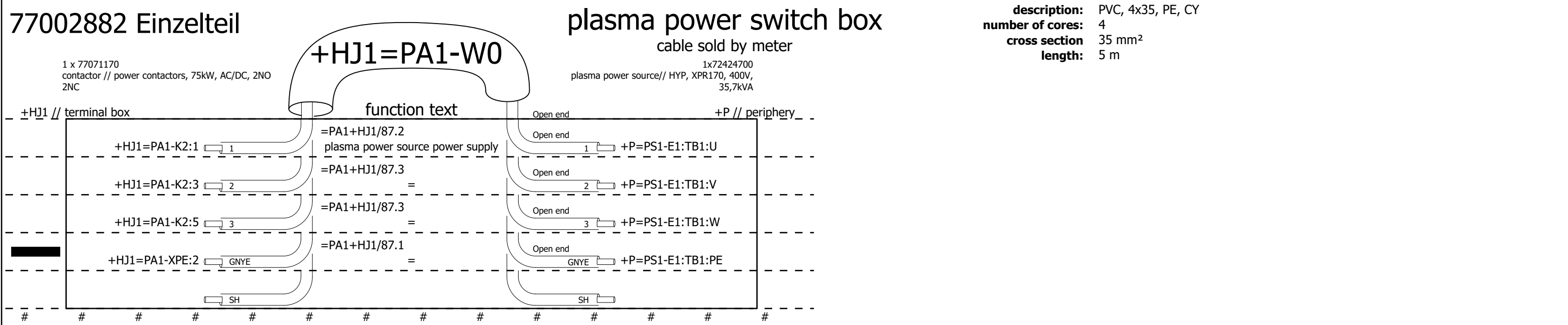


Quantity	Designation	Part number	
	cable sold by meter PVC, 16x0,34+5G0,75, PE, CY	77004655 operation panel battery pack	A A BC D
Stück 1	plug accessory M23, 16+3-pole, plug, 0°	77002794 =	
Stück 1	cable accessory cable gland, PG13,5, EMV	0546050 =	
Stück 1	plug accessory M23, 16+3-pole, socket, 90°	77070966 =	
Stück 1	cable accessory cable gland, PG13,5, EMV	0546050 =	

operation panel



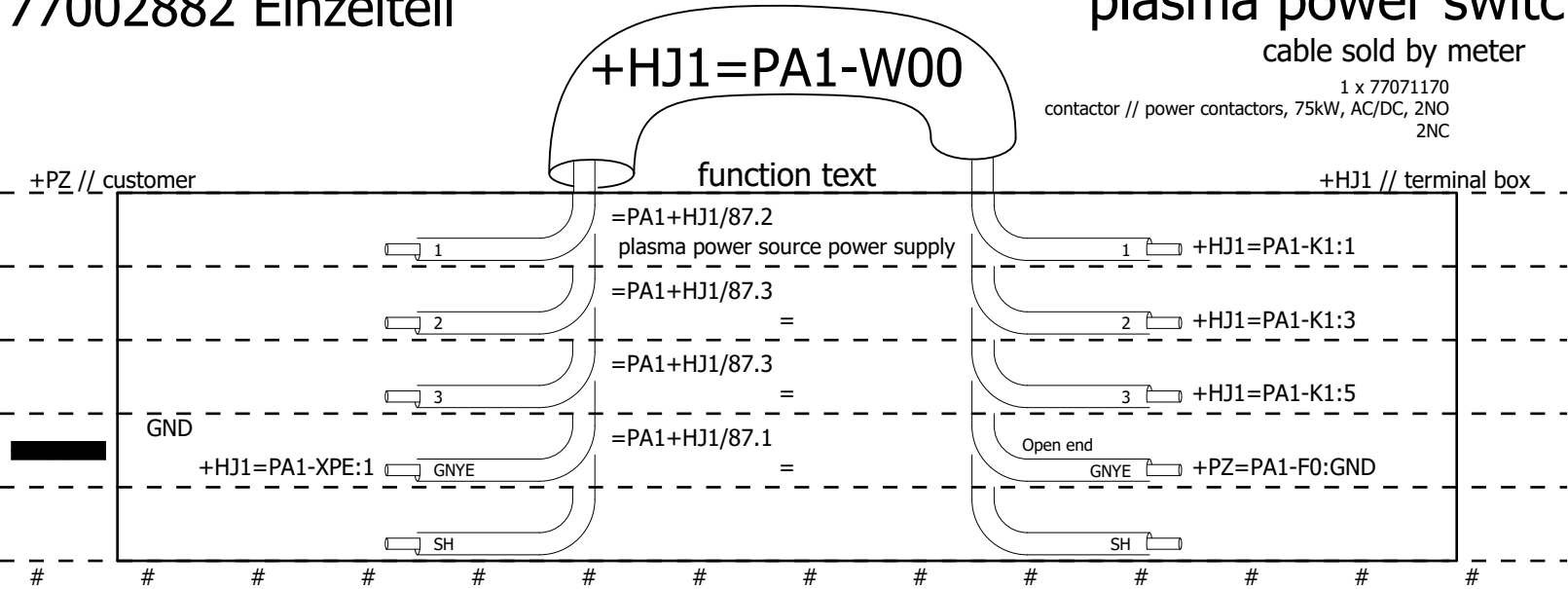
Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x16	77070130 panel grounding	A
1	cable accessory cable lug, M6, 16mm²	556018 =	
1	cable accessory cable lug, M10, 16mm²	556027 =	



77002882 Einzelteil

plasma power switch box

description: PVC, 4x35, PE, CY
number of cores: 4
cross section: 35 mm²
length: 8 m



0571423 Einzelteil

plasma power switch box

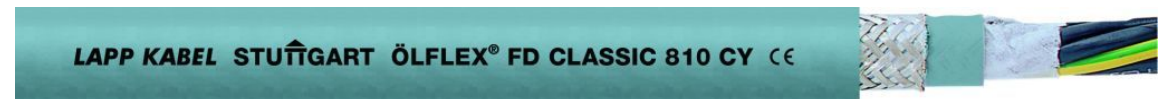
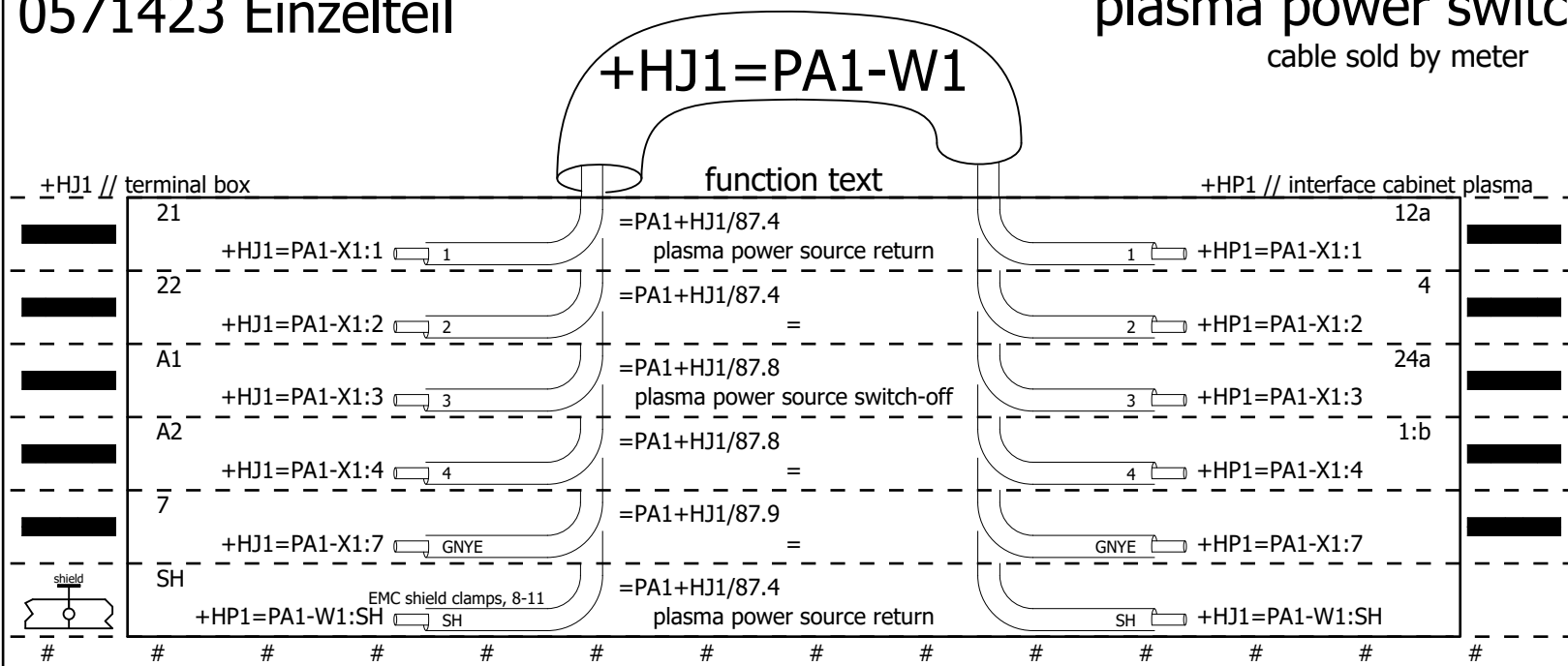
cable sold by meter

description: PVC, 5x 1,5, PE, CY

number of cores: 5

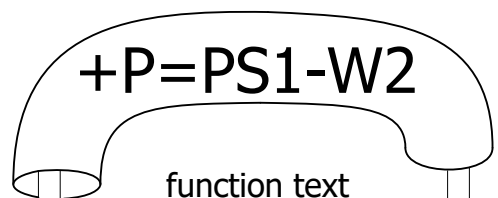
cross section: 1 mm²

length: 9 m



0551042 Einzelteil

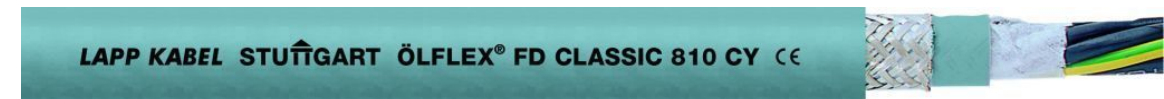
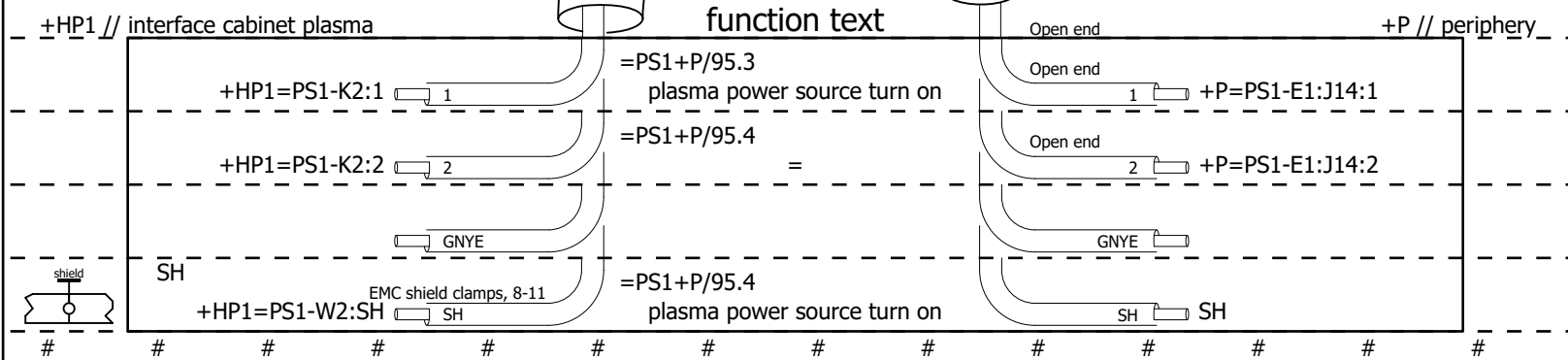
1x77011138
bus terminal// digital, relay, 2x NO , 230 V AC/30 V
DC



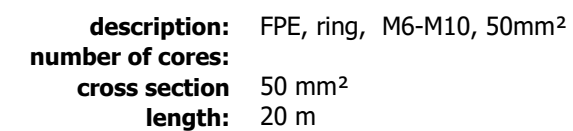
plasma source

1x72424700
plasma power source// HYP, XPR170, 400V,
35,7kVA

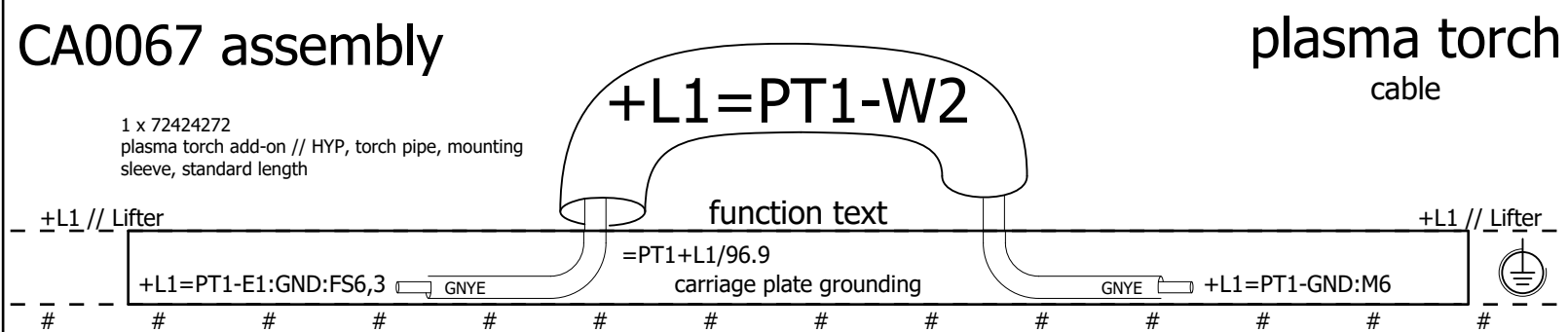
description: PVC, 3x 1, PE, CY
number of cores: 3
cross section: 1 mm²
length: 5 m



plasma source

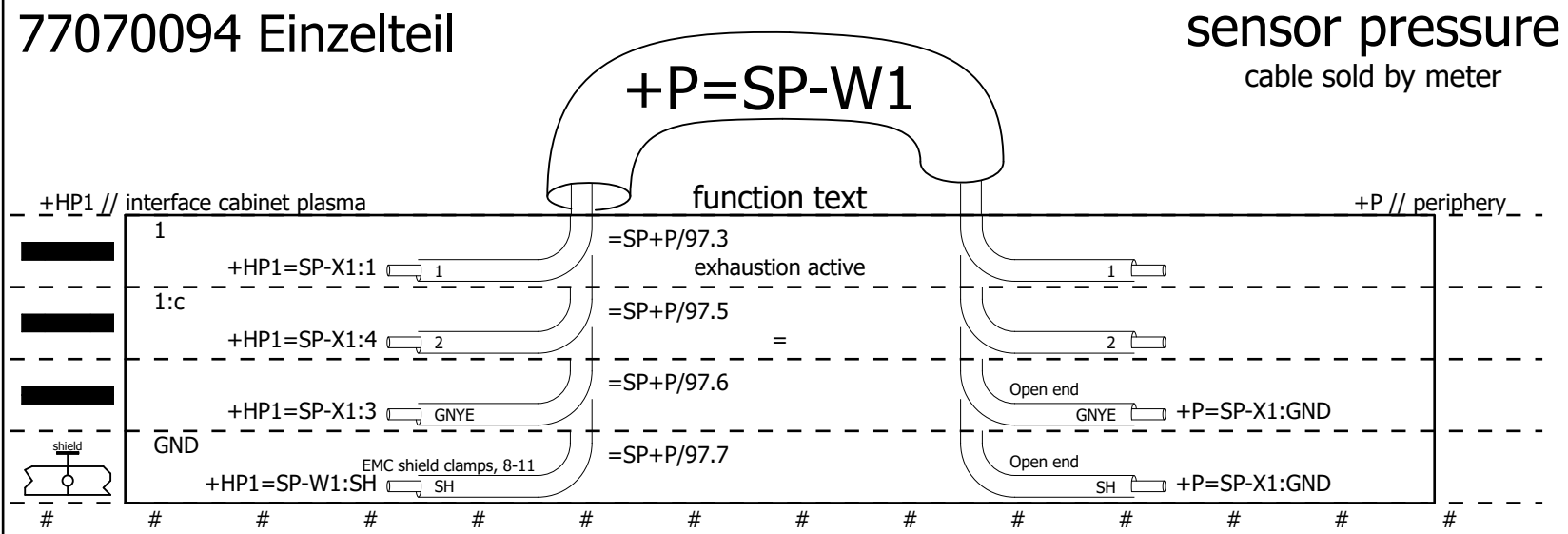


Quantity	Designation	Part number	
	cable sold by meter PVCT, 1x50	G10000792 plasma power source functional ground	A
Stück 1	cable accessory cable lug, M6, 50mm²	G10000672 =	
1	cable accessory cable lug, M10, 50mm²	556083 =	

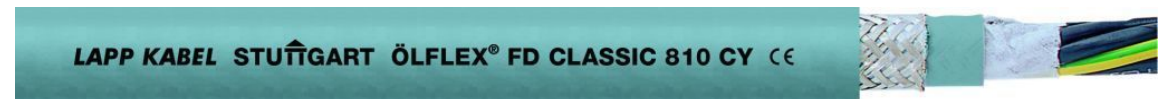


description: M6-POC, 1-core, ring-socket
number of cores:
cross section 4 mm²
length: 2 m

Quantity	Designation	Part number	
	cable sold by meter PVC, 1x 4	570130	A B
		carriage plate grounding	
Stück 1	cable accessory flat receptacle, 6,3mm, 4-6mm²	77070930 =	
1	cable accessory ring cable lug, M6, 6mm²	0556126 =	



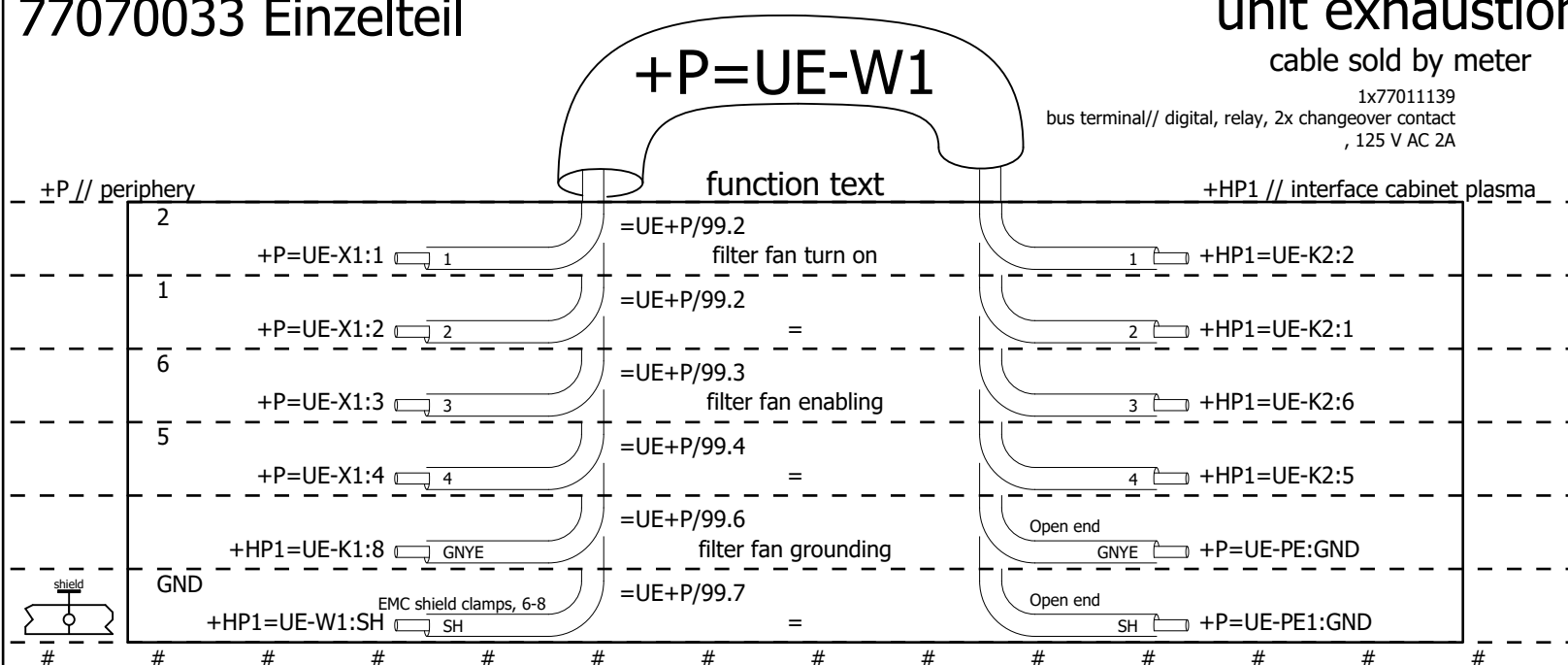
description: PVC, 3x 0,5, PE, CY
number of cores: 3
cross section: 0 mm²
length: 8 m

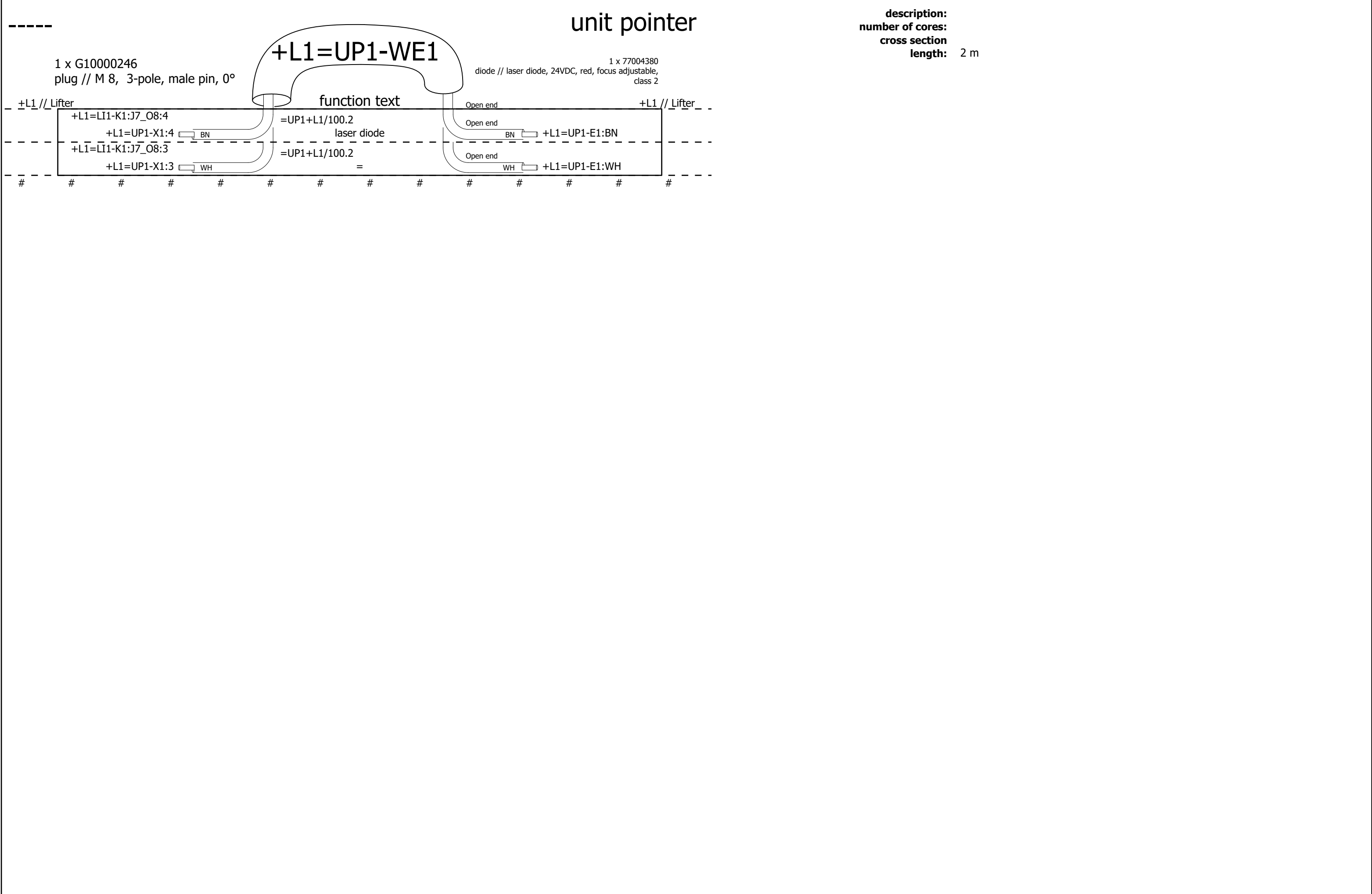


cable sold by meter

bus terminal// digital, relay, 2x changeover contact
, 125 V AC 2A

description: PVC, 5x 0,75, PE, CY
number of cores: 5
cross section 0 mm²
length: 8 m





description:
number of cores:
cross section
length: 2 m

G10000957 Einzelteil

unit temperature

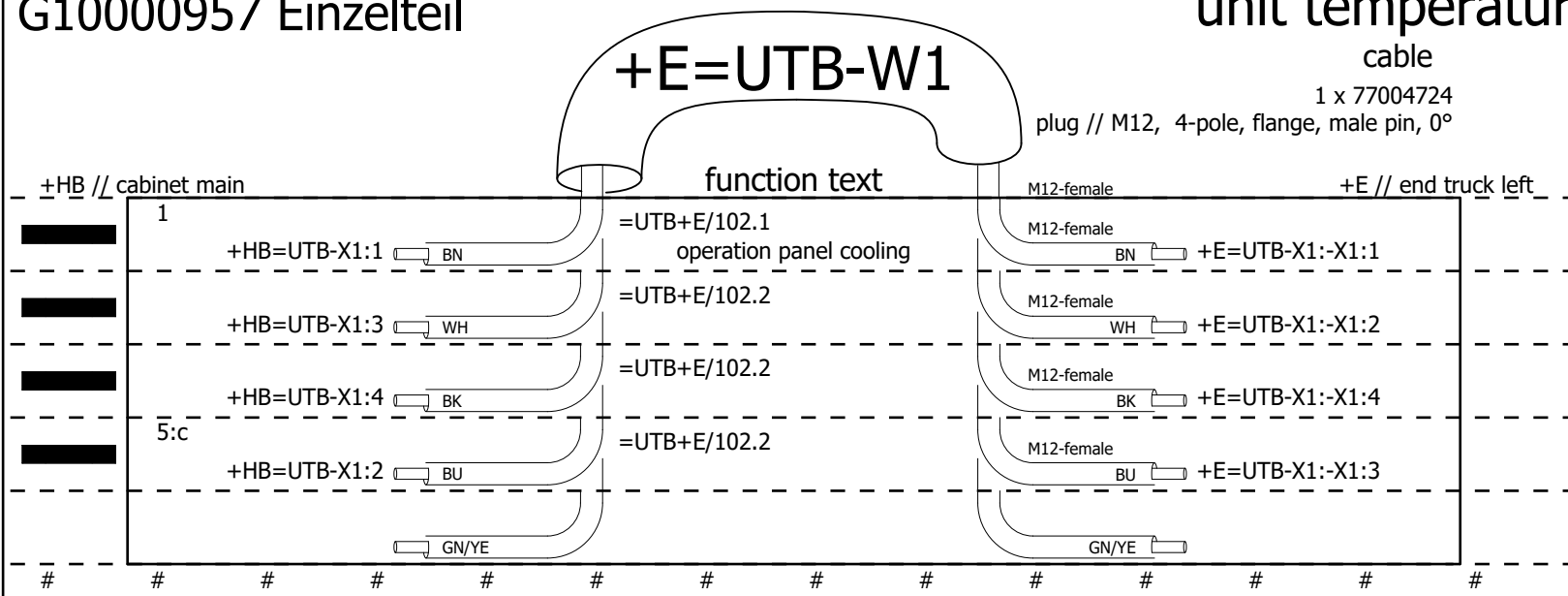
description:

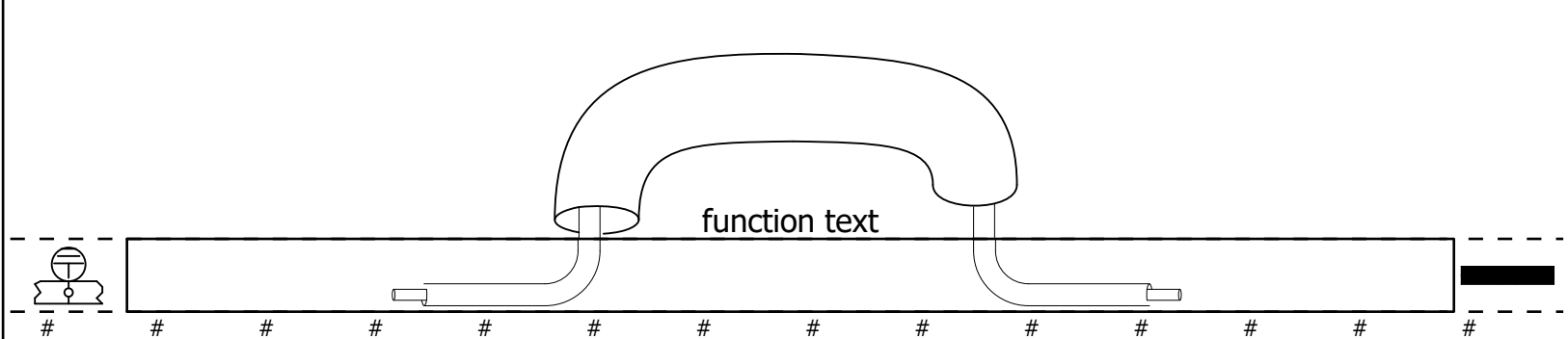
number of cores:

cross section

length:

OE-M12, 5-polig, Buchse, 10m
5
0 mm²
10 m





description:
number of cores:
cross section
length:

